



# **BHARATI VIDYAPEETH**

**(Deemed to be University), Pune**

**'A+' Accreditation (Third Cycle) by 'NAAC' in 2017**

**Category-I Deemed to be University Graded by UGC**

**'A' Grade University Status by MHRD Govt. of India**

**Ranked 76<sup>th</sup> by NIRF – 2022**

## **FACULTY OF MANAGEMENT STUDIES**

### **BACHELOR OF COMPUTER APPLICATION DEGREE**

**( THREE YEARS) / HONORS ( FOUR YEARS)**

**FRAMED AS PER NATIONAL EDUCATION POLICY (NEP 2020)**

### **SYLLABUS**

**Applicable with effect from 2022-23**

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#### **I. Preamble :**

The Bachelor of Computer Application (Honors) Programme is a full time four year programme offered by Bharati Vidyapeeth (Deemed to be University), Pune and conducted in Regular mode at its management institutes located in New Delhi, Pune, Navi Mumbai, Kolhapur, Sangli, Karad and Solapur. All the seven institutes have excellent faculty members, computer laboratories, Libraries, and other facilities to provide proper learning environment to the students. The University is accredited by NAAC with 'A+' grade. The expectations and requirements of the Software Industry, immediately and in the near future, are considered while designing the BCA programme. While designing the BCA Programme, the above facts are considered and the requirements for higher studies and immediate employment are visualized. This effort is reflected in the Vision and Mission statements of the BCA programme, the statements also embody the spirit of the vision of Dr. Patangraoji Kadam, the Founder of Bharati Vidyapeeth — “Social Transformation Through Dynamic Education”

#### **II. Vision:**

Preparing the Students to cope with the rigor of Post Graduate Programmes in global and creating high caliber solution architects for software development, who will also be sensitive to societal concerns.

#### **III. Mission:**

- We aim to drive transformation, technology and innovation through problem solving approach and research development.
- We aim to provide students with the IT tools to become productive and lifelong learner.

#### **IV. Aims:**

- To impart quality computer education to enhance logical computing and programming skills.
- To implement innovative techniques and process in leading-learning and evaluation.
- To further creativity and pursuit of excellence in computer applications.

## V. Learning Outcome Based Curriculum Framework -

### 1. Programme Education Objectives:

The Bachelor of Computer Application (Honors) Four Years degree programme has the following objectives...

- I. To prepare the youth to take up positions as system analysts, system engineers, software engineers and programmers.
- II. To aim at developing 'systems thinking' 'abstract thinking', 'skills to analyze and synthesize', and 'skills to apply knowledge', through 'extensive problem solving sessions', 'hands on practice under various hardware/software environments' and 'projects developed'.
- III. To prepare students with 'social interaction skills', 'communication skills', 'life skills', 'entrepreneurial skills', and 'research skills' which are necessary for career growth and for leading quality life are also imparted.

### 2. Programme Outcomes (POs) :

On completion of BCA (Honors) Four Year Degree Programme the expected programme outcomes that a student should be able to demonstrate are the following:

**PO1. Computational Knowledge:** Understand and apply mathematical foundation, computing and domain knowledge for the conceptualization of computing models from defined problems.

**PO2. Problem Analysis:** Ability to identify, critically analyze and formulate complex computing problems using fundamentals of computer science and application domains.

**PO3. Design / Development of Solutions:** Ability to transform complex business scenarios and contemporary issues into problems, investigate, understand and propose integrated solutions using emerging technologies.

**PO4. Conduct Investigations of Complex Computing Problems:** Ability to devise and conduct experiments, interpret data and provide well informed conclusions.

**PO5. Modern Tool Usage:** Ability to select modern computing tools, skills and techniques necessary for innovative software solutions

**PO6. Professional Ethics:** Ability to apply and commit professional ethics and cyber regulations in a global economic environment.

**PO7. Life-long Learning:** Recognize the need for and develop the ability to engage in continuous learning as a Computing professional.

**PO8. Project Management:** Ability to understand management and computing principles with computing knowledge to manage projects in multidisciplinary environments.

**PO9. Communication Efficacy:** Communicate effectively with the computing community as well as society by being able to comprehend effective documentations and presentations.

**PO10. Societal & Environmental Concern:** Ability to recognize economical, environmental, social, health, legal, ethical issues involved in the use of computer technology and other consequential responsibilities relevant to professional practice.

**PO11. Individual & Team Work:** Ability to work as a member or leader in diverse teams in multidisciplinary environment.

**PO12. Innovation and Entrepreneurship:** Identify opportunities, entrepreneurship vision and use of innovative ideas to create value and wealth for the betterment of the individual and society.

### 3. Programme Specific Outcomes (PSOs) :

After the completion of the course, a student is able to

**PSO1:** Ability to learn the various programming languages with database concepts along with development environment

**PSO2 :** Ability to apply theoretical and practical knowledge to solve business problems through data communication technology concepts.

**PSO3 :** Flourish the innovation and research attitude to develop IT artifact.

**PSO4:** Foster analytical and critical thinking abilities for efficient programming

**PSO5:** Demonstrate and apply the programming knowledge to develop effective software solution.

**PSO6:** Enrich the knowledge in the areas of Advanced technologies and business practices.

**PSO7:** Maintain the personality with environmental and social concerns

### 4. Graduate Attributes:

After completing BCA (Honors) Four Year Degree programme the students will be able to acquire following attributes and skills to groom the overall personality.

**GA01:** Competence (strong foundational knowledge, skills and attitudes) in providing professional service

**GA02:** Ability to make decisions based upon critical thinking and reasoning

**GA03:** Readiness to identify, assess and respond to the needs of individuals, organizations and society

**GA04:** Talent and attitude to ethically conduct research

**GA05:** Service within the ethical, professional and legal framework

**GA06:** Readiness to lead and be led to provide service as a professional, as a researcher, as a manager, as an educator and as an advocate of best practices

**GA07:** Technology user in professional, educational and research work.

**GA08:** Sensitivity and commitment to environmental conservation and sustainability in the professional and personal spheres

**GA09:** Valuing the diversity of Indian culture, ethos and knowledge system

**GA10:** Self-directed and lifelong learner for continuous professional and personal development

**GA11:** Effective communicator while providing professional service

## VI. Duration of the programme:

The duration of the BCA Bachelor's degree Program having six semesters and BCA (Honors) Degree Program is of four years spread across Eight Semesters with multiple entry and exit options. Student should complete the 4 years degree programme within 7 years.

a) Following EXIT options are available with the students:

| Exit Option                                                                                                                                                                             | Minimum Credits Requirements | NSQF Level | Bridge course                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Under graduate Certificate</b> - After successful completion of First Year                                                                                                           | 40                           | 5          | 10 credits bridge course(s) lasting two months including at least 06 credits job specific internship that would help the learner to acquire job ready competencies to enter the workforce. |
| <b>Under graduate Diploma</b> - After successful completion of Second Year                                                                                                              | 80                           | 6          |                                                                                                                                                                                            |
| <b>Bachelor's Degree</b> - After successful completion of Third Year                                                                                                                    | 120                          | 7          |                                                                                                                                                                                            |
| <b>Bachelor's Degree with Honors</b> - After successful completion of Fourth Year<br>OR<br><b>Bachelor's Degree with Honors (Research)</b> - After successful completion of Fourth Year | 150<br><br>152               | 8          |                                                                                                                                                                                            |

Note : Student is free to complete some interdisciplinary courses from other institutes provided he/she should earn 50% required credits from home HEI.

Student should complete the core disciplinary courses from home University (HEI) to get exit option for UG certificate/ UG diploma/ Bachelor Degree.

b) Following Entry options are available with the students :

- Student who opt Exit option at the end of 1<sup>st</sup> / 2<sup>nd</sup> / 3<sup>rd</sup> year, can reenter the same programme within three years from Exit.
- Student with Bachelors Degree can opt for Bachelor degree with Honors
- Student with Bachelors Degree can opt for Bachelor degree with Honors (Research) if the student secure CGPA  $\geq 7.5$

### National Skills Qualifications Framework (NSQF) Levels :

| Option                    | NSQF Level | Professional Knowledge                                                                                                | Skill                                                                                                                    |
|---------------------------|------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| At the end of first year  | 5          | Knowledge of facts, principles, processes, concepts in a field of work or study                                       | The student will have fundamental knowledge of computation, problem solving ability and basic website designing ability. |
| At the end of Second year | 6          | Factual and theoretical knowledge in the broad context within a field of work or study                                | Additionally the student will have advanced programming skills along with system development ability                     |
| At the end of Third year  | 7          | Wide ranging factual and theoretical knowledge in the broad context within a field of work or study                   | Additionally, student will have skills of Web Application development with Technical Writing and Report Generation.      |
| At the end of Fourth year | 8          | Comprehensive, cognitive theoretical knowledge and practical skills to develop creative solutions to abstract problem | Additionally, student will have skills of solving business application applying advanced technology                      |

### **VII. Academic Bank Of Credits (ABC) :**

As per the National Educational Policy (NEP) 2020, the Academic Bank of Credit offer the flexibility of curriculum framework and interdisciplinary /multidisciplinary academic mobility of students across Higher Educational Institutes (HEIs) with appropriate credit transfer mechanism. In furtherance to these guidelines the Faculty of Management Studies, Bharati Vidyapeeth (Deemed to be University) Pune has designed a four years undergraduate program offered at its constituent units.

As a pre-requisite a student should register in the Bharati Vidyapeeth (Deemed to be University) Academic Bank of Credit. The credits earned by the student/learner will be stored in it. A Student/learner would be required to complete the course as per the ABC (Academic Bank Credit) policy of UGC. The validity of the credits earned for a course is seven years only.

### **VIII. Eligibility Criteria for admission:**

A candidate applying for BCA(Honors) Four years programme should have passed higher secondary (10 + 2) or equivalent examination (10+3) of any recognized Board with satisfying the conditions to pass a common All India Entrance test (BU-MAT) conducted by Bharati Vidyapeeth (Deemed to be University), Pune. The final admission is based solely on the merit at the BU-MAT test.

## **IX. Grading System for Programmes under Management Studies:**

- **Grade Points** : The Faculty of Management Studies, Bharati Vidyapeeth (Deemed to be University) has suggested 10-point grading system for all programmes designed by its various Board of Studies. A grading system is a 10-point system if the maximum grade point is 10. The system is given in Table I below.

**Table I: The 10-point Grading System Adapted for Programmes under FMS**

|                        |          |           |          |           |          |          |          |
|------------------------|----------|-----------|----------|-----------|----------|----------|----------|
| Range of Percent Marks | [80,100] | [70,79]   | [60,69]  | [55,59]   | [50,54]  | [40,49]  | [00,39]  |
| Grade Point            | 10.0     | 9.0       | 8.0      | 7.0       | 6.0      | 5.0      | 0.0      |
| Grade                  | <b>O</b> | <b>A+</b> | <b>A</b> | <b>B+</b> | <b>B</b> | <b>C</b> | <b>D</b> |

Formula to calculate GP is as under:

Set  $x = \text{Max}/10$  where Max is the maximum marks assigned for the examination (i.e. 100)

Formula to calculate the individual evaluation

|                                  |                             |
|----------------------------------|-----------------------------|
| Range of Marks                   | Formula for the Grade Point |
| $8x \leq \text{Marks} \leq 10x$  | 10                          |
| $5.5x \leq \text{Marks} \leq 8x$ | Truncate (M/x) + 2          |
| $4x \leq \text{Marks} \leq 5.5x$ | Truncate (M/x) + 1          |

### ➤ **Scheme of Examination**

Courses having Internal Assessment (IA) and University Examinations (UE) shall be evaluated by the respective constituent units and the University at the term end for **40** and **60** Marks respectively. The total marks of IA and UE shall be 100 Marks and it will be converted into grade points and grades.

For Internal Assessment (IA) the subject teacher may use the following assessment tools:

- Attendance**
- Class Tests**
- Presentations**

- d) *Class Assignments*
- e) *Case studies*
- f) *Practical Assignments*
- g) *Mini Projects*
- h) *Oral*

#### **X) MOOCs Policy:-**

As per the guidelines provided by UGC each student have to complete TWO MOOCs (Massive Open Online Courses) as add on Course which provides wide access to the online learning. The student of regular programme should complete MOOCs prescribed by the institute in semester III, Sem IV, and / or Sem V. Each MOOC will be evaluated for TWO credits. The MOOC course fees should be borne by the respective student. On successful completion of MOOCs course, the student should produce the completion certificate to the institute on the basis of which additional Credits will be given to the students.

- Following are the sources from where students can undertake MOOCs

1. iimb.ac.in
2. swayam.gov.in
3. alison.com
4. edx.org
5. Coursera
6. harvardx.harvard.edu
7. udemy.com
8. futurelearn.com
9. Indira Gandhi National Open University (IGNOU)
10. National Council of Educational Research and Training (NCERT)
11. National Institute of Open Schooling (NIOS)
12. National Programme on Technology Enhanced Learning (NPTEL)
13. Any other sources offering online courses suggested by institute

## **XI. Standard of Passing:**

For all courses, both UE and IA constitute separate heads of passing. In order to pass in such courses and to earn the assigned credits, the student/learner must obtain a minimum grade point of 5.0 (40% marks) at UE and also a minimum grade point of 5.0 (40% marks) at IA.

If Student fails in IA, the learner passes in the course provided, he/she obtains a minimum 25% marks in IA and GPA for the course is at least 6.0 (50% in aggregate). The GPA for a course will be calculated only if the learner passes at UE.

A student who fails at UE in a course has to reappear only at UE as backlog candidate and clear the Head of Passing. Similarly, a student who fails in a course at IA he has to reappear only at IA as backlog candidate and clear the Head of Passing to secure the GPA required for passing.

The 10 point Grades and Grade Points according to the following table

| Range of Marks (%)              | Grade | Grade Point |
|---------------------------------|-------|-------------|
| $80 \leq \text{Marks} \leq 100$ | O     | 10          |
| $70 \leq \text{Marks} < 80$     | A+    | 9           |
| $60 \leq \text{Marks} < 70$     | A     | 8           |
| $55 \leq \text{Marks} < 60$     | B+    | 7           |
| $50 \leq \text{Marks} < 55$     | B     | 6           |
| $40 \leq \text{Marks} < 50$     | C     | 5           |
| Marks < 40                      | D     | 0           |

The performance at UE and IA will be combined to obtain GPA (Grade Point Average) for the course. The weights for performance at UE and IA shall be 60% and 40% respectively.

GPA is calculated by adding the UE marks out of 60 and IA marks out of 40. The total marks out of 100 are converted to grade point, which will be the GPA.

### **Formula to calculate Grade Points (GP)**

Suppose that “Max” is the maximum marks assigned for an examination or evaluation, based on which GP will be computed. In order to determine the GP, Set  $x = \text{Max}/10$  (since we have adopted 10 point system). Then GP is calculated by the following formulas

| Range of Marks                  | Formula for the Grade Point |
|---------------------------------|-----------------------------|
| $8x \leq \text{Marks} \leq 10x$ | 10                          |
| $5.5x \leq \text{Marks} < 8x$   | Truncate (M/x) +2           |
| $4x \leq \text{Marks} < 5.5x$   | Truncate (M/x) +1           |

Two kinds of performance indicators, namely the Semester Grade Point Average (SGPA) and the Cumulative Grade Point Average (CGPA) shall be computed at the end of each term. The SGPA measures the cumulative performance of a learner in all the courses in a particular semester, while the CGPA measures the cumulative performance in all the courses since his/her enrolment. The CGPA of learner when he/she completes the programme is the final result of the learner.

The SGPA is calculated by the formula

$$SGPA = \frac{\sum Ck * GPk}{\sum Ck}$$

where, Ck is the Credit value assigned to a course and GPk is the GPA obtained by the learner in the course. In the above, the sum is taken over all the courses that the learner has undertaken for the study during the Semester, including those in which he/she might have failed or those for which he/she remained absent. **The SGPA shall be calculated up to two decimal place accuracy.**

The CGPA is calculated by the following formula

$$CGPA = \frac{\sum C_k * GP_k}{\sum C_k}$$

where, Ck is the Credit value assigned to a course and GPk is the GPA obtained by the learner in the course. In the above, the sum is taken over all the courses that the learner has undertaken for the study from the time of his/her enrolment and also during the semester for which CGPA is calculated.

The CGPA shall be calculated up to two decimal place accuracy.

**The formula to compute equivalent percentage marks for specified CGPA:**

|                   |               |                                       |
|-------------------|---------------|---------------------------------------|
| % marks<br>(CGPA) | 10 * CGPA-10  | If $5.00 \leq \text{CGPA} < 6.00$     |
|                   | 5 * CGPA+20   | If $6.00 \leq \text{CGPA} < 8.00$     |
|                   | 10 * CGPA-20  | If $8.00 \leq \text{CGPA} < 9.00$     |
|                   | 20 * CGPA-110 | If $9.00 \leq \text{CGPA} < 9.50$     |
|                   | 40 * CGPA-300 | If $9.50 \leq \text{CGPA} \leq 10.00$ |

## **XII. Award of Honours:**

A student who has completed the minimum credits specified for the programme shall be declared to have passed in the programme. The final result will be in terms of letter grade only and is based on the CGPA of all courses studied and passed. The criteria for the award of honours are given below.

| Range of CGPA                    | Final Grade | Performance Descriptor | Equivalent Range of Marks (%)   |
|----------------------------------|-------------|------------------------|---------------------------------|
| $9.5 \leq \text{CGPA} \leq 10$   | O           | Outstanding            | $80 \leq \text{Marks} \leq 100$ |
| $9.0 \leq \text{CGPA} \leq 9.49$ | A+          | Excellent              | $70 \leq \text{Marks} < 80$     |
| $8.0 \leq \text{CGPA} \leq 8.99$ | A           | Very Good              | $60 \leq \text{Marks} < 70$     |
| $7.0 \leq \text{CGPA} \leq 7.99$ | B+          | Good                   | $55 \leq \text{Marks} < 60$     |
| $6.0 \leq \text{CGPA} \leq 6.99$ | B           | Average                | $50 \leq \text{Marks} < 55$     |
| $5.0 \leq \text{CGPA} \leq 5.99$ | C           | Satisfactory           | $40 \leq \text{Marks} < 50$     |
| CGPA below 5.0                   | F           | Fail                   | Marks below 40                  |

## **XIII. Rules of ATKT:**

- a) For admission to Semester V of BCA Third year, Students/Learners should pass all the courses under Sem I and II.
- ii) For admission to Semester VII of BCA Fourth year, Students/Learners should pass all the courses under Sem I, II, III and IV.

## **XIV. INTERNSHIP :**

At the end of Semester VI, each student shall undertake Internship in an Industry for 50 (Fifty Days). It is mandatory for the students to seek written approval from the Faculty Guide about the Topic & the Organisation before commencing the Internship.

During the Internship students are expected to take necessary guidance from the faculty guide allotted by the Institute. To do it effectively they should be in touch with their guide through e-mail or telecom. Internship Project should be a Computer Application to Real life business activity.

The learning outcomes and the utility to the organization must be highlighted in Internship Project Report.

**General chapterization of the report shall be as under:**

- 1) Introduction
- 2) Theoretical background
- 3) Company profile
- 4) Objectives of the study
- 5) System Requirements
- 6) System Analysis & Design
- 7) Implementation & Testing
- 8) Conclusion & Suggestions

References:

Annexure:

**TECHNICAL DETAILS:**

1. The report shall be printed on A-4 size white bond paper.
2. 12 pt. Times New Roman font shall be used with 1.5 line spacing for typing the report.
3. 1” margin shall be left from all the sides.
4. Considering the environmental issues, students are encouraged to print on both sides of the paper.
5. The report shall be hard bound as per the standard format of the cover page given by the Institute and shall be golden embossed.
6. The report should include a Certificate (on company’s letter head) from the company duly signed by the competent authority with the stamp.
7. The report shall be signed by the respective guide(s) & the Director of the Institute 10 (Ten) days before the viva-voce examinations.
8. Student should prepare two hard bound copies of the Summer Internship Project Report and submit one copy in the institute. The other copy of the report is to be kept by the student for their record and future references.
9. In addition to this, students should prepare two soft copies of their Summer IP reports & submit one each in Training & Placement Department of the Institute & Library

The Internship(804) shall be assessed out of 200 Marks. The breakup of these marks is as under;

|                          |                                |
|--------------------------|--------------------------------|
| Viva- voce examination = | 120 (One Hundred Twenty) Marks |
| Internship Report =      | + 80 (Eighty) Marks            |

-----  
200 (Two Hundred) Marks

The examiners' panel shall be decided as per the guidelines received from the University.

The viva –voce shall evaluate the project based on

- i. Actual work done by the student in the organization
- ii. Student's knowledge about the company & Business Environment
- iii. Learning outcomes for the student
- iv. Utility of the study to the organization

#### **XV. Project ( community Based/Software based )**

The project work would expose students to the socio-economic issues in society so that the theoretical learnings can be supplemented by actual life experiences to generate solutions to real-life problems.

As a part of Sem-VII (703), each student shall undertake Community based project related the areas of community engagement and service, environmental education, and value-based education.

It is mandatory for the students to seek written approval from the Faculty Guide about the Topic before commencing the project work. The topic may relate survey based or software based problem. The learning outcomes and the utility to the society must be highlighted in Project Report.

#### **XVI. Specializations:**

BCA three year degree programme and BCA(Hons.) four year degree programme 2022 offers specialization to the students/learners in the third year of both the programmes. The students/learner are required to select any one specialization from the list provided below.

| Sr. No. | Specialization Course       | Course No | Course Name                         |
|---------|-----------------------------|-----------|-------------------------------------|
| 01      | <b>Data Analysis</b>        | 504-1-A   | Data analysis using Excel           |
|         |                             | 604-1-B   | R Programming                       |
| 02      | <b>Information Security</b> | 504-2-A   | Information Security Concepts       |
|         |                             | 604-2-B   | Information Security Administration |
| 03      | <b>Data Science</b>         | 504-3-A   | Statistical Programming Using R     |
|         |                             | 604-3-B   | Introduction to Data Science        |
| 04      | <b>Information Systems</b>  | 504-4-A   | E-Commerce                          |
|         |                             | 604-4-B   | Knowledge Management                |

#### **Prerequisite for offering the specialization –**

- There must be minimum 10 (Ten) students for a particular specialization.

**XVII. Course Structure:****SEMESTER I**

| Course Number | Course Title                           | Course Type | Credits   | Hours / Week |          |          | IA         | UE         | Total Marks |
|---------------|----------------------------------------|-------------|-----------|--------------|----------|----------|------------|------------|-------------|
|               |                                        |             |           | L            | T        | P        |            |            |             |
| 101           | Fundamentals of Information Technology | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100         |
| 102           | C Programming                          | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100         |
| 103           | Organization of IT Business            | MDC         | 3         | 3            | 1        | -        | 40         | 60         | 100         |
| 104           | Discrete Mathematics                   | MDC         | 3         | 3            | 1        | -        | 40         | 60         | 100         |
| 105           | Lab on MS-Office Suite                 | DSC         | 2         | -            | -        | 4        | 40         | 60         | 100         |
| 106           | Lab on C Programming                   | DSC         | 2         | -            | -        | 4        | 40         | 60         | 100         |
| 107           | Human Universal Values                 | VBC         | 2         | 2            | -        |          | 50         | -          | 50          |
| 108           | Language – I                           | AEC         | 2         | 2            | -        | -        | 50         | -          | 50          |
| <b>Total</b>  |                                        |             | <b>20</b> | <b>16</b>    | <b>4</b> | <b>8</b> | <b>340</b> | <b>360</b> | <b>700</b>  |

## SEMESTER II

| Course Number | Course Title                            | Course Type | Credits   | Hours / Week |          |          | IA         | UE         | Total      |
|---------------|-----------------------------------------|-------------|-----------|--------------|----------|----------|------------|------------|------------|
|               |                                         |             |           | L            | T        | P        |            |            |            |
| 201           | Web Development Technology              | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 202           | DBMS I                                  | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 203           | Data Structures using C                 | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 204           | Financial Accounting                    | MDC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 205           | Lab on Data Structures using C          | DSC         | 2         | -            | -        | 4        | 40         | 60         | 100        |
| 206           | Lab on Web Development Technology       | DSC         | 2         |              |          | 4        | 40         | 60         | 100        |
| 207           | Environmental Studies                   | VBC         | 2         | 2            | -        | -        | 50         | -          | 50         |
| 208           | Community Work (Swaccha Bharat Abhiyan) | VBC         | 2         | 2            | -        | -        | 50         | -          | 50         |
| <b>Total</b>  |                                         |             | <b>20</b> | <b>16</b>    | <b>4</b> | <b>8</b> | <b>340</b> | <b>360</b> | <b>700</b> |

### SEMESTER III

| Course Number | Course Title         | Course Type | Credits   | Hours / Week |          |          | IA         | UE         | Total      |
|---------------|----------------------|-------------|-----------|--------------|----------|----------|------------|------------|------------|
|               |                      |             |           | L            | T        | P        |            |            |            |
| 301           | Operating Systems    | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 302           | Software Engineering | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 303           | Java Programming     | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 304           | Statistics           | MDC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 305           | Lab on Oracle        | DSC         | 2         | -            | -        | 4        | 40         | 60         | 100        |
| 306           | Lab on Java          | DSC         | 2         | -            | -        | 4        | 40         | 60         | 100        |
| 307           | Start-up Management  | AEC         | 2         | 2            | -        | -        | 50         | -          | 50         |
| 308           | Yoga & Meditation    | VBC         | 2         | 2            | -        | -        | 50         | -          | 50         |
| <b>Total</b>  |                      |             | <b>20</b> | <b>16</b>    | <b>4</b> | <b>8</b> | <b>340</b> | <b>360</b> | <b>700</b> |

The student should complete TWO MOOCs (Massive Open Online Courses ) as add on Course which provides wide access to the online learning. The student will complete MOOCs prescribed by the institute in semester III, Sem IV, and / or Sem V. Additional Credits will be given to the student as per MOOCs Policy

### SEMESTER IV

| Course Number | Course Title                                        | Course Type | Credits   | Hours / Week |          |          | IA         | UE         | Total      |
|---------------|-----------------------------------------------------|-------------|-----------|--------------|----------|----------|------------|------------|------------|
|               |                                                     |             |           | L            | T        | P        |            |            |            |
| 401           | Computer Networks                                   | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 402           | Advanced JAVA                                       | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 403           | Advanced HTML with Javascript and CSS               | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 404           | Optimization Techniques                             | MDC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 405           | Lab on Advanced JAVA                                | DSC         | 2         | -            | -        | 4        | 40         | 60         | 100        |
| 406           | Lab on HTML, Javascript and CSS & Minor Project - I | DSC         | 2         | -            | -        | 4        | 40         | 60         | 100        |
| 407           | Cyber security                                      | SEC         | 2         | 2            | -        | -        | 50         | -          | 50         |
| 408           | Mathematical Aptitude                               | AEC         | 2         | 2            | -        | -        | 50         | -          | 50         |
| <b>Total</b>  |                                                     |             | <b>20</b> | <b>16</b>    | <b>4</b> | <b>8</b> | <b>340</b> | <b>360</b> | <b>700</b> |

### SEMESTER V

| Course Number | Course Title                 | Course Type | Credits   | Hours / Week |          |          | IA         | UE         | Total      |
|---------------|------------------------------|-------------|-----------|--------------|----------|----------|------------|------------|------------|
|               |                              |             |           | L            | T        | P        |            |            |            |
| 501           | Basic Python Programming     | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 502           | Dot Net programming using C# | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 503           | Entrepreneurship Development | MDC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 504           | Elective I                   | DSE         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 505           | Lab on Python                | DSC         | 2         | -            | -        | 4        | 40         | 60         | 100        |
| 506           | Lab on Dot Net and C#        | DSC         | 2         | -            | -        | 4        | 40         | 60         | 100        |
| 507           | IT based Aptitude            | AEC         | 2         | 2            | -        | -        | 50         | -          | 50         |
| 508           | Human Rights                 | VBC         | 2         | 2            | -        | -        | 50         | -          | 50         |
| <b>Total</b>  |                              |             | <b>20</b> | <b>16</b>    | <b>4</b> | <b>8</b> | <b>340</b> | <b>360</b> | <b>700</b> |

### SEMESTER VI

| Course Number | Course Title                        | Course Type | Credits   | Hours / Week |          |          | IA         | UE         | Total      |
|---------------|-------------------------------------|-------------|-----------|--------------|----------|----------|------------|------------|------------|
|               |                                     |             |           | <b>L</b>     | <b>T</b> | <b>P</b> |            |            |            |
| 601           | Data warehousing and Data Mining    | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 602           | Web Programming (PHP)               | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 603           | Software Project Management         | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 604           | Elective II                         | DSE         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 605           | Lab on Web programming with Project | DSC         | 2         | -            | -        | 4        | 40         | 60         | 100        |
| 606           | Lab on Data Visualization           | DSC         | 2         | -            | -        | 4        | 40         | 60         | 100        |
| 607           | Digital marketing                   | SEC         | 2         | 2            | -        | -        | 50         | -          | 50         |
| 608           | Indian Culture                      | VBC         | 2         | 2            | -        | -        | 50         | -          | 50         |
| <b>Total</b>  |                                     |             | <b>20</b> | <b>16</b>    | <b>4</b> | <b>8</b> | <b>340</b> | <b>360</b> | <b>700</b> |

## Fourth year of BCA Honors Programme

### SEMESTER VII

| Course Number | Course Title                             | Course Type | Credits   | Hours / Week |          |          | IA         | UE         | Total      |
|---------------|------------------------------------------|-------------|-----------|--------------|----------|----------|------------|------------|------------|
|               |                                          |             |           | <b>L</b>     | <b>T</b> | <b>P</b> |            |            |            |
| 701           | Introduction to AI and ML                | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 702           | Object Oriented Analysis and Design      | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 703           | Project (Community based/Software based) | DSC         | 4         | -            | 1        | 4        | 60         | 90         | 150        |
| 704           | Mobile Application Development with Lab  | DSC         | 4         | 2            | 1        | 4        | 60         | 90         | 150        |
| <b>Total</b>  |                                          |             | <b>14</b> | <b>8</b>     | <b>4</b> | <b>8</b> | <b>200</b> | <b>300</b> | <b>500</b> |

### SEMESTER VIII

| Course Number | Course Title                 | Course Type | Credits   | Hours / Week |   |   | IA         | UE         | Total      |
|---------------|------------------------------|-------------|-----------|--------------|---|---|------------|------------|------------|
|               |                              |             |           | L            | T | P |            |            |            |
| 801           | Cloud Computing              | DSC         | 3         | 3            | 1 | - | 40         | 60         | 100        |
| 802           | Enterprise Resource Planning | DSC         | 3         | 3            | 1 | - | 40         | 60         | 100        |
| 803           | Block Chain Technology       | DSC         | 3         | 3            | 1 | - | 40         | 60         | 100        |
| 804           | Internship Project           | DSC         | 5         | -            | - | 8 | 80         | 120        | 200        |
| 805           | Research Publication         | DSC         | 2         | 3            | - | - | 100        | -          | 100        |
| <b>Total</b>  |                              |             | <b>16</b> | 12           | 3 | 8 | <b>300</b> | <b>300</b> | <b>600</b> |

## Fourth year of BCA Honors Programme with Research

### SEMESTER VII

| Course Number | Course Title                             | Course Type | Credits   | Hours / Week |          |          | IA         | UE         | Total      |
|---------------|------------------------------------------|-------------|-----------|--------------|----------|----------|------------|------------|------------|
|               |                                          |             |           | L            | T        | P        |            |            |            |
| 701           | Introduction to AI and ML                | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 702           | Object Oriented Analysis and Design      | DSC         | 3         | 3            | 1        | -        | 40         | 60         | 100        |
| 703           | Project (Community based/Software based) | DSC         | 4         | -            | 1        | 4        | 60         | 90         | 150        |
| 704           | Mobile Application Development with Lab  | DSC         | 4         | 2            | 1        | 4        | 60         | 90         | 150        |
| 705           | Research Publication-I                   | DSC         | 2         | -            | -        | -        | 100        | -          | 100        |
| <b>Total</b>  |                                          |             | <b>16</b> | <b>8</b>     | <b>4</b> | <b>8</b> | <b>300</b> | <b>300</b> | <b>600</b> |

### SEMESTER VIII

| Course Number | Course Title                                              | Course Type | Credits   | Hours / Week |   |    | IA         | UE         | Total      |
|---------------|-----------------------------------------------------------|-------------|-----------|--------------|---|----|------------|------------|------------|
|               |                                                           |             |           | L            | T | P  |            |            |            |
| 801           | Dissertation                                              | DSC         | 12        | -            | - | 12 | 100        | 300        | 400        |
| 802           | Seminar on Literature Review based on Recent Trends In IT | DSC         | 2         | 2            | - | -  | 100        | -          | 100        |
| 803           | Research Publication-II                                   | DSC         | 2         | -            | 2 | -  | 100        | -          | 100        |
| <b>Total</b>  |                                                           |             | <b>16</b> | 2            | 2 | 12 | <b>300</b> | <b>300</b> | <b>600</b> |

#### Abbreviations Expanded

- **DSC** - Discipline Specific Course
- **DSE** - Discipline Specific Elective
- **MDC** – Minor Disciplinary Course
- **SEC** - Skill Enhancement Course
- **VBC** - Value Based Course
- **AEC** - Ability Enhancement Course

**XVIII. Question Paper Patterns for University Examination:**

The pattern of question paper for the courses having University Examinations (**Regular mode**) will be as follows :

**Title of the Course****Day:****Total Marks: 60****Date:****Time: 03 Hours****Instructions:**

1. Section I **Question No 1** is **Compulsory** based on MCQ. Each question carries 01 marks
2. Attempt any **FIVE** questions from Section II. Each question carries 08 Marks.
3. Attempt any **ONE** from Section III. Each question carries 10 marks

| SECTION – I                                                                                                                                                     |            |                                                      |                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                 |            | CO<br>(CO number to be mentioned:<br>Refer Syllabus) | BL<br>(Bloom's Taxonomy Level to be mentioned viz.<br>Create (1);<br>Evaluate (2);<br>Analyze (3);<br>Apply (4);<br>Understand(5);<br>Remember (6)) |
| Q 1. Includes 10 objective type sub questions covering all units of course, each sub question carries 1 mark. (Each question should be mapped with the CO & BL) | (10 marks) | Each objective questions to be mapped with CO & BL   |                                                                                                                                                     |
| SECTION – II                                                                                                                                                    |            |                                                      |                                                                                                                                                     |
| <i>It should contain 6 questions covering the syllabus. Questions should be set uniformly from all the units.</i>                                               |            | CO<br>(CO number to be mentioned:<br>Refer Syllabus) | BL                                                                                                                                                  |
| Question                                                                                                                                                        | Marks      | CO                                                   | BL                                                                                                                                                  |
| Q.2 .....                                                                                                                                                       | (8 marks)  |                                                      |                                                                                                                                                     |
| Q.3 .....                                                                                                                                                       | (8 marks)  |                                                      |                                                                                                                                                     |
| Q.4 .....                                                                                                                                                       | (8 marks)  |                                                      |                                                                                                                                                     |
| Q.5.....                                                                                                                                                        | (8 marks)  |                                                      |                                                                                                                                                     |
| Q.6.....                                                                                                                                                        | (8 marks)  |                                                      |                                                                                                                                                     |

|                                                                                                                                                                                                                                    |            |    |    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----|----|
| Q.7 Write Short Notes on ANY TWO<br>a.<br>b.<br>c.                                                                                                                                                                                 | (8 marks)  |    |    |
| <b>SECTION – III</b>                                                                                                                                                                                                               |            |    |    |
| <i>This section should be based on case-study, problem solving and would carry 10 marks. Questions in this section should be designed to evaluate the higher levels of Bloom's Taxonomy viz. Create, Evaluate, Analyze, Apply.</i> |            | CO | BL |
| Q.8.....                                                                                                                                                                                                                           | (10 marks) |    |    |
| Q.9.....                                                                                                                                                                                                                           | (10 marks) |    |    |

Note:

1. Answer book for the Section I will be separate and student should return this answer book within first half an hour.
2. Answers to Section II and III should be written in the SAME ANSWER BOOK.
3. The question paper should be relevant to the set of course outcome.
4. Question Papers shall be prepared to incorporate varying levels of difficulty such as:
  - i. Must know – Vital (60% weightage)
  - ii. Should know – Essential (20% weightage)
  - iii. Could know – Desirable (20% weightage)
5. The length of the question-reasonably feasible for an average student to answer with in the stipulated time.



Dr. Pallavi  
Jamsandekar  
Chairperson  
Board of Studies  
Computer Applications and System Studies

| Programme: BCA-CBCS–RevisedSyllabusw.e.f.-Year2022–2023                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                                       |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Course Code | Course Title                          |       |
| I                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 101         | Fundamental of Information Technology |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Prepared by | Dr. Bhaskar V. Patil                  |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Credits     | Evaluation                            | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3           | UE:IE                                 | 60:40 |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                                       |       |
| <ul style="list-style-type: none"> <li>The main objective is to introduce IT in a simple language to all undergraduate students, regardless of their specialization. It will help them to pursue specialized programs leading to technical and professional careers and certifications in the IT industry. The focus of the subject is on introducing skills relating to IT basics, computer applications, programming, interactive medias, Internet basics.</li> </ul> |             |                                       |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                                       |       |
| <p>After completing the course the students shall be able to</p> <p><b>CO1:</b> Understand basic concepts and types of Computers, memory devices and software.<br/> <b>CO2:</b> Remember types of computers and its peripherals<br/> <b>CO3:</b> Demonstrating MS-office tools for data processing, mathematical operations in worksheets, presentations.<br/> <b>CO4:</b> Analyse the use of various components of computer</p>                                        |             |                                       |       |

| Unit | Content                                                                                                                                                                                                                                                                                                                                                      | Sessi<br>ons<br>(Hrs<br>) | COs<br>Number | Teaching<br>Methodolog<br>y          | Cognitio<br>nLevel | Evaluatio<br>nTools                  |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------|--------------------------------------|--------------------|--------------------------------------|
| 1    | <ul style="list-style-type: none"> <li>Computer-Definition, Characteristics, Concept of Hardware, Software, Evolution of computer and Generations</li> <li>Types of Computers – Analog and Digital computers, Hybrid Computers, General Purpose and Special Purpose Computer Limitations of Computer, Applications of Computer in Various Fields.</li> </ul> | 9                         | CO1           | As per individual faculty discretion | Remember           | As per individual faculty discretion |
| 2    | <ul style="list-style-type: none"> <li>Input Device – Keyboard, Mouse, Scanner, MICR, OMR.</li> <li>Output Devices – VDU, Printers – Dot Matrix, Daisywheel, Inkjet, Laser, Line Printers and</li> </ul>                                                                                                                                                     | 8                         | CO1, CO2      | As per individual faculty discretion | Understand         | As per individual faculty discretion |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |     |                                      |         |                                      |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|--------------------------------------|---------|--------------------------------------|
|   | Plotters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |     |                                      |         |                                      |
| 3 | <ul style="list-style-type: none"> <li>Memory Concept, Memory Cell, Memory Organisation, Semiconductor Memory – RAM, ROM, PROM, EPROM</li> <li>Secondary Storage Devices – Magnetic Tape, Magnetic Disk (Floppy Disk and Hard Disk.), Compact Disk.</li> </ul>                                                                                                                                                                                                                                                                                                                                                          | 8  | CO2 | As per individual faculty discretion | Analyze | As per individual faculty discretion |
| 4 | <ul style="list-style-type: none"> <li>Software and its needs, Types of S/W. System Software: Operating System, Utility Programs</li> <li>Programming Language: Machine Language, Assembly Language, High Level Language their advantages &amp; disadvantages.</li> <li>Application S/W and its types: Word Processing, Spread Sheets Presentation, Graphics, DBMS s/w.</li> <li>Concept of Network and its Type, Basic Elements of a Communication System, Data Transmission Media, Topologies</li> </ul>                                                                                                              | 8  | CO4 | As per individual faculty discretion | Create  | As per individual faculty discretion |
| 5 | <ul style="list-style-type: none"> <li><b>MS Office:</b> Introduction to MS Office, Components and Features.</li> <li><b>MS Word:</b> Creating Letter, Table, Fonts, Page Layout Document, Formatting, Spell Check, Print Preview, Template, Color, Mail Merge, Auto Text, Inserting Picture, Word Art.</li> <li><b>MS Excel:</b> Introduction to Excel, Sorting, Queries, Graphs, Scientific Functions.</li> <li><b>PowerPoint:</b> Introduction to PowerPoint, Creation of Slides, Inserting Pictures, Preparing Slide Show with Animation.</li> <li><b>MS Access:</b> Creation and Manipulation of Files.</li> </ul> | 12 | CO3 | As per individual faculty discretion | Create  | As per individual faculty discretion |

**Reference Books**

| <b>Sr.No.</b> | <b>NameoftheAuthor</b> | <b>TitleoftheBook</b>    | <b>Year<br/>Edition</b>         | <b>Publisher<br/>Company</b> |
|---------------|------------------------|--------------------------|---------------------------------|------------------------------|
| 1             | Dromey                 | How to solve<br>computer | 2015,3 <sup>rd</sup><br>edition | PHI Publication              |
| 2             | P. K. Sinha            | Computer<br>Fundamentals | 12 <sup>th</sup> edition        | PBP Publication              |
| 3             | V. Rajaraman           | Computer<br>Fundamentals | 6TH EDN.<br>2014                | PHI Publication              |

**Online Resources**

| <b>OnlineResourcesNo.</b> | <b>Websiteaddress</b>                                  |
|---------------------------|--------------------------------------------------------|
| 1                         | <a href="http://www.edx.com">www. edx.com</a>          |
| 2                         | <a href="http://www.coursera.com">www.coursera.com</a> |

**MOOCs:**

| <b>ResourcesNo.</b> | <b>Websiteaddress</b> |
|---------------------|-----------------------|
| 1                   | Alisons               |
| 2                   | Swayam                |

| Programme: BCA CBCS– Revised Syllabus w.e.f.-Year 2022 –2023                                                                                                                                                                                                                                                                                           |             |                 |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                               | Course Code | CourseTitle     |       |
| I                                                                                                                                                                                                                                                                                                                                                      | 102         | C Programming   |       |
|                                                                                                                                                                                                                                                                                                                                                        | Prepared by | Dr. A.R.Mujawar |       |
| Type of Course                                                                                                                                                                                                                                                                                                                                         | Credits     | Evaluation      | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                    | 3           | UE(60)+IE(40)   | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                              |             |                 |       |
| Objectives : <ul style="list-style-type: none"> <li>To learn Procedure Oriented Programming Language C.</li> <li>Emphasise on process of learning a computer language.</li> <li>Focus on semantics and problem solving.</li> </ul>                                                                                                                     |             |                 |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                |             |                 |       |
| After completing the course the students shall be able to<br><br><b>CO1:</b> To understand problem solving approach using procedural technique.<br><b>CO2:</b> To understand the basics of C Programming.<br><b>CO3:</b> To understand various statements, operators in C.<br><b>CO4:</b> To develop various C program using constructs in C language. |             |                 |       |

| Unit                              | Content                                                                                                                                                                                                                                                                                              | Sessions | COs Number | Teaching Methodology | Cognition Level | Evaluation Tools      |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|----------------------|-----------------|-----------------------|
| <b>Introduction to Algorithm</b>  | <ul style="list-style-type: none"> <li>Concept, of Problem, Procedure and Algorithm</li> <li>Algorithm Representation through Pseudo -Code and Flow - Charts</li> <li>Tracing of Algorithms Such as Swapping, Counting, Finding the Sum, Product, maximum, minimum, of a list of numbers.</li> </ul> | 5        | CO1        | Lecture              | Understand      | Quiz<br>Short Answers |
| <b>Introduction to C Language</b> | <ul style="list-style-type: none"> <li>History</li> <li>Structure of C Programming, Function as building blocks</li> </ul>                                                                                                                                                                           | 5        | CO2        | Lectures with PPTs   | Understand      | Quiz<br>Short Answers |

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                      |    |     |                    |            |                    |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|--------------------|------------|--------------------|
|                             | <ul style="list-style-type: none"> <li>• Language Fundamentals, Character set, C Tokens, Keywords, Identifiers, Variables, Constant,</li> <li>• Data Types, Comments</li> </ul>                                                                                                                                                                                                                                      |    |     |                    |            |                    |
| <b>Operators</b>            | <ul style="list-style-type: none"> <li>• Types of operators, Operator Precedence and Associativity</li> <li>• Expression, Statement and types of statements</li> <li>• Built in Operators and functions</li> <li>• Console based I/O and related built in I/O function- printf(), scanf(), getch(), getchar(), putchar(),</li> <li>• Concept of header files, Preprocessor directives - #include, #define</li> </ul> | 6  | CO3 | Lectures with PPTs | Understand | Quiz Short Answers |
| <b>Control Structures</b>   | <ul style="list-style-type: none"> <li>• Basic Control Structures</li> <li>• Decision making structures - if statement, if-else statement, Nested if-else statement, switch statement</li> <li>• Loop Control structures - while loop, do-while loop, for loop, Nested for loop</li> <li>• Other statements - break keyword, continue keyword, goto keyword, exit function</li> </ul>                                | 8  | CO4 | Lectures with PPTs | Create     | Quiz Short Answers |
| <b>Functions and Arrays</b> | <ul style="list-style-type: none"> <li>• Introduction</li> <li>• Purpose of function, Function declaration/ Function prototype, Function definition, Function call, return statement</li> <li>• Function parameters</li> <li>• Types of functions</li> <li>• Call by value</li> <li>• Storage classes</li> <li>• Recursion, Examples on recursive function</li> <li>• Introduction to one-</li> </ul>                | 13 | CO4 | Lectures with PPTs | Create     | Quiz Short Answers |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |     |                    |        |                    |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|--------------------|--------|--------------------|
|                                        | <p>dimensional Array, Definition, Declaration, Initialization, Accessing and displaying array elements</p> <ul style="list-style-type: none"> <li>• Arrays and functions</li> <li>• Introduction to two-dimensional Array, Definition, Declaration, Initialization, Accessing and displaying array elements</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |     |                    |        |                    |
| <b>Strings, Structure and Pointers</b> | <ul style="list-style-type: none"> <li><input type="checkbox"/> Introductions to Strings, Definition, Declaration, Initialization</li> <li><input type="checkbox"/> Input, output statements for strings</li> <li><input type="checkbox"/> Standard String library functions with example</li> <li><input type="checkbox"/> Structure – User defined data types, Concept of structure, Union; Member access operator</li> <li><input type="checkbox"/> Introduction to pointer, Definition, Declaring and Initializing pointer variable</li> <li><input type="checkbox"/> Indirection operator and address of operator, Accessing variable through its pointer, Pointer arithmetic</li> <li><input type="checkbox"/> Dynamic memory allocation</li> </ul> | 8 | CO4 | Lectures with PPTs | Create | Quiz Short Answers |

**Reference Books:**

| Sr.No. | Name of the Author            | Title of the Book            | Year Edition | Publisher Company |
|--------|-------------------------------|------------------------------|--------------|-------------------|
| 1      | Yashwant Kanetkar             | Let us C                     | 2018         | BPB Publications  |
| 2      | B.W.Kernighan,<br>D.M.Ritchie | The 'C' programming language | 1998         | PHI               |
| 3      | Balaguruswami                 | Programming in ANSIC         | 2019         | TMH               |

**Online Resources:**

| Online Resources No. | Website address                                                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.tutorialspoint.com/cprogramming">https://www.tutorialspoint.com/cprogramming</a>                               |
| 2                    | <a href="https://www.javatpoint.com/c-programming-language-tutorial">https://www.javatpoint.com/c-programming-language-tutorial</a> |
| 3                    | <a href="https://www.w3schools.in/c">https://www.w3schools.in/c</a>                                                                 |

**MOOCs:**

| Resources No. | Website address                                        |
|---------------|--------------------------------------------------------|
| 1             | NPTEL / Swayam                                         |
| 2             | <a href="http://www.edx.com">www.edx.com</a>           |
| 3             | <a href="http://www.coursera.com">www.coursera.com</a> |

| Programme: BCA CBCS –Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                                                                                                                                                                                                                       |             |                             |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                              | Course Code | Course Title                |       |
| I                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 103         | Organization of IT Business |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Prepared by | Dr.Mukund Kulkarni          |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Credits     | Evaluation                  | Marks |
| MDC                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3           | UE:IE                       | 60:40 |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                             |             |                             |       |
| To acquaint students with fundamentals of Business Organization and management systems as a body of knowledge.                                                                                                                                                                                                                                                                                                                                        |             |                             |       |
| <b>Course Outcomes :</b>                                                                                                                                                                                                                                                                                                                                                                                                                              |             |                             |       |
| <b>CO1:</b> To know about business and its structure and its various forms.<br><b>CO2:</b> To Apply and enlighten with nature and scope of IT business organization.<br><b>CO3:</b> To make them understand the office function and its significance on office Layout.<br><b>CO4:</b> To understand the complexities associated with management of human resources in the IT organizations and integrate the learning in handling these complexities. |             |                             |       |

| Unit | Content                                                                                                                                                                                                                                                                                                                                                                                                 | Sessions (Hrs) | COs Number | Teaching Methodology                 | Cognition Level | Evaluation Tools                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|--------------------------------------|-----------------|--------------------------------------|
| 1    | <b>Nature and Evolution of Business-</b><br>Concept of Business – Meaning, Definition, Nature and Scope, Characteristics of Business. Business as an Economic Activity. Objectives of Business. Structure of Business (Classification of Business Activities. Requisites for Success in Modern Business. Beginning and development of Commerce, Evolution of Industry, Industrial Revolution, Beginning | 10             | CO 1       | As per individual faculty discretion | Understand      | As per individual faculty discretion |

|   |                                                                                                                                                                                                                                                                                                    |    |             |                                      |                 |                                      |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------|--------------------------------------|-----------------|--------------------------------------|
|   | and growth of Indian Business, Industrialization in India                                                                                                                                                                                                                                          |    |             |                                      |                 |                                      |
| 2 | <b>Forms of Business Ownership-</b> Introduction to various forms – Factors affecting choices of an deal form of ownership, features Merits and Demerits of Sole Proprietorship – Joint Hindu Family Business – Partnership – Joint Stock Company – Co-operative Organization, Public Enterprises. | 10 | CO 1 & CO 2 | As per individual faculty discretion | Apply (Analyse) | As per individual faculty discretion |
| 3 | <b>Formation of a Company-</b> Stages in formation and incorporation of a company (e Promotion – incorporation and registration – Capital Subscription - Commencement of Business. - Documents of a Company i.e. Memorandum of Association – Articles of Association – Prospectus                  | 10 | CO 2        | As per individual faculty discretion | Apply           | As per individual faculty discretion |
| 4 | <b>The Impact of information technology on the Business-</b> Modern Organizations- IT runs the Airlines, Technology Transforms, Securities Industry, Creating New Types of Organization- Examples of Designs using IT Variables, Adding peoples to the design.                                     | 10 | CO2 & CO3   | As per individual faculty discretion | Evaluate        | As per individual faculty discretion |

|   |                                                                                                                                                                                                                                                                                                                                                                           |   |     |                                               |         |                                               |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|-----------------------------------------------|---------|-----------------------------------------------|
|   |                                                                                                                                                                                                                                                                                                                                                                           |   |     |                                               |         |                                               |
| 5 | <b>Strategic Issues of Information Technology-</b><br>IT and Corporate Strategy-<br>Some examples of<br>Technology strategy, value<br>chain, A framework for the<br>strategic use of IT. Creating<br>and sustaining a<br>Competitive edge- Using<br>resource to advantage,<br>protecting an IT innovation.<br>Integrating Technology<br>with the Business<br>Environment. | 5 | CO4 | As per<br>individual<br>faculty<br>discretion | Analyse | As per<br>individual<br>faculty<br>discretion |

#### Reference Books

| Sr. No. | Name of the Author  | Title of the Book                                 | Year Edition   | Publisher Company          |
|---------|---------------------|---------------------------------------------------|----------------|----------------------------|
| 1       | S.A. Sherlekar      | Modern Business Organization and Management       | latest edition | Himalaya Publishing House) |
| 2       | Y.K. Bhushan        | Fundamental of Business Organization & Management | latest edition | S Chand Publishers         |
| 3       | C. R. Basu          | Business Organization and Management              | 1998           | Tata McGraw Hill           |
| 4       | Lucas Henry C. , Jr | Information Technology for Management             | latest edition | Tata McGraw Hill           |

|   |            |                                                                    |                |                 |
|---|------------|--------------------------------------------------------------------|----------------|-----------------|
| 5 | S.S. Dubey | IT Services Business Management: Concepts, Processes and Practices | latest edition | PHI Publication |
|---|------------|--------------------------------------------------------------------|----------------|-----------------|

### Online Resources

| Online Resources No. | Web site address                                       |
|----------------------|--------------------------------------------------------|
| 1                    | NPTEL                                                  |
| 2                    | Swayam                                                 |
| 3                    | <a href="http://www.edx.com">www.edx.com</a>           |
| 4                    | <a href="http://www.coursera.com">www.coursera.com</a> |

| Programme:BCA CBCS –Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                                                                                                                                                                                                                           |             |                       |       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Course Code | Course Title          |       |
| I                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 104         | Discrete Mathematics  |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Prepared By | Dr. D.V.Sahasrabuddhe |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Credits     | Evaluation            | Marks |
| MDC                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3           | UE(60)+IE(40)         | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                |             |                       |       |
| <p>Get familiar with discrete structures of mathematics and its application in Business.</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Model the given data in set structure also Set relation among data descriptors.</li> <li><input type="checkbox"/> Define the function and identify the types of function</li> <li><input type="checkbox"/> Represent the facts in logic statements and resolve the given problem</li> </ul> |             |                       |       |
| <b>Course Outcomes :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                       |       |
| <p><b>CO1:</b> To understand discrete structures like sets, matrix, relations etc.(Understand)<br/> <b>CO2 :</b> To solve problems by carrying out various operations on structures (Apply)<br/> <b>CO3 :</b> To apply proper structure for representing given data (Apply)<br/> <b>CO4 :</b> To construct logic circuits for given Boolean expression (Create)<br/> <b>CO5 :</b> To test truthiness of the statement (Analyse)</p>                      |             |                       |       |

| Unit | Content                                                                                                                                                                                                                                               | Sessions (Hrs) | COs Number       | Teaching Methodology            | CognitionLevel    | EvaluationTools                       |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------|---------------------------------|-------------------|---------------------------------------|
| 1    | <b>Set Theory</b><br>Definition of a set, Representation of elements of sets, Methods of representing sets, types of sets, operations on sets , cardinality of a set, Principle of Inclusion and Exclusion, Venn Diagram, Proof by using Venn diagram | 8              | CO 1, CO 2, CO 3 | Lecture, problem solution, Quiz | Understand, Apply | Short Answers, Problem Solving Skills |
| 2    | <b>Functions and Relations</b><br>Definition of Function, Types of Functions ,Composite Function, Relation definition, representation of relations                                                                                                    | 8              | CO 1, CO 2, CO 3 | Lecture, problem solution, Quiz | Understand, Apply | Short Answers, Problem Solving Skills |
| 3    | <b>Logic</b>                                                                                                                                                                                                                                          | 9              | CO 1, CO 2,      | Lecture,                        | Understand,       | Short                                 |

|   |                                                                                                                                                                                                                                                                                                                                                                            |    |                 |                                 |                        |                                       |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----------------|---------------------------------|------------------------|---------------------------------------|
|   | Propositions, Logic Operations-Negation, Disjunction, Conjunction, Conditional and Biconditional, Truth Tables of compound propositions, Translating English sentences in to logical statements and vice versa, Logic gates and circuits                                                                                                                                   |    | CO 4, CO 5      | problem solution, Quiz          | Apply, Analyze, Create | Answers, Problem Solving Skills       |
| 4 | <b>Matrices</b><br>Matrix Definition, General Form, Representation of matrix in computers, Types of matrices, Operations on matrices: Addition, Subtraction and Multiplication, transpose , row / column transformations , Inverse of the matrix by Co-factor and Adjoint method, solutions to three variable problems by using matrices, application problems of matrices | 10 | CO1, CO 2, CO 3 | Lecture, problem solution, Quiz | Understand, Apply      | Short Answers, Problem Solving Skills |
| 5 | <b>Permutations, Combinations and Probability</b><br>Concept- Permutation, Combination, Sum and Product rules, problems on Permutation and combination (with wording atleast, atmost, neither nor, any one etc.)<br>Concept and problem solving, general probability, conditional probability, partitions, Bayes Theorem                                                   | 10 | CO1, CO 2, CO 3 | Lecture, problem solution, Quiz | Understand, Apply      | Short Answers, Problem Solving Skills |

**Reference Books**

| <b>Sr. No.</b> | <b>Name of the Author</b>       | <b>Title of the Book</b>                                         | <b>Year Edition</b> | <b>Publisher Company</b> |
|----------------|---------------------------------|------------------------------------------------------------------|---------------------|--------------------------|
| 1              | Kenneth Rosen                   | Discrete Mathematics & its Applications, 6 <sup>th</sup> Edition | 2007                | Tata Mc Graw Hill        |
| 2              | Semyour Lipschutz & Marc Lipson | Discrete Mathematics, 2 <sup>nd</sup> Edition                    | Reprint 2010        | Tata Mc Graw Hill        |

**MOOCs:**

| <b>Resources No.</b> | <b>Web site address</b>                                |
|----------------------|--------------------------------------------------------|
| 1                    | NPTEL Swayam                                           |
| 2                    | <a href="http://www.coursera.com">www.coursera.com</a> |
| 3                    | www. edx.com                                           |

| Programme: BCA – CBCS – Revised Syllabus w.e.f.- Year2022–2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                        |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Course Code | Course Title           |       |
| I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 105         | Lab on MS-Office Suite |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Prepared by | Dr.Bhaskar Patil       |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Credits     | Evaluation             | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2           | UE:IE                  | 60:40 |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |                        |       |
| <ul style="list-style-type: none"> <li>The objective of this course is to help the student gain proficiency in text editing and formatting, spreadsheet and database management, and presentation preparation. An additional objective of the course is for the student to gain basic knowledge of modern-day computing technology.</li> </ul>                                                                                                                                                                                                                                                                                                                                 |             |                        |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |                        |       |
| <p><b>CO1:</b> Students are able to prepare documentation using MS-Word</p> <p><b>CO2:</b> Demonstrate an advanced knowledge of the Word Processing package to design &amp; create effective and structured documents like technical reports, letters, brochures, etc.,</p> <p><b>CO3:</b> Demonstrate the skills in the appropriate use of various features of the spread sheet package MS Excel to create useful spreadsheet applications like tabulated statements, balance sheets, statistical charts, business statements, etc.</p> <p><b>CO4:</b> Demonstrate the skills in making an effective presentation with audio and video effects using them. MS Power Point</p> |             |                        |       |

| Unit | Content                                                                                                                                                                                                                                                                                                                                                                                                               | Sessions (Hrs) | COs Number | Teaching Methodology                 | CognitionLevel | EvaluationTools                      |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|--------------------------------------|----------------|--------------------------------------|
| 1    | Verify the components of a typical computer system, Explore, maintain files, andcustomize the Windows operating system, Review using the Internet Explorer.                                                                                                                                                                                                                                                           | 4              | CO1        | As per individual faculty discretion | Remember       | As per individual faculty discretion |
| 2    | Introduction to MS Word, Menus, Shortcuts, Document types Working with Documents: <ul style="list-style-type: none"> <li>Opening Files, formatting page and Setting Margins, converting files to different formats, Editing text documents, Using Toolbars, Ruler, Icons and help</li> <li>Formatting Documents: Setting Font Styles, Setting Paragraph style, Setting Page Style, Setting Document Styles</li> </ul> | 8              | CO1, CO2   | As per individual faculty discretion | Understand     | As per individual faculty discretion |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |           |                                      |         |                                      |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------|--------------------------------------|---------|--------------------------------------|
|   | <ul style="list-style-type: none"> <li>• Creating Tables: Table settings, Borders, Alignments, insertion, deletion, Merging, Splitting, Sorting, Formula</li> <li>• Drawing: Inserting Pictures/Files etc., Drawing Pictures, Formatting &amp; Editing pictures, Grouping and ordering, Rotating Tools: Word Completion, Spell Checks, Macros, Mail merge, Templates, Using Wizards, Tracking, Changes, Security</li> </ul>                                                                                                                                                                                                                                                                                                                                 |   |           |                                      |         |                                      |
| 3 | <ul style="list-style-type: none"> <li>○ Introduction: Opening new Presentation, Different presentation templates, setting backgrounds, Selecting presentation layouts</li> <li>○ Creating a presentation: Setting presentation style, Adding Text to the presentation</li> <li>○ Formatting a presentation: Adding style, Color, gradient fills, arranging objects, Adding Header &amp; Footer, Slide background, Slide layout</li> <li>○ Adding Graphics to the presentation: Inserting pictures, movies, tables, etc into the presentation, Drawing Pictures using Draw Adding effects to the presentation: Setting Animation &amp; transition effect, Adding audio and video Printing Handouts and Generating standalone presentation viewer</li> </ul> | 6 | CO3 & CO4 | As per individual faculty discretion | Analyze | As per individual faculty discretion |
| 4 | <ul style="list-style-type: none"> <li>○ Introduction: Spreadsheet &amp; its applications, opening spreadsheet, Menus &amp; Toolbars &amp; icons, Shortcuts, Using help</li> <li>○ Working with Spreadsheets:</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 4 | CO3       | As per individual faculty discretion | Create  | As per individual faculty discretion |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |     |                                      |        |                                      |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|--------------------------------------|--------|--------------------------------------|
|   | <p>Opening a File, Saving Files, Setting Margins, converting files to different formats: Importing, Exporting and Sending files to others, Spreadsheet addressing, Entering and Editing Data:</p> <ul style="list-style-type: none"> <li>○ Computing data: Setting Formula, finding total in a column or row, Mathematical Operations (Addition, Subtraction, Multiplication, Division, Exponentiation), Using other Formula.</li> <li>○ Formatting Spreadsheets: Formatting – Cell, row, column Headers, Row Height, Column Width, Visibility – Row, Column, Sheet, worksheet Security</li> <li>○ Formatting – worksheet: Sheet Formatting &amp; style - background, color, Borders &amp; shading, Anchoring objects, Formatting layout for Graphics, Clipart etc.,</li> <li>○ Working with sheets : Sorting, Filtering, Validation, Consolidation, Subtotal , Creating Charts, selecting charts, Formatting charts, label, scaling etc.,</li> <li>○ Using Tools: Error Checking, Spell Checks, Macros, Formula Auditing, Creating &amp; using Templates, Tracking changes, customization, printing worksheet</li> </ul> |   |     |                                      |        |                                      |
| 5 | <ul style="list-style-type: none"> <li>• Concept of Functions, commonly used functions: Sum, Max, Min, Average, Count, Today, Now, dated if, Count if, CountA, Count Blank, Round, Roundup, Round Down, ABS, Sign,</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 8 | CO3 | As per individual faculty discretion | Create | As per individual faculty discretion |

|  |                                                                                                                                                                                                                                                                  |  |  |  |  |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  | Ceiling, Floor, Trim, Value,<br>Clean, sqrt, if, sum if<br><b>MS Access:</b><br>What is an Access<br>Database, opening a<br>Database File, Create<br>Table, Create and<br>modify fields of tables,<br>construct simple queries,<br>Saving and Running<br>Queries |  |  |  |  |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|

### Reference Books

| Sr.No.          | NameoftheAuthor | TitleoftheBook           | Year<br>Edition                 | Publisher<br>Company |
|-----------------|-----------------|--------------------------|---------------------------------|----------------------|
| 1 International | Dromey          | How to solve<br>computer | 2015,3 <sup>rd</sup><br>edition | PHI Publication      |
| 2 National      | P. K. Sinha     | Computer<br>Fundamentals | 12 <sup>th</sup> edition        | PBP Publication      |

|            |              |                          |                  |                 |
|------------|--------------|--------------------------|------------------|-----------------|
| 3 National | V. Rajaraman | Computer<br>Fundamentals | 6TH EDN.<br>2014 | PHI Publication |
|------------|--------------|--------------------------|------------------|-----------------|

### Online Resources

| OnlineResourcesNo. | Websiteaddress                                                                                        |
|--------------------|-------------------------------------------------------------------------------------------------------|
| 1                  | <a href="http://www.bretlsimmons.com">www.bretlsimmons.com</a>                                        |
| 2                  | <a href="https://www.youtube.com/watch?v=Jla7vP3gyL4">https://www.youtube.com/watch?v=Jla7vP3gyL4</a> |
| 3                  | <a href="http://www.positivesharing.com">www.positivesharing.com</a>                                  |
| 4                  | <a href="https://www.youtube.com/watch?v=r2Xv9Am7PWQ">https://www.youtube.com/watch?v=r2Xv9Am7PWQ</a> |

### MOOCs:

| ResourcesNo. | Websiteaddress |
|--------------|----------------|
| 1            | Alisons        |
| 2            | Swayam         |

| Programme: BCA CBCS – Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |                      |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Course Code | Course Title         |       |
| I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 106         | Lab on C Programming |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Prepared By | Dr.Ayesha Mujawar    |       |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Credits     | Evaluation           | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2           | UE(60)+IE(40)        | 100   |
| <b>Course Objectives :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                      |       |
| <ul style="list-style-type: none"> <li>To make students practice on the procedure oriented programming using C</li> <li>To train the students for programming logic development</li> </ul>                                                                                                                                                                                                                                                                                                                                                                 |             |                      |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |                      |       |
| <p><b>CO1:</b> To develop skills to write simple programming concepts using C language.</p> <p><b>CO2:</b> To develop an application using Decision making and looping And Make use of proper operators to solve the problem.</p> <p><b>CO3:</b> To apply efficient use of functions and storage classes.</p> <p><b>CO4:</b> To apply use of Arrays and pointers efficiently and handling strings.</p> <p><b>CO5:</b> To understand the dynamic memory allocation and pointers in C. Able to define new data types using enum, structures and typedef.</p> |             |                      |       |

| Unit             | Content                                                                                                                                                                                                                                                                                                                                                                  | Sessions | COs Number | Teaching Methodology | Cognition Level | Evaluation Tools |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|----------------------|-----------------|------------------|
| <b>Operators</b> | <p>Compilation and Executing programs Arithmetic operations</p> <p>Use of Symbolic constants</p> <p>Demonstrating the following gcc options -o, -c, -D, -l, -I, -g, -E<br/>Programs to demonstrate use of operators and Input/ output</p> <p><b><i>gcc or an equivalent compiler is assumed.</i></b></p> <p>Compilation and Executing programs Arithmetic operations</p> | 5        | CO1        | Live Demo            | Understand      | Quiz             |

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |     |           |        |      |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|-----------|--------|------|
| <b>Selection<br/>&amp;Iteration<br/>Construct</b> | Program to demonstrate thefollowing<br>- Branching<br>- Nested Branching<br>- Looping Selection.                                                                                                                                                                                                                                                                                                                                                                                               | 7 | CO2 | Live Demo | Create | Quiz |
| <b>Function<br/>and<br/>Storage<br/>Classes</b>   | Working with functions<br>- Writing function prototype<br>and definition<br>- Using functions to solve<br>problems (Calling a<br>function )<br>- Using recursion Storage<br>classes - Using register, extern and<br>static                                                                                                                                                                                                                                                                     | 6 | CO3 | Live Demo | Create | Quiz |
| <b>Arrays and<br/>Strings</b>                     | Arrays and Strings<br>1D - Linear Search, Binary Search,<br>Bubble Sort, Selection Sort, Insertion<br>Sort 2 D - Matrix operations<br>Strings: program to do operations on<br>string using library and user defined<br>functions<br><br>Finding length of string, String<br>concatenation, removing extra spaces,<br>get substring, check whether second<br>string is part of another, converting<br>string to<br>lowercase, uppercase etc..                                                   | 7 | CO4 | Live Demo | Create | Quiz |
| <b>Structures &amp;<br/>Pointers</b>              | Structures<br>Making use of structures to define<br>new types(user defined types) Arrays<br>of structure, display all elements of<br>array and sorting of them.<br>Pointers,<br>Programs to demonstrate working of<br>pointer; need of pointer, Pointer as<br>parameter to function<br>Comparison of pointer with arrays and<br>using pointer to refer an arrayCreating<br>pointer dynamically by using dynamic<br>memory allocation<br>Array of Pointers, Ragged Arrays,<br>Function pointer. | 5 | CO5 | Live Demo | Create | Quiz |

## Reference Books

| Sr.No. | Name of the Author         | Title of the Book            | Year Edition | Publisher Company |
|--------|----------------------------|------------------------------|--------------|-------------------|
| 1      | Yashwant Kanetkar          | Let us C                     | 2018         | BPB Publications  |
| 2      | B.W.Kernighan, D.M.Ritchie | The 'C' programming language | 1998         | PHI               |
| 3      | Balaguruswami              | Programming in ANSIC         | 2019         | TMH               |

## Online Resources

| Online Resources No. | Website address                                                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.tutorialspoint.com/cprogramming">https://www.tutorialspoint.com/cprogramming</a>                               |
| 2                    | <a href="https://www.javatpoint.com/c-programming-language-tutorial">https://www.javatpoint.com/c-programming-language-tutorial</a> |
| 3                    | <a href="https://www.w3schools.in/c">https://www.w3schools.in/c</a>                                                                 |

## MOOCs:

| Resources No. | Website address                                        |
|---------------|--------------------------------------------------------|
| 1             | NPTEL / Swayam                                         |
| 2             | <a href="http://www.edx.com">www.edx.com</a>           |
| 3             | <a href="http://www.coursera.com">www.coursera.com</a> |

| Programme: BCA CBCS –Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                                                                                                                                                                                                                     |             |                        |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                            | Course Code | Course Title           |       |
| I                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 107         | Universal Human Values |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Prepared by | Dr.Deepali Gala        |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                | Credits     | Evaluation             | Marks |
| VBC                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2           | IA                     | 50    |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                           |             |                        |       |
| <ul style="list-style-type: none"> <li>To help the student to see the need for developing a holistic perspective of life.</li> <li>To sensitize the student about the scope of life – individual, family, society and nature/existence.</li> <li>Strengthening self-reflection.</li> <li>To develop more confidence and commitment to understand, learn and act accordingly</li> </ul>                                                              |             |                        |       |
| <b>Course Outcomes :</b>                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                        |       |
| <p><b>CO1:</b> Provide an overview of Prerequisites to Human Values</p> <p><b>CO2:</b> Understand the role of a human being in ensuring harmony in self and society</p> <p><b>CO3:</b> Analyze ethical dilemma while discharging duties in professional life.</p> <p><b>CO4:</b> Evaluate ethical and unethical decisions and take a right stand</p> <p><b>CO5:</b> Develop a harmonious environment for holistic development of self and body.</p> |             |                        |       |

| Unit | Content | Sessions<br>(Hrs) | COs Number | Teaching<br>Methodology | Cognition<br>Level | Evaluation<br>Tools |
|------|---------|-------------------|------------|-------------------------|--------------------|---------------------|
|      |         |                   |            |                         |                    |                     |

|   |                                                                                                                                                                                                                                                                                                                                                                     |    |          |                                      |            |                                      |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----------|--------------------------------------|------------|--------------------------------------|
| 1 | <p>1. Value Education, Definition, Concept and Need for Value Education.</p> <p>2. Self exploration as a means of Value Education.</p>                                                                                                                                                                                                                              | 3  | CO 1     | As per individual faculty discretion | Remember   | As per individual faculty discretion |
| 2 | <p>1. Human Being is more than just the Body.</p> <p>2. Harmony of the Self ('I') with the Body - happiness and physical facility</p> <p>3. Understanding Myself as Co-existence of the Self and the Body.</p> <p>4. Understanding Needs of the Self and the needs of the Body.</p> <p>Understanding the activities in the Self and the activities in the Body.</p> | 7  | CO2, CO5 | As per individual faculty discretion | Understand | As per individual faculty discretion |
| 3 | <p>1. Family as a basic unit of Human Interaction and Values in Relationships.</p> <p>2. The Basics for Respect and today's Crisis: Affection, e, Guidance, Reverence, Glory, Gratitude, Prosperity and Love.</p> <p>3. Comprehensive Human Goal: The Five Dimensions of Human</p>                                                                                  | 10 | CO 3     | Lecture with PPTs<br>Case Study      | Analyse    | As per individual faculty discretion |

|    |                                                                                                                                                                                                                |   |     |                                      |        |                                      |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|--------------------------------------|--------|--------------------------------------|
|    | <p>Endeavour.</p> <p>4. Harmony in Nature: The Four Orders in Nature.</p> <p>5. The Holistic Perception of Harmony in Existence.</p>                                                                           |   |     |                                      |        |                                      |
| 10 | <p>1. Value based Life and Profession.</p> <p>2. Professional Ethics and Right Understanding.</p> <p>3. Competence in Professional Ethics.</p> <p>4. Issues in Professional Ethics – The Current Scenario.</p> | 8 | CO4 | As per individual faculty discretion | Create | As per individual faculty discretion |

#### Reference Books

| Sr. No. | Name of the Author | Title of the Book                  | Year Edition | Publisher Company  |
|---------|--------------------|------------------------------------|--------------|--------------------|
| 1       | Bertrand Russell   | Human Society in Ethics & Politics | 2015         | Taylor and Francis |
| 2       | I.C. Sharma        | Ethical Philosophy of India        | 1965         | Johnsen            |

#### Online Resources

| Online Resources No. | Web site address                                                                                                                                                                                |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://fdp-si.aicte-india.org/verifiedProgramDetailsList.php">https://fdp-si.aicte-india.org/verifiedProgramDetailsList.php</a>                                                       |
| 2                    | <a href="https://citizenchoice.in/course/Universal-Human-Values/Unit%201/Happiness-and-Prosperity">https://citizenchoice.in/course/Universal-Human-Values/Unit%201/Happiness-and-Prosperity</a> |

#### MOOCs:

| Resources No. | Web site address                                                          |
|---------------|---------------------------------------------------------------------------|
| 1             | <a href="https://swayam.gov.in">Swayam.gov.in</a>                         |
| 2             | <a href="https://epgp.inflibnet.ac.in/">https://epgp.inflibnet.ac.in/</a> |

| Programme:BCA CBCS – Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                                                                                                                                                                                           |             |                     |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                  | Course Code | Course Title        |       |
| I                                                                                                                                                                                                                                                                                                                                                                                                                         | 108         | Language-I          |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                           | Prepared by | Dr. Amarja Nargunde |       |
| Type of Course                                                                                                                                                                                                                                                                                                                                                                                                            | Credits     | Evaluation          | Marks |
| AEC                                                                                                                                                                                                                                                                                                                                                                                                                       | 2           | IE (50)             | 50    |
| <b>Course Objectives :</b>                                                                                                                                                                                                                                                                                                                                                                                                |             |                     |       |
| To make students to:                                                                                                                                                                                                                                                                                                                                                                                                      |             |                     |       |
| <ol style="list-style-type: none"> <li>1. Participate actively in discussions &amp; debates</li> <li>2. Give impromptu speeches and prepared presentations</li> <li>3. Read, comprehend and summarize articles</li> <li>4. Learn typical formats for writing and practice writing skills</li> <li>5. Prepare power-point presentations</li> <li>6. Receive extensive feedback on their oral and written skills</li> </ol> |             |                     |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                   |             |                     |       |
| After completing the course the students shall be able to<br><b>CO1:</b> Understand and read English better<br><b>CO2:</b> Write accurately and speak fluently.<br><b>CO3:</b> Participate actively in discussions and debates<br><b>CO4:</b> Give presentations.                                                                                                                                                         |             |                     |       |

| Unit | Content                                                                                                                                                                                                                                                                                                 | Sessions | CO Number | Teaching Methodology | Cognition Level      | Evaluation Tools |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|----------------------|----------------------|------------------|
| 1    | Construction of sentences with there is, there are, it is etc.<br>Usage of articles, tenses and prepositions etc.<br>Translation of sentences, & passages from mother tongue to English<br>General errors in Sentence Constructions<br>Synonyms, Antonyms, use of appropriate words<br>Idioms & Phrases | 6        | CO1, CO2  | Lectures, Videos     | Understand and Apply | Quizzes          |

|   |                                                                                                                                                                                                                      |   |          |                                                   |                         |                                        |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------|---------------------------------------------------|-------------------------|----------------------------------------|
| 2 | Reading short passages aloud and discussion<br>Listening of conversations and answering questions<br>Comprehension of Short Passages<br>Comprehensions of texts, judgments and other passages of more general nature | 6 | CO2      | Practical- Reading by Students                    | Understand and Evaluate | Class Exercises<br>Evaluation          |
| 3 | Introducing oneself<br>Conversations between two student on a given topic/role play<br>Impromptu speech on a given topics<br>Debates and Logical reasoning                                                           | 6 | CO2, CO3 | Practical- Role Play, speeches and debates        | Create                  | Class Exercises<br>Evaluation          |
| 4 | Writing correctly (Grammar, Punctuation)<br>Paragraph Writing<br>Letters – Structure & Layout (Business & Official letters)<br>Essay writing<br>Resume writing                                                       | 6 | CO2      | Lecture and practical writing exercise            | Create                  | Long Assignments                       |
| 5 | Preparing PowerPoint presentations<br>Preparing for class-room presentations                                                                                                                                         | 6 | CO4      | Lectures and students giving actual presentations | Create                  | PPT making and Presentation evaluation |

**Reference Books:**

| Sr. No. | Name of the Author   | Title of the Book                                                                     | Year Edition   | Publisher Company                                             |
|---------|----------------------|---------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------|
| 1       | B.M. Sheridan        | Speaking and Writing in English                                                       | 2017           | The Readers Paradise                                          |
| 2       | Ellen Kaye           | Maximize Your Presentation Skills: How to Speak, Look, and Act on Your Way to the Top | 2002           | Currency                                                      |
| 3       | Thomson and Martinet | A practical English Grammar                                                           | 1970           | The English Language Book Society and Oxford University Press |
| 4       | Wren and Martin,     | English Grammar and Composition                                                       | latest edition | S. Chand, Delhi                                               |
| 5       | Mike Gould           | Cambridge Grammar and Writing Skills Learner's Book 8                                 | 2019           | Cambridge University Press                                    |

**Online Resources:**

| Online Resources No. | Web site address                                                                                                                                  |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.passporttoenglish.com">https://www.passporttoenglish.com</a>                                                                 |
| 2                    | <a href="https://www.youtube.com/user/EnglishLessons4U">https://www.youtube.com/user/EnglishLessons4U</a>                                         |
| 3                    | <a href="http://www.5minuteenglish.com/grammar.htm">http://www.5minuteenglish.com/grammar.htm</a>                                                 |
| 4                    | <a href="https://learnenglish.britishcouncil.org/skills/writing/a1-writing">https://learnenglish.britishcouncil.org/skills/writing/a1-writing</a> |
| 5                    | <a href="https://www.skillsyouneed.com/presentation-skills.html">https://www.skillsyouneed.com/presentation-skills.html</a>                       |

**MOOCs:**

| Resources | Web site address                                                                                                                                                                          |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1         | <a href="https://www.my-mooc.com/en/mooc/english-grammar-style-uqx-write101x-3/">https://www.my-mooc.com/en/mooc/english-grammar-style-uqx-write101x-3/</a>                               |
| 2         | <a href="https://www.my-mooc.com/en/mooc/business-english-making-presentations/">https://www.my-mooc.com/en/mooc/business-english-making-presentations/</a>                               |
| 3         | <a href="https://www.my-mooc.com/en/mooc/english-for-effective-business-speaking/">https://www.my-mooc.com/en/mooc/english-for-effective-business-speaking/</a>                           |
| 4         | <a href="https://www.my-mooc.com/en/mooc/english-for-business-and-entrepreneurship/">https://www.my-mooc.com/en/mooc/english-for-business-and-entrepreneurship/</a>                       |
| 5         | <a href="https://www.my-mooc.com/en/mooc/english-doing-business-asia-writing-hkustx-eba102x-1/">https://www.my-mooc.com/en/mooc/english-doing-business-asia-writing-hkustx-eba102x-1/</a> |

|                                                                                                                                                                                                                                                                                                       |                    |                                   |              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------|--------------|
|                                                                                                                                                                                                                                                                                                       |                    |                                   |              |
| <b>Semester</b>                                                                                                                                                                                                                                                                                       | <b>Course Code</b> | <b>Course Title</b>               |              |
| <b>II</b>                                                                                                                                                                                                                                                                                             | <b>201</b>         | <b>Web Development Technology</b> |              |
|                                                                                                                                                                                                                                                                                                       | <b>Prepared by</b> | <b>Dr.Suvarna Patil</b>           |              |
| <b>Type</b>                                                                                                                                                                                                                                                                                           | <b>Credits</b>     | <b>Evaluation</b>                 | <b>Marks</b> |
| DSC                                                                                                                                                                                                                                                                                                   | 3                  | UE:IE                             | 60:40        |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                             |                    |                                   |              |
| To make students to: <ul style="list-style-type: none"> <li>• To get proficiency in Website designing</li> <li>• To learn Wordpress as Content Management System</li> <li>• To get familiar to use all setting and components of Wordpress</li> </ul>                                                 |                    |                                   |              |
| <b>Course Outcomes :</b>                                                                                                                                                                                                                                                                              |                    |                                   |              |
| <b>CO1:</b> To understand Wordpress as a Content Management System<br><b>CO2 :</b> To understand Hosting,Website layout, Admin Panel<br><b>CO3:</b> To understand use of Themes and Templates, PlugIn in Wordpress<br><b>CO4:</b> To apply Themes and Templates, PlugIn in Web Page to create Website |                    |                                   |              |

| <b>Unit</b> | <b>Content</b>                                                                                                                                                                                                                                            | <b>Sessions (Hrs )</b> | <b>COs Number</b> | <b>Teaching Methodology</b> | <b>CognitionLevel</b> | <b>EvaluationTools</b> |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------|-----------------------------|-----------------------|------------------------|
| 1           | Elements of website -<br>Domain ,Hosting ,<br>Content Management System (Wordpress),<br>Domain – Registration ,<br>Manage DNS ,<br>Nameserver and Domain Forward<br>Hosting – Understand the Difference in Shared Hosting , Cloud Hosting and VPS Hosting | 9                      | CO 1              | Lecture with Ppts           | Understand            | Quiz Short Answers     |

|   |                                                                                                                                                                                                                                                                                                                                                                        |    |      |                    |            |                    |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------|--------------------|------------|--------------------|
|   | WordPress - Installation of WordPress , MySQL Secuirty Certificate – Understand the use of SSL using Free and Paid Service Providers                                                                                                                                                                                                                                   |    |      |                    |            |                    |
| 2 | <b>Website Configuration</b><br>Header and Footer Configuration<br>General Configuration – Font / Forecolor / Button Type / Backcolor<br>Site Configuration – Logo , Site Icon , Site Name<br>Home page Setting , Website layout Setting                                                                                                                               | 9  | CO 2 | Lecture with Ppts  | Understand | Quiz Short Answers |
| 3 | <b>Admin Panel Understanding</b><br><br>Change Settings- General Writing Reading, Discussion , media, permalinks and privacy<br>Import and Export website data<br>Add / modify Themes<br>Install – Activate Plugin                                                                                                                                                     | 9  | CO2  | Lecture with PPTs  | Understand | Quiz Short Answers |
| 4 | <b>WordPress Themes And Working with Content</b><br><br>Basics of Themes, Downloading, installing, and activating themes, Installing themes from Dashboard<br><br>WordPress Plugin: Basics of Plugin, Downloading, installing, and activating free and Paid Plugin<br><br>WordPress Templates: Basics of Templates, Downloading, installing, and activating Templates, | 10 | CO3  | Lectures with PPTs | Understand | Quiz Short Answers |

|   |                                                                                                                                                                                                                                                                                                                                                              |   |     |         |        |                    |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|---------|--------|--------------------|
|   | <p>Design Pages using Template</p> <p>Posts Vs Pages, Adding Hyperlinks, Playing with Media content, Previewing and Editing Posts, Previewing and Editing Pages, Page Order, Creating a post, Adding Media files to content –images and videos, Using Categories and Tags, Creating Pages, Page Hierarchy</p>                                                |   |     |         |        |                    |
| 5 | <p><b>Case Study –Online Sales Website</b></p> <p>Design Page using Elementor plugin<br/>Demonstrate the use of WooCommerce plugin<br/>Add WhatsApp Chat button to website for communication<br/>Integrate Shipping solution to website using (shiprocket / instashipin ) plugin<br/>Integrate Payment gateway to website using (payu / razorpay) plugin</p> | 8 | CO4 | Lecture | Create | Quiz Short Answers |

#### Reference Books

| Sr. No. | Name of the Author  | Title of the Book                                                                          | Year Edition | Publisher Company        |
|---------|---------------------|--------------------------------------------------------------------------------------------|--------------|--------------------------|
| 1       | Lisa Sabin - Wilson | Wordpress Web Design for Dummies                                                           | 2015         | For Dummies              |
| 2       | Lisa Sabin- Wilson  | Wordpress All in One for Dummies                                                           | 2017         | John Wiley & Sons        |
| 3       | Sayyed Majid        | Wordpress Web Development:Basic to Advance                                                 | 2021         | Code Academy, Aurangabad |
| 4       | Joseph Joyner       | Wordpress For Beginners: How to Create and Set Up Your Own Website or Blog Using Wordpress | 2015         | Mihails Konoplovs        |
| 5       | Dr. Ritesh Kumar    | Learn WordPress in Easy Way                                                                | 2019         | Ganpati Book Centre      |

### Online Resources

| Online Resources No. | Web site address                                                                                          |
|----------------------|-----------------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.tutorialspoint.com/wordpress">https://www.tutorialspoint.com/wordpress</a>           |
| 2                    | <a href="https://www.javatpoint.com/wordpress-tutorial">https://www.javatpoint.com/wordpress-tutorial</a> |
| 3                    | <a href="https://www.w3schools.in/wordpress">https://www.w3schools.in/wordpress</a>                       |

### MOOCs:

| Resources No. | Web site address |
|---------------|------------------|
| 1             | NPTEL            |
| 2             | Swayam           |
| 3             | edx.com          |
| 4             | coursera.com     |

| Programme :BCA CBCS– Revised Syllabus w.e.f.-Year2022 –2023                                                                                                                                                                                                                                                                                                                                                                                                               |             |                    |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Course Code | Course Title       |       |
| II                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 202         | DBMS - I           |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Prepared by | Dr. Ayesha Mujawar |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Credits     | Evaluation         | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3           | IE(40) + UA(60)    | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                    |       |
| To make students to: <ul style="list-style-type: none"> <li>• Get familiar with basic concepts of DBMS.</li> <li>• To impart knowledge of the concepts related to database and operations on databases.</li> <li>• To manage database in various environments with emphasis on security measures and concurrency.</li> </ul>                                                                                                                                              |             |                    |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |                    |       |
| After completing the course the students shall be able to<br><b>CO1:</b> To understand the basic concepts in database management system.<br><b>CO2:</b> To design the database by applying data model like Entity relational model.<br><b>CO3:</b> To apply the keys and normalization technique while designing the database.<br><b>CO4:</b> To understand and apply various SQL Components.<br><b>CO5:</b> To understand the concept of transaction and its operations. |             |                    |       |

| Unit                                              | Content                                                                                                                                                                                                                                                                                         | Sessions (in Hrs) | COs Number | Teaching Methodology | Cognition Level | Evaluation Tools      |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------|----------------------|-----------------|-----------------------|
| <b>Introduction Of Database Management System</b> | Basic Concepts of DBMS (Data Vs. Information), Data Processing, Definition of DBMS, Characteristic of Database<br><b>Database architecture:</b> Levels of Abstraction, Database schema and instances<br>3 tier architecture of DBMS Data Independence, Database users, Types of Database System | 8                 | CO1        | Lecture with Ppt     | Understand      | Quiz<br>Short Answers |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                             |    |     |                  |                 |                    |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|------------------|-----------------|--------------------|
| <b>Data Modeling</b> | <b>Logical Data Modeling:</b><br>Hierarchical Data Model,<br>Network Data Model,<br>Relational Data Model.<br><b>Conceptual Data Modeling:</b> Entity Relationship Model, Entities, Attributes, Types of Attributes, Relationships, Relationship set, Degree of relationship Set, Mapping Cardinalities, ER Diagram Notations<br><b>Roles Participation:</b> Total and Partial, Strong and Weak Entity Set. | 8  | CO2 | Lecture with Ppt | Apply (Analyse) | Case Study         |
| <b>Normalization</b> | Codd's Rules for RDBMS<br><b>Keys:</b> Primary key, Foreign key, Candidate key, Super key, Unique key. Simple Key, Composite key<br>Normalization: Concept of normalization, Decomposition, Lossy and Lossless Decomposition, Functional Dependencies. Normal Form: First NF, Second NF, Third NF, Case Studies on Normalization                                                                            | 11 | CO3 | Lecture with Ppt | Apply           | Quiz Short answers |

|                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                       |    |     |                    |            |                      |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|--------------------|------------|----------------------|
| <b>Introduction to Database Languages and Basic concepts of SQL</b> | <p>Database Languages: Introduction of SQL, features, SQL data types.</p> <p>DDL commands: create table, describe table, alter table, and drop table commands.</p> <p>DML Commands: insert, delete, update command</p> <p>DQL commands: All select commands, and order by clause.</p>                                                                                                                 | 8  | CO4 | Lecture with ppt   | Create     | Quiz                 |
| <b>Transaction management and Concurrency control</b>               | <p>Transaction management: Definition of transaction, State of Transaction, ACID properties, Schedules, Serializability of schedules</p> <p>Concurrency control: Lock based concurrency control (2PL), Strict 2PL, Time stamping method.</p> <p>Deadlock and its handling: Definition, Wait-Die and Wound-Wait methods.</p> <p>Database Recovery: Log Based Recovery, Check points, Shadow Paging</p> | 10 | CO5 | Lectures with PPTs | Understand | Quiz<br>Short Answer |

**Reference Books:**

| Sr.No. | Name of the Author                   | Title of the Book                   | Year                            | Publisher Company |
|--------|--------------------------------------|-------------------------------------|---------------------------------|-------------------|
| 1      | Ramez Elmasri,<br>S.Navathe          | Fundamentals of Database Systems    | 6th Edition<br>2010             | Pearson Education |
| 2      | A Silberschatz, H Korth, S Sudarshan | Database System and Concepts        | 6th Edition<br>2010             | McGraw-Hill.      |
| 3      | C.J.Date                             | An Introduction to Database Systems | 3 <sup>rd</sup> Edition<br>2006 | Addison Wesley    |

**Online Resources:**

| Online Resources No. | Website address                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.javatpoint.com/dbms-tutorial">https://www.javatpoint.com/dbms-tutorial</a> |
| 2                    | <a href="https://www.tutorialspoint.com/dbms">https://www.tutorialspoint.com/dbms</a>           |
| 3                    | <a href="https://www.w3schools.in/dbms">https://www.w3schools.in/dbms</a>                       |

**MOOCs:**

| Resources No. | Website address                                        |
|---------------|--------------------------------------------------------|
| 1             | NPTEL / Swayam                                         |
| 2             | <a href="http://www.edx.com">www.edx.com</a>           |
| 3             | <a href="http://www.coursera.com">www.coursera.com</a> |

| Programme: BCA CBCS– Revised Syllabus w.e.f.-Year 2022–2023                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |            |                           |                 |                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------|---------------------------|-----------------|----------------------------------------------------|
| Semester                                                                                                                                                                                                                                                                                                                                                | Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                               | Course Title            |            |                           |                 |                                                    |
| II                                                                                                                                                                                                                                                                                                                                                      | 203                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Data Structures using C |            |                           |                 |                                                    |
|                                                                                                                                                                                                                                                                                                                                                         | Prepared by                                                                                                                                                                                                                                                                                                                                                                                                                                               | Mr. B. D. Patil         |            |                           |                 |                                                    |
| Type                                                                                                                                                                                                                                                                                                                                                    | Credits                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Evaluation              | Marks      |                           |                 |                                                    |
| DSC                                                                                                                                                                                                                                                                                                                                                     | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                         | IE&UA                   | 40+60      |                           |                 |                                                    |
| Course Objectives:                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |            |                           |                 |                                                    |
| <ul style="list-style-type: none"><li>To provide the knowledge of basic data structures and their implementations.</li><li>To understand importance of data structures in context of writing efficient programs.</li><li>To develop skills to apply appropriate data structures in problem solving.</li><li>To understand file handling in C.</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |            |                           |                 |                                                    |
| Course Outcomes:                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |            |                           |                 |                                                    |
| After completing the course the students shall be able to                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |            |                           |                 |                                                    |
| CO1: Learn the basic types for data structure, implementation and application.                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |            |                           |                 |                                                    |
| CO2: Know the strength and weakness of different data structures.                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |            |                           |                 |                                                    |
| CO3: Use the appropriate data structure in context of solution of given problem..                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |            |                           |                 |                                                    |
| CO4: Develop programming skills which require solving given problem.                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |            |                           |                 |                                                    |
| Unit                                                                                                                                                                                                                                                                                                                                                    | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Sessions (Hrs)          | COs Number | Teaching Methodology      | Cognition Level | Evaluation Tools                                   |
| 1                                                                                                                                                                                                                                                                                                                                                       | <b>Introduction to Data Structure -</b> <ul style="list-style-type: none"><li>Data type and data object</li><li>Abstract Data Type (ADT)</li><li>Type of data structure</li><li>Applications of data structures in real life</li><li>Array as a data structure</li><li>Sorting techniques with time complexity: Bubble sort, Selection sort, Insertion sort and Quick sort</li><li>Searching techniques with time complexity: Linear search and</li></ul> | 10                      | CO 1       | Lecture with Ppts<br>Quiz | Understand      | Quiz<br>End Term<br>Internals:<br>Short<br>Answers |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |      |                                              |                    |                                                                                         |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------|----------------------------------------------|--------------------|-----------------------------------------------------------------------------------------|
|   | Binary search                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |      |                                              |                    |                                                                                         |
| 2 | <b>Linked List -</b> <ul style="list-style-type: none"> <li>• Definition and Memory representation of linked list</li> <li>• Types of Linked List- singly, doubly and circular</li> <li>• Basic Operations of linked list</li> <li>• Applications of linked list</li> </ul>                                                                                                                                                                                                                   | 10 | CO 2 | Lecture with Ppts<br>Case Study Applications | Apply (Analyse)    | Case Study , Business cases<br>End Term: Applied Questions                              |
| 3 | <b>Stack and Queue</b><br><b>Stack:</b> <ul style="list-style-type: none"> <li>• Definition</li> <li>• Stack operations</li> <li>• Array implementation of stack</li> <li>• Linked list implementation of stack</li> <li>• Applications of stack</li> </ul> <b>Queue:</b> <ul style="list-style-type: none"> <li>• Definition</li> <li>• Queue operations</li> <li>• Array implementation of queue</li> <li>• Linked list implementation of queue</li> <li>• Applications of queue</li> </ul> | 12 | CO 3 | Lecture with PPTs<br>Case Study Applications | Analyse            | Case Study with Presentations<br>End Term Exams: Case based Questions/Applied Questions |
| 4 | <b>Tree</b> <ul style="list-style-type: none"> <li>• Concept of tree</li> <li>• Tree terminologies</li> <li>• Binary Tree</li> <li>• Types of binary tree</li> <li>• Types of traversal- Preorder, Inorder and Postorder</li> </ul>                                                                                                                                                                                                                                                           | 7  | CO 3 | Lectures with PPTs<br>Video Cases            | Evaluate           | Group Activity<br><br>End Term Exam: Short business cases and situation based questions |
| 5 | <b>File Handling</b> <ul style="list-style-type: none"> <li>• Concept of file</li> <li>• Types of File</li> </ul>                                                                                                                                                                                                                                                                                                                                                                             | 6  | CO 4 | Lecture with ppt<br>Case study on real life  | Analyze / Evaluate | Case Presentation Activity<br>End Term:                                                 |

|  |                                                                                                                                                                                                                                                                             |  |  |                              |  |                                |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|------------------------------|--|--------------------------------|
|  | <ul style="list-style-type: none"> <li>• Operations on file</li> <li>• File modes</li> <li>• file management functions-fopen(), fclose(),fprintf (), fscanf(), getc(), putc (), getw(), putw ()</li> <li>• Random access functions-fseek(), ftell() and rewind()</li> </ul> |  |  | applications<br><br>Activity |  | Theory<br>Applied<br>Questions |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|------------------------------|--|--------------------------------|

### Reference Books

| Sr.No. | Name of the Author | Title of the Book               | Year Edition   | Publisher Company       |
|--------|--------------------|---------------------------------|----------------|-------------------------|
| 1      | Yashavant Kanetkar | Data Structures Through C       | 2009 Second    | BPB Publications        |
| 2      | Reema Thareja      | Programming in c                | 2011 First     | Oxford University Press |
| 3      | Aaron Tenenbaum    | Data Structures using C and C++ | Second Edition | Pearson Education       |

### Online Resources:

| Online Resources No. | Website address                                                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.mygreatlearning.com/blog/data-structures-using-c/">https://www.mygreatlearning.com/blog/data-structures-using-c/</a> |
| 2                    | <a href="https://www.edureka.co/blog/c-data-structures/">https://www.edureka.co/blog/c-data-structures/</a>                               |
| 3                    | <a href="https://www.programiz.com/dsa">https://www.programiz.com/dsa</a>                                                                 |
| 4                    | <a href="https://www.javatpoint.com/data-structure-tutorial">https://www.javatpoint.com/data-structure-tutorial</a>                       |

### MOOCs:

| ResourcesNo. | Website address                                        |
|--------------|--------------------------------------------------------|
| 1            | NPTEL / Swayam                                         |
| 2            | <a href="http://www.edx.com">www.edx.com</a>           |
| 3            | <a href="http://www.coursera.com">www.coursera.com</a> |

| Programme : BCA CBCS– RevisedSyllabusw.e.f.-Year2022 –2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                      |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Course Code | Course Title         |       |
| II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 204         | Financial Accounting |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Prepared by | Dr.A.B.Nadaf         |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Credits     | Evaluation           | Marks |
| MDC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3           | IE (40) + UA(60)     | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |                      |       |
| <ul style="list-style-type: none"> <li>▪ To get familiar with basics of accounting concepts.</li> <li>▪ To learn journal entries and prepare financial statements</li> <li>▪ To get acquainted with computerised accounting system</li> </ul>                                                                                                                                                                                                                                                                                                                         |             |                      |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                      |       |
| <p>After completing the course, the students shall be able to</p> <p><b>CO1</b> : Remember the basic numerical operations and pass book entries.</p> <p><b>CO2</b> :Understand the basics of financial accounting and accounting principles</p> <p><b>CO3</b> : Apply the rules of journal entries for preparing journals, ledgers and trial balance.</p> <p><b>CO4</b> : Analyse the trial balance and transferring the accounts to respective financial statements.</p> <p><b>CO5</b> :Evaluate the adjustments and applying its effect on respective accounts.</p> |             |                      |       |

| Unit | Content | Sessions (Hrs) | COs No | Teaching Methodology | Cognition Level | Evaluation Tools |
|------|---------|----------------|--------|----------------------|-----------------|------------------|
|      |         |                |        |                      |                 |                  |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |            |                    |                                                 |                                                                                    |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------------|--------------------|-------------------------------------------------|------------------------------------------------------------------------------------|
| 1 | <p>Need for Accounting, Meaning and definition of book keeping, System of Book keeping. Financial Accounting- definition, Scope and objectives, Financial Accounting v/s Book Keeping, Limitations of Financial Accounting. End users of financial statements.</p> <p>Accounting principles- Accounting Concepts and Conventions , , Branches of accounting , concept of bad debts , depreciation ,methods of depreciation :Fixed and reducing, Examples on depreciation</p> | 10 | CO1<br>CO2 | Classroom Lectures | Understand                                      | Attentiveness of the students, End Term Exams                                      |
| 2 | <p>Journal-importance and utility, classification of accounts, journalizing of transactions. Ledger-meaning and utility, posting of journal entries to the ledgers ,closing the ledger accounts, Examples on journal entries of transactions and posting them to ledgers, closing ledger accounts</p>                                                                                                                                                                        | 10 | CO2        | Lecture Method     | Understand and Apply with the simple case study | <p>Case Study Discussion ,</p> <p>Class Test' End Term</p> <p>Class Assignment</p> |

|   |                                                                                                                                                                                                                                                           |    |     |                |                      |                                           |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|----------------|----------------------|-------------------------------------------|
| 3 | Simple Cash book, Cash Book with two columns, Cashbook with three columns, Petty Cash Book , Purchase book, Sales book, Purchase Return book , Sales return book . Trial Balance - meaning and purpose, Preparation of Trial Balance from ledger accounts | 09 | CO3 | Lecture Method | Understand and Apply | Case Study, Question and Answer, End Term |
| 4 | Meaning of final account, Need to prepare final account , Uses of Final account , Preparation of Final account of Sole Proprietorship : Trading and Profit, Loss Account and Balance Sheet of sole proprietary business with given adjustments            | 09 | CO4 | Lecture Method | Understand and Apply | Case Study, End Term                      |
| 5 | Need of accounting software, features of accounting packages , introduction to Tally package, various books maintained in Tally accounting package, atomized effect of one transaction in various books of accounting through accounting package.         | 07 | CO5 | Lecture Method | Understand           | End Term                                  |

**Reference Books:**

| <b>Sr. No.</b> | <b>Name of the Author</b>     | <b>Title of the Book</b>                        | <b>Year Edition</b> | <b>Publisher Company</b> |
|----------------|-------------------------------|-------------------------------------------------|---------------------|--------------------------|
| 1              | Dr. S. N. Maheshwari          | Financial Accounting For Management             | 2012                | Vikas Publishing House   |
| 2              | Robert Anthony, David Hawkins | Business Accounting                             | 2009                | Tata McGraw–Hill         |
| 3              | M.G.Patkar                    | Book-Keeping & Accountancy                      | 2006                | FYJC Commerce            |
| 4              | Anil Chowdhary                | Fundamentals of Accounting & Financial Analysis | 2007                | Pearson Education        |

**MOOC's:**

| <b>Sr.No.</b> | <b>Website address</b>                                                                                          |
|---------------|-----------------------------------------------------------------------------------------------------------------|
| 1             | <a href="https://in.coursera.org/courses?query=accounting">https://in.coursera.org/courses?query=accounting</a> |

| Programme: BCA CBCS– Revised Syllabus w.e.f.-Year 2022–2023                                                                                                                                                                                                                                                                                                                                                               |             |                                |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------|---------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                  | Course Code | Course Title                   |         |
| II                                                                                                                                                                                                                                                                                                                                                                                                                        | 205         | Lab on Data Structures using C |         |
|                                                                                                                                                                                                                                                                                                                                                                                                                           | Prepared by | Mr.B.D.Patil                   |         |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                      | Credits     | Evaluation                     | Marks   |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                       | 2           | IE&UA                          | 40 + 60 |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                 |             |                                |         |
| <input type="checkbox"/> To write and execute programs in C to solve problems using data structures such as arrays, linked list, stack, queues and trees.<br><input type="checkbox"/> To learn to write C programs to implement various sorting and searching algorithms.<br><input type="checkbox"/> To understand the basics of file handling and to write C programs to implement different file management functions. |             |                                |         |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                   |             |                                |         |
| <b>CO1:</b> Able to identify the appropriate data structures for solving real world problems.<br><b>CO2:</b> Able to implement various kinds of searching and sorting techniques.<br><b>CO3:</b> Able to implement data structures such as arrays, linked list, stack, queues and trees to solve various computing problems.<br><b>CO4:</b> Able to implement different file management functions.                        |             |                                |         |

| Unit | Content                                                                                                                                                                                                                                                                                                                                     | Sessions (Hrs) | COs Number | Teaching Methodology | Cognition Level | Evaluation Tools                                   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|----------------------|-----------------|----------------------------------------------------|
| 1    | <b>Introduction to Data Structure -</b><br>Write C programs for the following operations on Array.<br>(i) Creation (ii) insertion (iii) deletion (iv) traversal<br><br>Write C programs for implementing the following searching techniques.<br>1) Linear search<br>2) Binary search<br><br>Write C programs for implementing the following | 7              | CO 1       | Lab Demo , Quiz      | Understand      | Quiz<br>End Term<br>Internals:<br>Short<br>Answers |

|   |                                                                                                                                                                                                                                                                                                                                                                                                   |   |      |                             |                 |                                                                                         |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------|-----------------------------|-----------------|-----------------------------------------------------------------------------------------|
|   | sorting techniques to arrange a list of integers in ascending order.<br>1) Bubble sort<br>2) Insertion sort<br>3) Selection sort                                                                                                                                                                                                                                                                  |   |      |                             |                 |                                                                                         |
| 2 | <b>Linked List -</b><br>Write a C program for the following operations on Singly Linked List.<br>1) Creation 2) insertion 3) deletion 4) traversal 5) Searching<br><br>Write a C program to count number of items present in a singly linked list.<br><br>Write a C program for the following operations on Doubly Linked List.<br>1) Creation 2) insertion 3) deletion 4) traversal 5) Searching | 7 | CO 2 | Lab Demo , Quiz, Case study | Apply (Analyse) | Case Study , Business cases<br>End Term: Applied Questions                              |
| 3 | <b>Stack and Queue</b><br>Write a C program to implement stack using array.<br><br>Write a C program to implement stack using linked list.<br><br>Write a C program that convert infix expression into postfix form.<br><br>Write a C program to convert decimal to binary using stack.<br><br>Write a C program to check whether a string is a Palindrome or not using stack.                    | 7 | CO 3 | Lab Demo , Quiz, Case study | Analyse         | Case Study with Presentations<br>End Term Exams: Case based Questions/Applied Questions |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |      |                             |                    |                                                                                                |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------|-----------------------------|--------------------|------------------------------------------------------------------------------------------------|
|   | <p>Write a C program to convert an infix expression into prefix format.</p> <p>Write a C program to implement queue using array.</p> <p>Write a C program to implement queue using linked list.</p>                                                                                                                                                                                                                                                                                                              |   |      |                             |                    |                                                                                                |
| 4 | <p><b>Tree</b></p> <p>Write C program to demonstrate concept of tree.</p> <p>Write a C program to count number of leaf nodes and total number of nodes in a tree.</p>                                                                                                                                                                                                                                                                                                                                            | 4 | CO 3 | Lab Demo , Quiz, Case study | Evaluate           | <p>Group Activity</p> <p>End Term Exam: Short business cases and situation based questions</p> |
| 5 | <p><b>File Handling</b></p> <p>Write C programs to implement working of following file management functions:<br/>fprintf (), fscanf(), getc(), putc (), getw(), putw ()</p> <p>Write C programs to implement working of following Random access functions:<br/>fseek(), ftell() and rewind()</p> <p>Write a C program to display contents of a file in uppercase and lowercase letters.</p> <p>Write a C program to count characters, spaces, tabs and new lines in a file.</p> <p>Write a C program to copy</p> | 5 | CO 4 | Lab Demo , Quiz, Case study | Analyze / Evaluate | <p>Case Presentation Activity</p> <p>End Term: Theory Applied Questions</p>                    |

|  |                                                                                                                                                                                                                    |  |  |  |  |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  | <p>the contents of one file to another file.</p> <p>Write a C program to receive strings from keyboard and write them to a file.</p> <p>Write a program to read strings from a file and display them on screen</p> |  |  |  |  |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|

### Reference Books

| Sr.No. | Name of the Author | Title of the Book               | Year Edition   | Publisher Company       |
|--------|--------------------|---------------------------------|----------------|-------------------------|
| 1      | Yashavant Kanetkar | Data Structures Through C       | 2009 Second    | BPB Publications        |
| 2      | Reema Thareja      | Programming in c                | 2011 First     | Oxford University Press |
| 3      | Aaron Tenenbaum    | Data Structures using C and C++ | Second Edition | Pearson Education       |

### Online Resources:

| Online Resources No. | Website address                                                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.mygreatlearning.com/blog/data-structures-using-c/">https://www.mygreatlearning.com/blog/data-structures-using-c/</a> |
| 2                    | <a href="https://www.edureka.co/blog/c-data-structures/">https://www.edureka.co/blog/c-data-structures/</a>                               |
| 3                    | <a href="https://www.programiz.com/dsa">https://www.programiz.com/dsa</a>                                                                 |
| 4                    | <a href="https://www.javatpoint.com/data-structure-tutorial">https://www.javatpoint.com/data-structure-tutorial</a>                       |

### MOOCs:

| Resources No. | Website address                                        |
|---------------|--------------------------------------------------------|
| 1             | NPTEL / Swayam                                         |
| 2             | <a href="http://www.edx.com">www.edx.com</a>           |
| 3             | <a href="http://www.coursera.com">www.coursera.com</a> |

| Programme:BCA CBCS –Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                        |             |                            |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------|-------|
| Semester                                                                                                                                                                                                                                              | Course Code | Course Title               |       |
| II                                                                                                                                                                                                                                                    | 206         | Web Development Technology |       |
|                                                                                                                                                                                                                                                       | Prepared by | Dr.Suvarna Patil           |       |
| Type                                                                                                                                                                                                                                                  | Credits     | Evaluation                 | Marks |
| DSC                                                                                                                                                                                                                                                   | 2           | UE:IE                      | 60:40 |
| <b>Course Objectives:</b>                                                                                                                                                                                                                             |             |                            |       |
| To make students to: <ul style="list-style-type: none"> <li>To get proficiency in Website designing</li> <li>To learn Wordpress as Content Management System</li> <li>To get familiar to use all setting and components of Wordpress</li> </ul>       |             |                            |       |
| <b>Course Outcomes :</b>                                                                                                                                                                                                                              |             |                            |       |
| <b>CO1:</b> To demonstrate Hosting,Website layout, Admin Panel, Header, footer<br><b>CO2 :</b> To demonstrate general setting and use of Themes and Templates, PlugIn in Wordpress<br><b>CO3:</b> To create Website with Themes and Templates, PlugIn |             |                            |       |

| Unit | Content                                                                                                                                                     | Sessions (Hrs ) | COs Number | Teaching Methodology | CognitionLevel | EvaluationTools |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|----------------------|----------------|-----------------|
| 1    | Domain Hosting<br><br>Content Management System (Wordpress),<br>Domain – Registration ,<br>Manage DNS , Nameserver<br>WordPress - Installation of WordPress | 4               | CO 1       | Practical Demo       | Create         | Quiz            |
| 2    | Header and Footer Configuration<br>General Configuration –Site Configuration – Logo , Site Icon , Site Name                                                 | 5               | CO 2       | Practical Demo       | Apply          | Quiz            |

|   |                                                                                                                                                                                                                                                                                                                                                                                                       |    |     |                       |        |      |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|-----------------------|--------|------|
|   | Home page Setting ,<br>Website layout Setting                                                                                                                                                                                                                                                                                                                                                         |    |     |                       |        |      |
| 3 | General Writing Reading<br>,Discussion , media,<br>permalinks and privacy<br>data<br>Themes<br>Activate Plugin                                                                                                                                                                                                                                                                                        | 5  | CO2 | Practic<br>al<br>Demo | Create | Quiz |
| 4 | Themes, Downloading,<br>installing, and activating<br>themes, WordPress Plugin:<br><br>Downloading, installing,<br>and activating<br><br>Templates Downloading,<br>installing, and activating<br>Templates, Design Pages<br>using Template<br><br>Adding Hyperlinks, Playing<br>with Media content,<br>Previewing and Editing Pages,<br>Page Order, Creating a post,<br>Adding Media files to content | 6  | CO3 | Practic<br>al<br>Demo | Create | Quiz |
| 5 | Demonstrate the use of<br>WooCommerce plugin<br>Add WhatsApp Chat button<br>to website for<br>communication<br>Integrate Shipping solution<br>to website Integrate<br>Payment gateway to<br>website                                                                                                                                                                                                   | 10 | CO3 | Practic<br>al<br>Demo | Create | Quiz |

**Reference Books**

| <b>Sr. No.</b> | <b>Name of the Author</b> | <b>Title of the Book</b>                                                                 | <b>Year Edition</b> | <b>Publisher Company</b> |
|----------------|---------------------------|------------------------------------------------------------------------------------------|---------------------|--------------------------|
| 1              | Lisa Sabin - Wilson       | Wordpress Web Design forDummies                                                          | 2015                | For Dummies              |
| 2              | Lisa Sabin- Wilson        | Wordpress All in One forDummies                                                          | 2017                | John Wiley & Sons        |
| 3              | Sayyed Majid              | Wordpress Web Development:Basic to Advance                                               | 2021                | Code Academy, Aurangabad |
| 4              | Joseph Joyner             | Wordpress For Beginners: Howto Create and Set Up Your OwnWebsite or Blog Using Wordpress | 2015                | Mihails Konoplovs        |
| 5              | Dr. Ritesh Kumar          | Learn WordPress in Easy Way                                                              | 2019                | Ganpati BookCentre       |

**Online Resources**

| <b>Online Resources No.</b> | <b>Web site address</b>                                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------|
| 1                           | <a href="https://www.tutorialspoint.com/wordpress">https://www.tutorialspoint.com/wordpress</a>           |
| 2                           | <a href="https://www.javatpoint.com/wordpress-tutorial">https://www.javatpoint.com/wordpress-tutorial</a> |
| 3                           | <a href="https://www.w3schools.in/wordpress">https://www.w3schools.in/wordpress</a>                       |

**MOOC:**

| <b>ResourcesNo.</b> | <b>Web site address</b> |
|---------------------|-------------------------|
| 1                   | NPTEL                   |
| 2                   | Swayam                  |
| 3                   | edx.com                 |
| 4                   | coursera.com            |

**Programme :BCA CBCS– Revised Syllabus w.e.f.-Year2022 –2023**

| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Course Code        | Course Title                  |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------|-------|
| <b>II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>207</b>         | <b>Environmental Studies</b>  |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Prepared by</b> | <b>Dr.Pallavi Jamsandekar</b> |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Credits            | Evaluation                    | Marks |
| VBC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2                  | IE(50)                        | 50    |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                               |       |
| <ul style="list-style-type: none"> <li>To Understand the nature and function of the natural environment affecting society.</li> </ul>                                                                                                                                                                                                                                                                                                                                                   |                    |                               |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                    |                               |       |
| <p><b>CO1:</b> Understand the importance of Environment in the life of living things.<br/> <b>CO2:</b> Apply the awareness knowledge in taking eco-friendly actions in society.<br/> <b>CO3:</b> Judge what is right and wrong for the environment in day to day life.<br/> <b>CO4:</b> Analyse the impact of activities on environment and its effect.<br/> <b>CO5:</b> Understand the need and way of sustainable development and will pass the knowledge to the next generation.</p> |                    |                               |       |

| Unit                                                       | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Sessions (Hrs ) | COs Number | Teaching Methodology | Cognition Level | Evaluation Tools |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|----------------------|-----------------|------------------|
| <b>The multidisciplinary nature of environment studies</b> | <p>Definition, scope and importance-need of public awareness.</p> <p><b>Natural Resources:</b><br/> <b>Renewable and non-renewable resources:</b> Forest resources: Use and over-exploitation, deforestation.</p> <p>Case studies.</p> <p>Timber extraction, mining, dams and their effects on forest and tribal people.</p> <p><b>Water resources:</b> Use and over-utilization of surface and groundwater, floods, droughts, conflicts over water, dams-benefit and Problems.</p> | 8               | CO1        | Class Teaching       | Understanding   | Class Test       |

|  |                                                          |  |  |  |  |  |
|--|----------------------------------------------------------|--|--|--|--|--|
|  | <b>Mineral<br/>resources:</b><br>Use and<br>exploitation |  |  |  |  |  |
|--|----------------------------------------------------------|--|--|--|--|--|

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |         |                                 |               |         |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------|---------------------------------|---------------|---------|
|                  | <p>‘environmental effects of extracting and using mineral resources, case studies.</p> <p><b>Food resources:</b> World food problems, changes caused by agriculture. Fertilizer- pesticide problems, water logging, salinity, case studies.</p> <p><b>Energy resources:</b> Growing energy needs, renewable and non-renewable energy resources, use of alternative energy sources.</p> <p><b>Land resources:</b> Land as resources, land degradation, man induced landslides, desertification. Role of individual in conservation of natural resources. Equitable use of resources for sustainable lifestyles</p> |   |         |                                 |               |         |
| <b>Ecosystem</b> | <p>Concept of ecosystem, structure and function of an ecosystem, producers, consumers and decomposers .Energy flow in the ecosystem, Ecological succession, food chains, food webs and ecological pyramids, introduction, types, characteristics features structure and function of the following ecosystem, forest</p>                                                                                                                                                                                                                                                                                           | 8 | CO2,CO3 | Classroom Teaching and Projects | Understanding | Seminar |

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |              |                               |         |                     |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------------|-------------------------------|---------|---------------------|
|                                           | ecosystem ,grassland ecosystem, Desert ecosystem, Aquatic ecosystems, ponds, stream, lakes, rivers, estuaries                                                                                                                                                                                                                                                                                                                                                                                  |   |              |                               |         |                     |
| <b>Biodiversity and its conservations</b> | Introduction, Definition: genetic, species and ecosystem diversity, Biogeographically classification of India, value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option vales, India as a mega diversity nation, Hot-Spots of biodiversity, Threats to biodiversity: habitat loss, poaching of wildlife, Man wildlife conflicts, Endangered and endemic species of India, Conservation of biodiversity: In situ and Ex-situ conservation of biodiversity. | 6 | CO1,CO3,C O4 | Class Teaching and Field Work | Analyse | Quiz and Case Study |
| <b>Role for Environment Conservation</b>  | Social issues and environment - Unsustainable to sustainable Role of IT in Environment and human health.- Human population issue. E-waste – Impact and remedies<br><ul style="list-style-type: none"> <li>Climate Change-Green House gases effect</li> </ul>                                                                                                                                                                                                                                   | 8 | CO1,CO2,C O5 | Field Work and Project        | Apply   | Project             |

|  |                                                                                                                       |  |  |  |  |  |
|--|-----------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  | Project work- Each candidate has to go for field visit and complete a project work on Environmental issues in society |  |  |  |  |  |
|--|-----------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|

### References Books:

| Sr. No | Name of the Author        | Title of the Book                 | Publisher Company          |
|--------|---------------------------|-----------------------------------|----------------------------|
| 1      | Bharucha Erach            | The Biodiversity of India         | Mapin Publishing Pvt. Ltd. |
| 2      | Agrawal K.C               | Environmental Biology             | Nidhi Publishers Ltd(2001) |
| 3      | Jadhav H and Bhosale V.M. | Environmental Protection and Laws | Himalaya Publishing House. |
| 4      | Miller T.G. Jr.           | Environmental Science             | Wadsworth Publishing Co.   |

### MOOC

| Resource No. | Website Address |
|--------------|-----------------|
| 1            | NPTEL           |
| 2            | Swayam          |
| 3            | edx.com         |
| 4            | coursera.com    |

| Programme :BCA CBCS– Revised Syllabus w.e.f.-Year2022 –2023                                                                                     |             |                                        |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------|-------|
| Semester                                                                                                                                        | Course Code | Course Title                           |       |
| II                                                                                                                                              | 208         | Community Work – Swacch Bharat Abhiyan |       |
|                                                                                                                                                 | Prepared by | Prof. Dexter Woodward                  |       |
| Type                                                                                                                                            | Credits     | Evaluation                             | Marks |
| VBC                                                                                                                                             | 2           | IE(50)                                 | 50    |
| <b>Course Objectives:</b>                                                                                                                       |             |                                        |       |
| <ul style="list-style-type: none"> <li>This course aims to expose the students to Swacch Bharat Abhiyan initiative of the government</li> </ul> |             |                                        |       |
| <b>Course Outcomes:</b>                                                                                                                         |             |                                        |       |
| CO1:Students will be able to understand the details about the Swacch Bharat Abhiyanand its impact on society.                                   |             |                                        |       |

| Unit                     | Content                                                                                                                                                                           | Ses<br>sio<br>ns | CO<br>s<br>Nu<br>mb<br>e<br>r | Teaching<br>Methodolog<br>y                                                                                                      | Cognitio<br>nLevel | Evaluatio<br>nTools                                                   |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------|
| 1.Swacch Bharat Abhiyan  | Swacch Bharat Abhiyan : History, meaning, Roots of Swacch Bharat Abhiyan, Goals of Cleanliness initiatives.                                                                       | 8                | CO 1                          | Lecture with Ppts                                                                                                                | Understand         | Quiz/ GD on solutions to disposal of garbage                          |
| 2. Cleanliness           | Initiators of cleanliness drive in India. Sant Ghadage Baba, Mahatma Gandhi, Efforts taken towards the Swachh Bharat Abhiyan, Swachh Bharat Mission, Role of NGO's in Cleanliness | 6                | CO 1                          | Lecture with Ppts<br>Guest lectured by eminent personalities of society like Govt Officials / NGO's involved in Cleanliness work | Apply (Analyse)    | Case Study with Presentations<br>End Term Exams: Case based Questions |
| 3. Impact of Cleanliness | Impact of Cleanliness initiatives and sanitation awareness. Social Awareness, Case Studies- COVID-19, Swachh Toycathon Initiative, Mumbai Municipality Slum                       | 6                | CO 1                          | Lecture with Ppts<br>Guest Lectures by Doctors<br>Visit to Slums                                                                 | Analyse            | Case Study<br>Field Visit                                             |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                     |    |      |            |            |                                                                          |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------|------------|------------|--------------------------------------------------------------------------|
|                           | Sanitation Program India.                                                                                                                                                                                                                                                                                                                                                                           |    |      |            |            |                                                                          |
| <b>4. Community Hours</b> | Internship of 15 days (100 hours) to be undertaken<br>Submit a report on a particular type of community involvement undertaken<br>Topics may be related to:<br>Sanitation, Waste Management, Digital Innovations, Green Practices, Involvement in Public Infrastructure Cleanliness, Animations, Videos creating awareness about Swacch Bharat, Designing innovative Swachh Toycathon toy/game etc. | 10 | CO 1 | Field Work | Evaluation | Field Study Project Report<br><br>Presentation Viva<br><br>End Term Exam |

#### References Websites:

|    |                                                                                                                                                                                                                                                                                                     |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | <a href="http://www.swachhbharaturban.in/">www.swachhbharaturban.in/</a>                                                                                                                                                                                                                            |
| 2. | <a href="https://en.wikipedia.org/wiki/Municipal_solid_waste">https://en.wikipedia.org/wiki/Municipal_solid_waste</a>                                                                                                                                                                               |
| 3. | <a href="https://swachhbharatmission.gov.in/sbmcms/index.htm">https://swachhbharatmission.gov.in/sbmcms/index.htm</a>                                                                                                                                                                               |
| 4. | <a href="https://innovateindia.mygov.in/swachh-toycathon/">https://innovateindia.mygov.in/swachh-toycathon/</a>                                                                                                                                                                                     |
| 5. | <a href="https://www.susana.org/_resources/documents/default/2-1925-india-draft--en-susana-cs-india-mumbai-slumsanitationprogram-2010doc-anlage.pdf">https://www.susana.org/_resources/documents/default/2-1925-india-draft--en-susana-cs-india-mumbai-slumsanitationprogram-2010doc-anlage.pdf</a> |

## MOOC

| Resource No. | Website Address |
|--------------|-----------------|
| 1            | NPTEL           |
| 2            | Swayam          |
| 3            | edx.com         |
| 4            | coursera.com    |

| Programme: BCA CBCS– Revised Syllabus w.e.f.-Year 2022 –2023                                                                                                                                                                                                                                                                                                                                           |             |                    |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                               | Course Code | Course Title       |       |
| III                                                                                                                                                                                                                                                                                                                                                                                                    | 301         | Operating Systems  |       |
|                                                                                                                                                                                                                                                                                                                                                                                                        | Prepared by | Dr. Prashant Patil |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                   | Credits     | Evaluation         | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                    | 3           | IE(40) + UA(60)    | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                              |             |                    |       |
| To make students to: <ul style="list-style-type: none"> <li>To acquire knowledge regarding structure and working of the major operating system components</li> <li>To learn and apply different process and memory scheduling algorithms and synchronization techniques to achieve better performance of computer system.</li> <li>To understand structure and organisation of file system.</li> </ul> |             |                    |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                |             |                    |       |
| After completing the course the students shall be able to <p><b>CO1:</b> Understand functioning and working of Operating System</p> <p><b>CO2:</b> Explain the concepts of process scheduling, memory and file management</p> <p><b>CO3:</b> Understand I/O System</p>                                                                                                                                 |             |                    |       |

| Unit | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Session (Hrs.) | COs Number | Teaching Methodology      | Cognition Level | Evaluation Tools                           |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|---------------------------|-----------------|--------------------------------------------|
| 1    | <b>Introduction to operating System</b> Definition and concept of OS, History of OS, Importance and function of Operating system. Types of OS<br>-Batch System, timesharing, Multitasking, Multiprogramming, multi-processing, online operating system, real time, distributed operating system. Views- command language users view, system call users view, structure of OS- simple, monolithic system and layered system, client server model.<br>User operating - system interface: command line interface, GUI, system calls | 7              | CO1        | Lecture with PPTs<br>Quiz | Understand      | End Term<br>Internals:<br>Short<br>Answers |

|   |                                                                                                                                                                                       |    |     |                            |                       |                                   |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|----------------------------|-----------------------|-----------------------------------|
| 2 | <b>Process Management -</b><br>Processconcept, Process Control<br>Block OS services for Process management, scheduling and typesof schedulers, scheduling algorithm- First come first | 10 | CO2 | Lecture with PPTs<br>Video | Understand & Evaluate | End Term Internals: Short Answers |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|----------------------------|-----------------------|-----------------------------------|

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |     |                                   |                       |                                                              |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|-----------------------------------|-----------------------|--------------------------------------------------------------|
|   | served, shortest job first, shortest remaining time next, time slice scheduling, priority-based scheduling, multilevel queue, multilevel queue with feedback                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |     |                                   |                       |                                                              |
| 3 | <b>Storage Management</b> - Basic concept of storage management, logical and physical address space, swapping, contiguous allocation, non-Contiguous allocation, fragmentation, segmentation, paging, demand paging, virtual memory, page replacement algorithms- FIFO, Optimal page replacement algorithm, least recently page replacement algorithm, clock page replacement algorithm, design issue of paging, thrashing.                                                                                                                                                                                | 10 | CO2 | Lecture with PPTs<br>Video        | Understand & Evaluate | Assignments<br>End Term<br>Internals:<br>Short<br>Answers    |
| 4 | <b>Inter-process communication and synchronization</b> - Need, Mutual Exclusion, Semaphore, Busy-wait Implementation, characteristics of semaphore, queuing implementation of semaphore, producer consumer problem, critical region and conditional critical area. What is deadlock? Conditions to occur the deadlock, deadlock prevention, deadlock avoidance- banker's algorithm. Resource request, resource release.                                                                                                                                                                                    | 8  | CO2 | Lecture with PPTs<br>Quiz         | Analyze               | Classroom test<br>End Term<br>Internals:<br>Short<br>Answers |
| 5 | <b>File Systems and I/O System :</b><br><b>File System :</b> Files-basic concept, file attributes, operations, file types, file structure, access methods, Directory- structure- single level directory system, two level directory system, hierarchical directory system, directory operations, protection, security, allocation method.<br><b>Input/output System:</b> Principles of I/O hardware, I/O devices, device controller, DMA, Principles of I/O software- goals, interrupt handler, device driver. Mass storage structure-disk structure, disk scheduling (FCFS, SSTF, SCAN, LOOK, C- SCAN, C- | 10 | CO3 | Lecture with PPTs<br>Case Studies | Understand & Apply    | Quiz<br>End Term<br>Internals:<br>Short<br>Answers           |

|  |       |  |  |  |  |  |
|--|-------|--|--|--|--|--|
|  | LOOK) |  |  |  |  |  |
|--|-------|--|--|--|--|--|

### Reference Books:

| Sr.No. | Name of the Author                      | Title of the Book                                 | Year                     | Publisher Company              |
|--------|-----------------------------------------|---------------------------------------------------|--------------------------|--------------------------------|
| 1      | Silber Schatz,<br>Galvin,Gagne          | Operating System<br>Concepts                      | 11 <sup>th</sup> Edition | Wiley Publication              |
| 2      | Milan Milinkovic                        | Operating Systems<br>Concept and Design           | 2 <sup>nd</sup> Edition  | McGraw Hill<br>Education India |
| 3      | Andrew Tanenbaum and<br>Albert Woodhull | Operating Systems<br>Design and<br>Implementation | 3 <sup>rd</sup> Edition  | Pearson                        |

### Online Resources:

| Online Resources No. | Website address                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.studytonight.com/operating-system/">https://www.studytonight.com/operating-system/</a>                       |
| 2                    | <a href="https://www.tutorialspoint.com/operating_system/index.htm">https://www.tutorialspoint.com/operating_system/index.htm</a> |
| 3                    | <a href="https://www.youtube.com/watch?v=WJ-UaAaumNA">https://www.youtube.com/watch?v=WJ-UaAaumNA</a>                             |
| 4                    | <a href="https://www.youtube.com/watch?v=zFnrUVqtiOY">https://www.youtube.com/watch?v=zFnrUVqtiOY</a>                             |

### MOOCs:

| Resources No. | Website address                                        |
|---------------|--------------------------------------------------------|
| 1             | NPTEL/ Swayam                                          |
| 2             | <a href="http://www.edx.com">www.edx.com</a>           |
| 3             | <a href="http://www.coursera.com">www.coursera.com</a> |

| Semester                                                                                                                                                                                                                                                                                                                                                                                                | Course Code | CourseTitle          |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------|-------|
| III                                                                                                                                                                                                                                                                                                                                                                                                     | 302         | Software engineering |       |
|                                                                                                                                                                                                                                                                                                                                                                                                         | Prepared by | Dr. Smita Gambhire   |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                    | Credits     | Evaluation           | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                     | 3           | IA(40) + UE(60)      | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                               |             |                      |       |
| <p>To make students to:</p> <ul style="list-style-type: none"> <li>• To make students familiar with basic concepts of Software Engineering.</li> <li>• To introduce the methodologies involved in the development and maintenance of Software over its entire life cycle.</li> </ul>                                                                                                                    |             |                      |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                 |             |                      |       |
| <p>After completing the course, the students shall be able to</p> <p><b>CO1:</b> Understand life cycle models, Requirement elicitation techniques, understand the concept of Analysis and Design of software.</p> <p><b>CO2:</b> Develop SRS as per any of the existing standards.</p> <p><b>CO3:</b> Implement software engineering concepts in software development to develop quality software..</p> |             |                      |       |

| Unit | Contents                                                                                                                                                                                                                                                                                                 | Sessions (Hrs) | COs Number   | Teaching Methodology                              | Cognition Level               | Evaluation Tools                            |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|---------------------------------------------------|-------------------------------|---------------------------------------------|
| 1.   | <b>1. Introduction to Software Engineering:</b> Software, Program vs Software, software characteristics, Definition of Software Engineering, importance, principles of software engineering, Difference between software engineering and software programming, Members involved in software development. | 8              | CO1,CO3      | Lecture with Ppts, Discussion                     | Understand                    | Discussion                                  |
| 2.   | <b>2. Software process and Feasibility study:</b> Need of Feasibility study, types of Feasibility study, Cost Benefit Analysis.<br>General software development life cycle with all phases. Overview of software models (Waterfall, Prototyping, and Spiral and Rapid Application Development model).    | 8              | CO1,CO2, CO3 | Lecture with Ppts, Practical sessions on computer | Understand and calculate      | Understand and calculate cost of project    |
| 3.   | <b>3. Requirement Engineering Concepts and Methods :</b> What is Requirement Engineering, Types of requirements, Requirement elicitation techniques- Traditional methods and Modern methods, Verification and validation process. Principles of Requirement Specification,                               | 11             | CO1,CO3      | Lecture with PPTs, Case Studies                   | Understand and data gathering | Analyze and apply engineering steps for it. |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |     |                                      |                                  |                                                           |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|--------------------------------------|----------------------------------|-----------------------------------------------------------|
|    | Software Requirement Specification document<br>Outline Characteristics of good SRS: - correct, complete, unambiguous, consistent, modifiable, traceable, Understandable                                                                                                                                                                                                                                                                 |    |     |                                      |                                  |                                                           |
| 4. | <b>Analysis and Design Tools:</b> Entity-Relationship Diagrams, Decision Tree and Decision Table, Data Flow Diagrams (DFD), Data Dictionary, Elements of DD Advantage of DD, Pseudocode, Input and Output Design<br><b>Structured System Design:</b><br>Modules Concepts and Types of Modules Structured Chart, Qualities of Good Design, Coupling, Types of Coupling, Cohesion, Types of Cohesion, CASE STUDIES (Based on Above Topic) | 8  | CO3 | Lectures with PPTs, and Case Studies | Evaluate                         | Formulate and practice the case studies on various topics |
| 5. | <b>Software Testing, Quality Control and Software Maintenance</b><br>:Definition, Test characteristics, Types of testing:<br>Black-Box Testing, White-Box Testing, Unit testing,                                                                                                                                                                                                                                                        | 10 | CO3 | Lectures with PPTs,                  | Design Quality Control mechanism | Use quality control and maintenance mechanism             |

|  |                                                                                                                                                                                                                                                                                 |  |  |  |  |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  | Integration testing<br><b>Quality concept:</b><br>Quality, SQA Plan,<br>Software<br>Configuration<br>Management<br>Formal Technical<br>review: Review<br>meeting, review<br>reporting and review<br>guidelines Software<br>Configuration<br>Process.<br>software<br>maintenance |  |  |  |  |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|

#### ReferenceBooks:

| Sr.No. | NameoftheAuthor                 | Title ofthe Book                                       | Year               | PublisherCompa<br>ny                 |
|--------|---------------------------------|--------------------------------------------------------|--------------------|--------------------------------------|
| 1      | Roger S. Pressman               | SOFTWARE<br>ENGINEERING A<br>PRACTITIONERS<br>APPROACH | seventh<br>edition | McGraw Hill<br>International Edition |
| 2      | Sommerville                     | Software Engineering                                   | seventh<br>edition | Pearson Education                    |
| 3      | K.K. Aggarwal &<br>Yogesh Singh | Software Engineering                                   | -                  | New Age<br>International             |

#### Online Resources:

| OnlineResourcesNo. | Websiteaddress                                                                                        |
|--------------------|-------------------------------------------------------------------------------------------------------|
| 1                  | <a href="https://www.youtube.com/watch?v=Z6f9ckEElsU">https://www.youtube.com/watch?v=Z6f9ckEElsU</a> |
| 2                  | <a href="https://www.youtube.com/watch?v=4b1D1QFEel0">https://www.youtube.com/watch?v=4b1D1QFEel0</a> |

#### MOOCs:

| ResourcesNo. | Websiteaddress                                                                                                                                                              |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1            | <a href="https://onlinecourses.nptel.ac.in/noc19_cs69/preview">https://onlinecourses.nptel.ac.in/noc19_cs69/preview</a>                                                     |
| 2            | <a href="https://www.classcentral.com/course/introduction-to-software-engineering-98973">https://www.classcentral.com/course/introduction-to-software-engineering-98973</a> |

| Programme: BCA CBCS– Revised Syllabus w.e.f.-Year2022 –2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |                  |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CourseCode  | Course Title     |       |
| III                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 303         | Java Programming |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Prepared by | Dr. Rahul Jadhav |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Credits     | Evaluation       | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3           | IA(40) + UA(60)  | 100   |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |                  |       |
| <p>After completing the course the students shall be able to:</p> <p><b>CO1:</b> To develop proficiency in creating console based applications using the Java Programming Language.</p> <p><b>CO2:</b> To interpret the concepts of object oriented Programming Language and easily use Java.</p> <p><b>CO3:</b>Design interfaces, abstract and concrete classes</p> <p><b>CO4:</b>Use concurrent programming, Java Collections and utility classes</p> <p><b>CO5:</b> To understand and implement File Handling in Java.</p> <p><b>CO6:</b>Get the main features of Java Programming for Business Applications</p> |             |                  |       |

| Unit | Contents                                                                                                                                                                                                                                                                                                                                          | Sessions (Hrs) | COs Number | Teaching Methodology    | Cognition Level | Evaluation Tools |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|-------------------------|-----------------|------------------|
| 1    | Features of Java, Java compiler,JVM, Garbage collection, Data types, concept of class and object, control structures in java, arrays in java, array of objects.                                                                                                                                                                                   | 8              | CO1, CO2   | PowerPoint Presentation | Understanding   | Short answer     |
| 2    | Concepts of OOP, Defining a class, creating objects from class, adding attributes and methods to the class, using constructors, Passing values to the functions – pass by value, pass by reference, Function overloading. Modifiers – public, private, protected, default, static, final, Concept of package, Introduction to Exception Handling. | 10             | CO2, CO6   | Lab Demonstration       | Applying        | Short answer     |
| 3    | Concept and importance of inheritance, is-a relationship, types of inheritance, Polymorphism – function overriding, dynamic method dispatch. Using abstract and final keywords with class declaration, Concept of interface and class.                                                                                                            | 8              | CO2 CO3    | Lab Demonstration       | Applying        | Short answer     |
| 4    | Concept of streams, types of streams – byte streams, character streams. The Console: System.out, System.in, and System.err, InputStream class,                                                                                                                                                                                                    | 8              | CO4 CO5    | Lab Demonstration       | Applying        | Short answer     |

|   |                                                                                                                                                                                                                                 |    |     |                                             |          |              |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|---------------------------------------------|----------|--------------|
|   | OutputStream class, File class, FileInputStreams, File OutputStream, Reader class, Writer class, FileReader, FileWriter.                                                                                                        |    |     |                                             |          |              |
| 5 | Introduction to GUI controls – Button, Lable, TextField, TextArea, List, Checkbox and RadioButtons, Scrollbar, Menu etc.<br><b>Applets:</b> Applet concept, creating basic applet, applet lifecycle, controlling applet content | 11 | CO6 | PowerPoint Presentation , Lab Demonstration | Creating | Short answer |

#### Reference Books:

| Sr.No. | Name of the Author                | Title of the Book           | Year                    | Publisher Company |
|--------|-----------------------------------|-----------------------------|-------------------------|-------------------|
| 1      | Herbert Schildt                   | The Complete Reference JAVA | 7 <sup>th</sup> Edition | McGraw-Hill       |
| 2      | Cay S. Horstmann and Gary Cornell | Core Java Volume-I          | 8 <sup>th</sup> Edition | Sun Core Series   |
| 3      | Bruce Eckel                       | Thinking In Java            | 4 <sup>th</sup> Edition | Printice Hall     |

#### Online Resources:

| Online Resources No. | Website address                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.w3schools.com/java/">https://www.w3schools.com/java/</a>                             |
| 2                    | <a href="https://www.javatpoint.com/java-tutorial">https://www.javatpoint.com/java-tutorial</a>           |
| 3                    | <a href="https://www.tutorialspoint.com/java/index.htm">https://www.tutorialspoint.com/java/index.htm</a> |
| 4                    | <a href="https://docs.oracle.com/javase/tutorial/">https://docs.oracle.com/javase/tutorial/</a>           |

#### MOOCs:

| Resources No. | Website address                                        |
|---------------|--------------------------------------------------------|
| 1             | NPTEL/ Swayam                                          |
| 2             | <a href="http://www.edx.com">www.edx.com</a>           |
| 3             | <a href="http://www.coursera.com">www.coursera.com</a> |

| Programme:BCA CBCS-RevisedSyllabusw.e.f.-Year2022 – 2023                                                                                                                                                                                                                                                                                                                                                                            |             |                      |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                            | Course Code | Course Title         |       |
| III                                                                                                                                                                                                                                                                                                                                                                                                                                 | 304         | Statistics           |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                     | Prepared by | Dr. Sheetal Deshmukh |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                | Credits     | Evaluation           | Marks |
| MDC                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3           | IE(40) + UA(60)      | 100   |
| Course Objectives:                                                                                                                                                                                                                                                                                                                                                                                                                  |             |                      |       |
| To make students to: <ul style="list-style-type: none"> <li>To understand the statistical concepts.</li> <li>To provide knowledge related to various tabulation methods and representation of data.</li> </ul> To learn and apply Measures of Central Tendencies, Measures o Dispersion, Regression and Correlation Analysis.                                                                                                       |             |                      |       |
| Course Outcomes :                                                                                                                                                                                                                                                                                                                                                                                                                   |             |                      |       |
| After completion of the course the students shall be able to<br><b>CO1:</b> Understand types of statistical data, data collection and representation of data.<br><b>CO2:</b> Explain the concepts of Measures of Central Tendencies, Measures o Dispersion, Regression and Correlation Analysis.<br><b>CO3:</b> Solve examples applying Measures of Central Tendencies, Measures o Dispersion, Regression and Correlation Analysis. |             |                      |       |

| Unit | Content                                                                                                                                                                                                                                                                                                                                                                                                                    | Sess<br>ions<br>(Hrs<br>) | COs Number           | Teaching<br>Methodolog<br>y   | Cognitio<br>nLevel | Evaluatio<br>nTools                    |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------|-------------------------------|--------------------|----------------------------------------|
| 1    | <b>Introduction to Statistics Data Collection and representation:</b><br>Definition of Statistics,<br>Importance of Statistics,<br>Scope of statistics,<br>Limitations of Statistics,<br>Advantages and Disadvantages of Statistics.<br><b>Types of data:</b> Primary and Secondary data,<br>Sources of Data collection,<br><b>Tabular Representation of data:</b> Ungrouped and grouped frequency distribution, Graphical | 13                        | CO 1<br>CO 2<br>CO 3 | Lecture with PPT, White board | Understand         | Quiz, Assignment Questions, Class Test |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |                      |                               |         |                                        |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------------|-------------------------------|---------|----------------------------------------|
|   | representation of data:<br>Histogram, frequency polygon and Curve, Cumulative frequency curves (ogive curves).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |                      |                               |         |                                        |
| 2 | <b>Measures of central tendency:</b><br>a) <b>Mean:</b> Definition, problems on mean for individual observations, ungrouped frequency distribution and grouped frequency distribution, merits and demerits, Examples.<br><br>b) <b>Median:</b> Definition, problems on median individual observations, ungrouped frequency distribution and grouped frequency distribution, merits and demerits, Examples.<br><br>c) <b>Mode:</b> Definition, problems on mode for individual observations, ungrouped frequency distribution and grouped frequency distribution, merits and demerits, Examples.          | 9 | CO 1<br>CO 2<br>CO 3 | Lecture with PPT, White board | Apply   | Quiz, Assignment Questions, Class Test |
| 3 | <b>Measures of Dispersion:</b><br>a) <b>Range:</b> Definition, problems on range for individual observations, ungrouped frequency distribution and grouped frequency distribution, merits and demerits of Range, Examples.<br><br>b) <b>Mean Deviation:</b> Definition, problems on mean deviation about mean for individual observations, ungrouped frequency distribution and grouped frequency distribution, merits and demerits, Examples.<br><br>c) <b>Standard Deviation:</b> Definition, problems on standard deviation for individual observations, ungrouped frequency distribution and grouped | 9 | CO 1<br>CO 2<br>CO 3 | Lecture with PPT, White board | Analyze | Quiz, Assignment Questions, Class Test |

|   |                                                                                                                                                                                                                                                                                                 |   |                      |                               |                       |                                        |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------------|-------------------------------|-----------------------|----------------------------------------|
|   | frequency distribution, merits and demerits. Coefficient of variation, coefficient of Determination and Standard error, Examples                                                                                                                                                                |   |                      |                               |                       |                                        |
| 4 | <b>Regression Analysis:</b><br>Introduction to Regression Analysis, Lines of Regression Equation:<br><br>A) Regression Equation of Y on X,<br>B) Regression Equation of X on Y , Properties of Regression co-efficient , problems on finding regression equations and estimations               | 7 | CO 1<br>CO 2<br>CO 3 | Lecture with PPT, White board | Analysis & Evaluation | Quiz, Assignment Questions, Class Test |
| 5 | <b>Correlation Analysis:</b><br>Introduction, Types of Correlation, Scatter Diagram , Karl Pearson's coefficient of correlation, Properties and Interpretation of Correlation coefficient, Merits and Demerits of Karl Pearson's Coefficient, Spearman's Rank correlation Coefficient, Examples | 7 | CO 1<br>CO 2<br>CO 3 | Lecture with PPT, White board | Create                | Quiz, Assignment Questions, Class Test |

### Reference Books

| Sr. No.    | Name of the Author | Title of the Book      | Year Edition             | Publisher Company |
|------------|--------------------|------------------------|--------------------------|-------------------|
| 1 National | S.P.Gupta          | Statistical Techniques | 46 <sup>th</sup> Edition | Pearson           |
| 2 National | Ranjeet Chitale    | Statistical Techniques | 1 <sup>st</sup> edition  | Nirali Prakashan  |

### Online Resources

| Online Resources No. | Web site address                                                                                                                      |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.geeksforgeeks.org/measures-of-central-tendency/">https://www.geeksforgeeks.org/measures-of-central-tendency/</a> |
| 2                    | <a href="https://www.cuemath.com/data/measures-of-dispersion/">https://www.cuemath.com/data/measures-of-dispersion/</a>               |

**MOOCs:**

| Resources No. | Web site address                                   |
|---------------|----------------------------------------------------|
| 1             | <a href="http://www.Swayam.Com">www.Swayam.Com</a> |
| 2             | <a href="http://www.nptel.com">www.nptel.com</a>   |

| Programme: BCA CBCS– Revised Syllabus w.e.f.-Year2022 –2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |                   |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Course Code | Course Title      |       |
| III                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 305         | Lab on Oracle     |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Prepared by | Dr. Hanmant Rense |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Credits     | Evaluation        | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2           | IA(40) + UA(60)   | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                   |       |
| <ul style="list-style-type: none"> <li>To provide a strong formal foundation in database concepts and implementation</li> <li>To provide practice to the participants to groom them into well informed database application developers.</li> <li>To understand the Architecture of Oracle database.</li> <li>To design and develop a relational database system with appropriate functionality to process data with integrity constraints and avoid data redundancy.</li> <li>To implement queries using SQL (Structured Query Language)</li> <li>To work with various objects of Oracle.</li> </ul>                                                                                 |             |                   |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |                   |       |
| <p>After completing the course the students shall be able to:</p> <p><b>CO1</b>-To provide a strong formal foundation in database concepts and implementation.</p> <p><b>CO2</b>-To provide practice to the participants to groom them into well informed database application developers.</p> <p><b>CO3</b>-To understand the Architecture of Oracle database.</p> <p><b>CO4</b>-To design and develop a relational database system with appropriate functionality to process data with integrity constraints and avoid data redundancy.</p> <p><b>CO5</b>-To implement queries using SQL (Structured Query Language)</p> <p><b>CO6</b>-To work with various objects of Oracle.</p> |             |                   |       |

| Unit | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Sessions (Hrs) | COs Number | Teaching Methodology    | Cognition Level | Evaluation Tools |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|-------------------------|-----------------|------------------|
| 1    | <p><b>Introduction to Oracle:</b> History, Architecture, Features, Versions of Oracle, Oracle File Management, Spool command</p> <p><b>SQL:</b><br/>SQL query Rules, Data types, Keywords, Delimiters, Literals.<br/>Defining a database in SQL.<br/><b>Components of SQL:</b> DDL, DML, DCL, DQL,<br/>DDL Commands – Defining a database in SQL, Creating table, changing table definition, removing table.<br/>DML Commands- Inserting, updating,</p> | 6              | CO1, CO3   | PowerPoint Presentation | Understanding   | Short answer     |

|   |                                                                                                                                                                                                                                                                             |   |          |                                             |          |              |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------|---------------------------------------------|----------|--------------|
|   | deleting data.<br>DQL Commands: Select Statement with all options. Renaming table, Describe Command, Distinct Clause, Sorting Data in a Table.                                                                                                                              |   |          |                                             |          |              |
| 2 | <b>Data Constraints</b><br>Primary key, Foreign Key, NOT NULL, UNIQUE, CHECK constraint.                                                                                                                                                                                    | 6 | CO2, CO4 | Lab Demonstration                           | Applying | Short answer |
| 3 | <b>Operators:-</b><br>Arithmetic, Logical, Relational, Range Searching, Pattern Matching, IN & NOT IN Predicate, all, % any, exists, not exists clauses,<br><br>Set Operations: Union, Union All, Minus, Intersect.                                                         | 6 | CO5      | Lab Demonstration                           | Applying | Short answer |
| 4 | <b>Joins and Oracle Functions:-</b><br>Join Concept. Simple join, equi join, non equi join, Self-join, Outer join, Sub queries, Aggregate Functions, Numeric Functions, String Functions, Conversion functions, Date conversion functions, and Date functions.              | 6 | CO5      | Lab Demonstration                           | Applying | Short answer |
| 5 | <b>Database Objects :-</b><br><b>Index:</b> Creating index, simple index, composite index, unique index, dropping indexes, multiple indexes on table<br><b>Sequence:</b> Creating sequence, altering sequence, dropping sequence.<br><b>Views:</b> Concept, creation, usage | 6 | CO6      | PowerPoint Presentation , Lab Demonstration | Creating | Short answer |

#### Reference Books:

| Sr.No. | Name of the Author            | Title of the Book                                         | Year Edition            | Publisher Company |
|--------|-------------------------------|-----------------------------------------------------------|-------------------------|-------------------|
| 1      | Ivan Bayross                  | SQL,PL/SQL The programming – Language of Oracle           | 3 <sup>rd</sup> Edition | BPB Publication   |
| 2      | Bob Bryla , Kevin Loney       | Oracle Database 12c The Complete Reference (Oracle Press) | 2 <sup>nd</sup> Edition | Oracle Press      |
| 3      | Sanjay Mishra & Alen Beaulieu | Mastering Oracle SQL                                      | --                      | O'Reilly          |

**Online Resources:**

| Online Resources No. | Web site address                                                                                                                                         |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                    | SQL Tutorial for Beginners:<br><a href="https://www.youtube.com/watch?v=wkOD6mbXc2M">https://www.youtube.com/watch?v=wkOD6mbXc2M</a>                     |
| 2                    | <a href="https://www.mygreatlearning.com/blog/sql-tutorial-for-beginners/">https://www.mygreatlearning.com/blog/sql-tutorial-for-beginners/</a>          |
| 3                    | SQL TUTORIALS FOR BEGINNERS:<br><a href="https://www.youtube.com/watch?v=zPes5jBZ62c">https://www.youtube.com/watch?v=zPes5jBZ62c</a>                    |
| 4                    | Learn SQL (Structured Query Language)   Edureka<br><a href="https://www.youtube.com/watch?v=BPHAr4QGGVE">https://www.youtube.com/watch?v=BPHAr4QGGVE</a> |

**MOOCs:**

| Resources No. | Web site address                                                                                              |
|---------------|---------------------------------------------------------------------------------------------------------------|
| 1             | <a href="https://www.classcentral.com/institution/oracle">https://www.classcentral.com/institution/oracle</a> |
| 2             | <a href="https://www.mooc-list.com/tags/oracle">https://www.mooc-list.com/tags/oracle</a>                     |
| 3             | <a href="https://in.coursera.org/courses?query=oracle">https://in.coursera.org/courses?query=oracle</a>       |

| Programme: BCA CBCS– Revised Syllabus w.e.f.-Year2022 –2023                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |                 |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Course Code | Course Title    |       |
| III                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 306         | Lab on JAVA     |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Prepared by | Dr.Rahul Jadhav |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Credits     | Evaluation      | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2           | IA(40) + UA(60) | 100   |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                 |       |
| After completing the course the students shall be able to:<br><b>CO1:</b> Provide foundation for programming and Enable the students to analyze and efficiently solve the problems using Java Programming<br><b>CO2:</b> To develop proficiency in creating console based applications using the Java Programming Language.<br><b>CO3:</b> To interpret the concepts of object oriented Programming Language.<br><b>CO4:</b> To develop logical abilities of students using Java Programming language |             |                 |       |

| Unit | Content                                                                                                                                                                                                                            | Sessions (Hrs) | COs Number | Teaching Methodology | Cognition Level | Evaluation Tools |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|----------------------|-----------------|------------------|
| 1    | Program to demonstrate the following:<br>1. Branching Statements<br>2. Looping Statements<br>3. Classes and objects<br>4. Arrays<br>5. Array of objects.                                                                           | 8              | CO1, CO2   | Lab Demonstration    | Applying        | Short answer     |
| 2    | Design Programs on following concepts:<br>1. Constructor<br>2. Constructor Overloading<br>3. Pass by value<br>4. Method Overloading<br>5. Package<br>6. Exception Handling                                                         | 10             | CO2,       | Lab Demonstration    | Applying        | Short answer     |
| 3    | Working with Inheritance and Interface:<br>1. Programs to demonstrate working of Inheritance, types of inheritance and Polymorphism – function overriding.<br>2. Making use of abstract and final keywords with class declaration. | 8              | CO2 CO3    | Lab Demonstration    | Applying        | Short answer     |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |     |                   |          |              |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|-------------------|----------|--------------|
|   | 3. Programs to demonstrate working of interface.                                                                                                                                                                                                                                                                                                                                                                                                     |    |     |                   |          |              |
| 4 | Program to demonstrate Java Input/Output :<br>1. Concept of streams, byte streams, character streams.<br>2. The Console: System.out, System.in, and System.err<br>3. Making use of InputStream class, OutputStream class, File class, FileInputStreams, File OutputStream, Reader class, Writer class, FileReader, FileWriter.<br>Buffered streams –<br>BufferedInputStream, BufferedOutputStream, BufferedReader, BufferedWriter.<br>Object Streams | 8  | CO4 | Lab Demonstration | Applying | Short answer |
| 5 | Write a java program that loads names and phone numbers from a text file where the data is organized as one line per record and each field in a record are separated by a tab (\t).it takes a name or phone number as input and prints the corresponding other value from the hash table(hint: use hash tables)<br>Implement the above program with database instead of a text file.                                                                 | 11 | CO4 | Lab Demonstration | Applying | Short answer |

#### ReferenceBooks:

| Sr.No. | Name of the Author                | Title of the Book           | Year                    | Publisher Company |
|--------|-----------------------------------|-----------------------------|-------------------------|-------------------|
| 1      | Herbert Schildt                   | The Complete Reference JAVA | 7 <sup>th</sup> Edition | McGraw-Hill       |
| 2      | Cay S. Horstmann and Gary Cornell | Core Java Volume-I          | 8 <sup>th</sup> Edition | Sun Core Series   |
| 3      | Bruce Eckel                       | Thinking In Java            | 4 <sup>th</sup> Edition | Printice Hall     |

#### Online Resources:

| Online Resources No. | Website address                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.w3schools.com/java/">https://www.w3schools.com/java/</a>                             |
| 2                    | <a href="https://www.javatpoint.com/java-tutorial">https://www.javatpoint.com/java-tutorial</a>           |
| 3                    | <a href="https://www.tutorialspoint.com/java/index.htm">https://www.tutorialspoint.com/java/index.htm</a> |
| 4                    | <a href="https://docs.oracle.com/javase/tutorial/">https://docs.oracle.com/javase/tutorial/</a>           |

**MOOCs:**

| <b>Resources No.</b> | <b>Website address</b>                                 |
|----------------------|--------------------------------------------------------|
| 1                    | NPTEL/ Swayam                                          |
| 2                    | <a href="http://www.edx.com">www.edx.com</a>           |
| 3                    | <a href="http://www.coursera.com">www.coursera.com</a> |

| Programme: BCA CBCS – Revised Syllabus w.e.f – 2022-2023                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                               |                    |             |                                            |                                            |                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------|--------------------------------------------|--------------------------------------------|-----------------------------------------------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                 | Course Code                                                                                                                                                                                                                                                                                                                                                   | Course Title       |             |                                            |                                            |                                               |
| III                                                                                                                                                                                                                                                                                                                                                                                      | 307                                                                                                                                                                                                                                                                                                                                                           | Startup Management |             |                                            |                                            |                                               |
|                                                                                                                                                                                                                                                                                                                                                                                          | Prepared by                                                                                                                                                                                                                                                                                                                                                   | Mr.Dexter Woodward |             |                                            |                                            |                                               |
| Type                                                                                                                                                                                                                                                                                                                                                                                     | Credits                                                                                                                                                                                                                                                                                                                                                       | Evaluation         | Marks       |                                            |                                            |                                               |
| AEC                                                                                                                                                                                                                                                                                                                                                                                      | 2                                                                                                                                                                                                                                                                                                                                                             | IA                 | 50          |                                            |                                            |                                               |
| Course Objectives:                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                               |                    |             |                                            |                                            |                                               |
| 1. To inspire the student Fraternity with entrepreneurial mind sets and encourage them to brainstorm ideas for a startup.<br>2. To identify various sources of funding and how one can raise capital for a startup.<br>3. To Outline various phases of the new ventures and help one to identify growing markets.<br>4. To acquire skills to overcome challenges one faces in a startup. |                                                                                                                                                                                                                                                                                                                                                               |                    |             |                                            |                                            |                                               |
| Course Outcomes:                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                               |                    |             |                                            |                                            |                                               |
| CO1: Students will get a better understanding of how to establish a startup and various options available for startup.<br>CO2: Better Understanding of capital raising and other legal requirements for a new venture.<br>CO3: Develop in students requisite qualities of an entrepreneur<br>CO4: Helps a student from the desire of a start up to a complete entrepreneur               |                                                                                                                                                                                                                                                                                                                                                               |                    |             |                                            |                                            |                                               |
| Unit                                                                                                                                                                                                                                                                                                                                                                                     | Cont ents                                                                                                                                                                                                                                                                                                                                                     | Sessi ons (H rs. ) | COs Num ber | Teaching Methodolog y                      | Cognitio n Level                           | Evaluation Tools                              |
| 1                                                                                                                                                                                                                                                                                                                                                                                        | <b>Introduction to Startup Management</b> <ul style="list-style-type: none"><li>• What is a startup</li><li>• Interception of a startup,idea generation.</li><li>• Business startup, venturechoice</li><li>• Startup prominence in theIndian Scenario</li><li>• Role of the Government in promotion of startups</li><li>• The six forces of change.</li></ul> | 7                  | CO1<br>CO2  | Lectures, Experts form Industry Case study | Understa nding<br>Remembe ring<br>Planning | Quiz<br><br>Class test                        |
| 2                                                                                                                                                                                                                                                                                                                                                                                        | <b>Venture capital and StatutoryEnvironment</b> <ul style="list-style-type: none"><li>• Identifying startup capital</li><li>• Sources of capital and</li></ul>                                                                                                                                                                                                | 8                  | CO2         | Lectures<br><br>Case Studies<br><br>Group  | Understa nding<br>ImPLYing<br>Analysin g   | Class Test<br>Online Quiz<br>Group Discussion |

|  |                                                           |  |  |  |  |  |
|--|-----------------------------------------------------------|--|--|--|--|--|
|  | <ul style="list-style-type: none"><li>• funding</li></ul> |  |  |  |  |  |
|--|-----------------------------------------------------------|--|--|--|--|--|

|   |                                                                                                                                                                                                                                                                                                                                                                                                             |   |            |                                                                     |                                                          |                            |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------|---------------------------------------------------------------------|----------------------------------------------------------|----------------------------|
|   | <ul style="list-style-type: none"> <li>• Estimation of fund requirement for a startup</li> <li>• Positioning of a new startup / Venture</li> <li>• Approval of new venture</li> <li>• Tax structure and tax discounts for new ventures</li> <li>• Legal environment for startups and new ventures</li> <li>• Case study</li> </ul>                                                                          |   |            | Discussion                                                          | Learning                                                 |                            |
| 3 | <b>Financial aspects at the start and during growth phase</b> <ul style="list-style-type: none"> <li>• Feasibility Analysis</li> <li>• Ways and means of raising funding's</li> <li>• Equity Funding</li> <li>• Crowd funding</li> <li>• Alliance and Partnership</li> <li>• Growth strategies and market growth.</li> <li>• Venture life patterns and reasons of failure.</li> <li>• Case Study</li> </ul> | 7 | CO3<br>CO4 | Lectures<br>Case studies<br>Presentation Evaluation<br>Field Visits | Understanding<br>Exploring<br>Implementation             | Online Tests<br>Internship |
| 4 | <b>Growth, Failure and Exit</b> <ul style="list-style-type: none"> <li>• Stages of Growth</li> <li>• Venture life partners</li> <li>• Failure and reason of failure</li> <li>• Preparing for change</li> <li>• Leadership successor</li> <li>• Dealing with bankruptcy</li> <li>• Exist strategies, sale of startup, being acquired /going public / liquidation</li> </ul>                                  | 8 | CO4        | Lectures<br>Case study                                              | Learning<br>Understanding<br>Exploring<br>Implementation | presentation               |

| Sr. No. | Name of the Author             | Title of the Book                                                   | Year Edition: | Publisher Company                                     |
|---------|--------------------------------|---------------------------------------------------------------------|---------------|-------------------------------------------------------|
| 1       | Anjan Raichaudhuri             | Managing New Ventures Concepts and Cases                            | 2010          | Prentice Hall International                           |
| 2       | S.R. Bhowmik and M. Bhowmik    | Entrepreneurship                                                    | 2011          | New Age International,                                |
| 3       | Vijay Sathe,                   | Corporate Entrepreneurship                                          | 2009          | Cambridge,                                            |
| 4       | Steven Fisher, Janna' Duane, , | The Startup Equation -A Visual Guidebook for Building Your Startup, | 2016          | Indian Edition Mc Graw Hill Education India Pvt. Ltd, |

|   |                     |                                 |      |                                  |
|---|---------------------|---------------------------------|------|----------------------------------|
| 5 | Peter F. Drucker () | Innovation and Entrepreneurship | 2007 | Classic Drucker Collection, 2007 |
|---|---------------------|---------------------------------|------|----------------------------------|

| Online Resources | Website address                                                                                                                                                                                                                                                                              |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                | 1 <a href="https://www.cloudways.com/blog/best-startup-tools/">https://www.cloudways.com/blog/best-startup-tools/</a> The 30 Best Startup Tools & Resources to Grow Your Business                                                                                                            |
| 2                | <a href="https://otm.illinois.edu/sites/default/files/StartUp%20Handbook%20for%20web.pdf">https://otm.illinois.edu/sites/default/files/StartUp%20Handbook%20for%20web.pdf</a><br>The Start-up Handbook                                                                                       |
| 3                | <a href="https://visme.co/blog/wp-content/uploads/24-Essential-Tools-and-Resourcesfor-Entrepreneurs-by-Visme.pdf">https://visme.co/blog/wp-content/uploads/24-Essential-Tools-and-Resourcesfor-Entrepreneurs-by-Visme.pdf</a> 24 Essential Tools and Resources for Startups and Entrepreneur |

| MOOCS | Website address                                                                                                                                   |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <a href="https://www.mooc-list.com/tags/startup">https://www.mooc-list.com/tags/startup</a>                                                       |
| 2     | <a href="https://www.mooc-list.com/course/entrepreneurial-mindset-coursera">https://www.mooc-list.com/course/entrepreneurial-mindset-coursera</a> |
| 3     | <a href="https://www.my-mooc.com/en/categorie/entrepreneurship">https://www.my-mooc.com/en/categorie/entrepreneurship</a>                         |

| Programme: BCA CBCS– Revised Syllabus w.e.f.-Year 2022–2023                                                                                                                                                                                                                  |             |                     |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------|-------|
| Semester                                                                                                                                                                                                                                                                     | Course Code | Course Title        |       |
| III                                                                                                                                                                                                                                                                          | 308         | Yoga and Meditation |       |
|                                                                                                                                                                                                                                                                              | Prepared by | Dr.Anita Patil      |       |
| Type                                                                                                                                                                                                                                                                         | Credits     | Evaluation          | Marks |
| VBC                                                                                                                                                                                                                                                                          | 2           | IA                  | 50    |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                    |             |                     |       |
| <ul style="list-style-type: none"> <li>To provide the basic knowledge of the theory and practice of yoga so that the students learn to practice asana</li> <li>To build awareness of yoga among student</li> <li>To promote positive health and holistic wellness</li> </ul> |             |                     |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                      |             |                     |       |
| <b>CO1:</b> Study Yogasana, Kriya, Bandhas, Mudra, Meditation and Pranayama<br><b>CO2:</b> Demonstrate and practice Yoga exercise for wellness.                                                                                                                              |             |                     |       |

| Unit No | Content                                                                                                                                                                                       | Session (hrs.) | CO Number   | Teaching Methodology   | Cognition (As per Bloom's Taxonomy) | Evaluation Tools                                       |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------|------------------------|-------------------------------------|--------------------------------------------------------|
| 1       | What is Yoga? Brief history and development of Yoga. The Fundamentals of Yoga Traditional Schools of Yoga Yogic practices for health and wellness General Guidelines for Yoga Practice Prayer | 5              | CO 1<br>CO2 | Lecture with PPTs      | Understand                          | End Term: Applied Questions                            |
| 2       | Preparatory Exercises I. Neck Bending II. Trunk Movement III. Knee Movement IV. Other movements Surya Namaskara and Benefits                                                                  | 5              | CO 1<br>CO2 | Lecture with Ppts Quiz | Apply (Analyse)                     | Quiz<br>End Term<br>Internals: Short Answers           |
| 3       | Definition, Benefits A. Standing Asana Tadasana ,Vrikṣasana , ArdhaChakrasana a Trikoṇasana,                                                                                                  | 10             | CO 1<br>CO2 | Lecture Case Activity  | Create                              | Case Presentation Activity<br>End Term: Theory Applied |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |             |                                           |         |                                            |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------|-------------------------------------------|---------|--------------------------------------------|
|   | Virasana B.<br>Sitting Asana<br>ArdhaUṣṭrasana,<br>Sanskarsana<br>Vakrasana,<br>Vajrasana C.<br>Pron Asana<br>Bhujangasana,<br>Shalabhasana<br>Dharunasan,<br>Makarasan D.<br>Supine Asana<br>Setubandhasana,<br>Pavanamuktasan<br>a Sarvangasana,<br>Savasana                                                                                                                                                                                                                                                                         |    |             |                                           |         |                                            |
| 4 | and Praṇayama<br>Meditative<br>Postures<br>: Sukhasana,<br>Swastikasana;<br>Vajrsan;<br>Ardhapadmasana,<br>Padmasana,<br>Siddhasana<br>Preparatory<br>Breathing<br>Practices<br>Sectional<br>Breathing<br>(Abdominal,<br>Thoracic and<br>Clavicular<br>Breathing)<br>Yogic Deep<br>Breathing<br>Concept of<br>Puraka, Rechaka<br>and Kumbhaka<br>OM Meditation<br>Shuddikriya<br>Definition,<br>Benefits,<br>Kapalbhati<br>Trataka<br>Praṇayama<br>Definition,<br>Benefits,<br>NadiSodhana /<br>AnulomaViloma<br>BhramariPraṇay<br>ama | 10 | CO 1<br>CO2 | Lectures<br>with PPTs<br>Case<br>Activity | Analyse | Activity<br>End Term:<br>Theory<br>Applied |

**Reference Books:**

| Sr.No. | Name of the Author          | Title of the Book               | Year Edition | Publisher Company                                        |
|--------|-----------------------------|---------------------------------|--------------|----------------------------------------------------------|
| 1      | Goyandka, Harikrishandass   | Yoga Darshan                    | 2010         | Geeta Press, Gorakhpur                                   |
| 2      | DhirendraBrahmachari        | Yogic SuksmaVyayma              | 1986         | Dhirendra Yoga Publications, New Delhi,                  |
| 3      | Joshi, K.S.                 | Yoga in daily life              | 1985         | Orient paper backs Delhi                                 |
| 4      | VishwasMandlik              | Yoga Parichay                   |              |                                                          |
| 5      | Saraswati, Swami Satyananda | Asana, Pranayama, Mudra, Bandha | 2006         | Yoga Publications Trust<br>Bihar School of Yoga, Munger, |

**Online Recources:**

| Sr.No | URL                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <a href="https://yoga.ayush.gov.in/public/assets/front/pdf/CYPEnglishLeaflet.pdf">https://yoga.ayush.gov.in/public/assets/front/pdf/CYPEnglishLeaflet.pdf</a> |

| Programme: BCA CBCS– Revised Syllabus w.e.f.-Year 2022 –2023                                                                                                                                                                                                                                                                                                                                                                  |             |                    |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                      | Course Code | Course Title       |       |
| IV                                                                                                                                                                                                                                                                                                                                                                                                                            | 401         | Computer Networks  |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                               | Prepared by | Mr. Prasanna Rasal |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                          | Credits     | Evaluation         | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                           | 3           | IE(40) +<br>UA(60) | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                     |             |                    |       |
| <ul style="list-style-type: none"> <li>To acquire a foundational understanding of computer network and communication technologies.</li> <li>To provide knowledge regarding various network protocols.</li> <li>To understand the Advanced Network Technologies and applications of Network.</li> </ul>                                                                                                                        |             |                    |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                       |             |                    |       |
| <p>After completing the course</p> <p><b>CO1:</b> Students will acquire a good knowledge of the computer network, its architecture and Operation.</p> <p><b>CO2:</b> Student will be able to pursue his study in advanced networking courses.</p> <p><b>CO3:</b> Students will be able to follow trends of computer networks. So, students will get exposure of advanced network technologies like MANET, WSN, 4G and 5G.</p> |             |                    |       |

| Unit                                     | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Session (Hrs.) | COs Number | Teaching Methodology      | Cognition Level | Evaluation Tools                           |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|---------------------------|-----------------|--------------------------------------------|
| <b>Introduction to Computer Networks</b> | <p>What is Computer Network? Network Goals and Motivations, Application of Networks, Network Topologies, Types of Networks.</p> <p>Network software: Network Protocols, Protocol Hierarchies, Connection Oriented and Connectionless Services.</p> <p>Network Models: The OSI Reference Model, The TCP/IP Reference Model, Comparison of OSI and TCP/IP Reference Model,</p> <p>Examples of some networks: Internet, X.25, ISDN, Frame relay, ATM, Ethernet, Wireless LAN-(Wi-Fi).</p> | 8              | CO1        | Lecture with PPTs<br>Quiz | Understand      | End Term<br>Internals:<br>Short<br>Answers |

|                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |     |                                   |                       |                                                         |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|-----------------------------------|-----------------------|---------------------------------------------------------|
| <b>Data Transmission and Physical Layer</b>                | <p>Signals: Analog and Digital Signals, Data Rate, Transmission Impairment, Signal Measurement: Throughput, Propagation Speed and Time, Wavelength, Frequency, Bandwidth, Spectrum</p> <p>Transmission Media &amp; its Characteristics: Guided and Unguided Media, Synchronous and Asynchronous Transmission, Multiplexing: FDM,WDM,TDM, Switching: Circuit, Message and Packet Switching,</p> <p>MobileTelephoneSystems:1G,2G,3G,4G,5G</p>                                                                    | 9  | CO2 | Lecture with PPTs<br>Video        | Understand & Evaluate | End Term Internals: Short Answers                       |
| <b>Network Layer: Design Issues and Routing Algorithms</b> | <p>Static/ Dynamic, Direct/ Indirect, Shortest Path Routing, Flooding, Distance Vector Routing , Link State Routing, Hierarchical Routing, BroadcastRouting, Multicast Routing, Congestion Control Algorithms: General Principal of Congestion Control, congestion prevention polices, Load shedding, Jitter Control, IP Addressing: IP-Protocol, IP-Address Classes (A, B, C, D,E), Broadcast address, Multicast address, Network Mask, Subnetting, Internet Control Protocol-ICMP, IGMP, Mobile-IP, IPv6</p> | 10 | CO2 | Lecture with PPTs<br>Video        | Understand & Evaluate | Assignments<br><br>End Term Internals: Short Answers    |
| <b>Transport and Application Support Protocols</b>         | <p>Transport service, Service Primitives, Internet, and Transport Protocols: TCP/UDP, Remote Procedure Calls, RTP Session Layer: Token Concept Presentation Layer: Data Encryption and Data Security, Message Authentication</p> <p>Application Layer: Domain Name Service, Telnet, FTP, SMTP,SNMP, MIME,POP,IMAP, WWW,HTTP</p>                                                                                                                                                                                | 8  | CO2 | Lecture with PPTs<br>Quiz         | Analyse               | Classroom test<br><br>End Term Internals: Short Answers |
| <b>Advance Networks and Internet</b>                       | <p>Concept of 5G Networks, Introduction of 802.16,802.20, Bluetooth, Infrared, MANET, Sensor Networks. Technical Issues of Advanced Networks.</p> <p>Mobile Ad-hoc Networks: Introductory concepts, Destination-Sequenced Distance Vector protocol, Ad-hoc On-Demand Distance Vector</p>                                                                                                                                                                                                                       | 10 | CO3 | Lecture with PPTs<br>Case Studies | Understand & Apply    | Quiz<br><br>End Term Internals: Short Answers           |

|  |                                                                                                                                                                                                                                                                                                     |  |  |  |  |  |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  | Protocol Wireless Sensor Networks:<br>Sensor networks overview:<br>Introduction, applications, design<br>issues, requirements. Internet Basics:<br>Concept and Characteristics of<br>Internet, Intranet, Extranet. Structure<br>of Internet, Application of Internet and<br>Concept of Domain name. |  |  |  |  |  |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|

**Reference Books:**

| Sr.No. | Name of the Author               | Title of the Book                 | Year                    | Publisher Company      |
|--------|----------------------------------|-----------------------------------|-------------------------|------------------------|
| 1      | A.S. Tanenbaum                   | Computer Networks                 | 6 <sup>th</sup> Edition | Prentice-Hall of India |
| 2      | W.Behrouz Forouzan and S.C.Fegan | Data Communication and Networking | 5 <sup>th</sup> Edition | McGraw Hill            |
| 3      | Uyless D. Black                  | Computer Networks                 | 8 <sup>th</sup> Edition | Prentice Hall          |

**Online Resources:**

| Online Resources No. | Website address                                                                                                                                                         |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.tutorialspoint.com/computer_fundamentals/computer_networking.htm">https://www.tutorialspoint.com/computer_fundamentals/computer_networking.htm</a> |
| 2                    | <a href="https://www.javatpoint.com/computer-network-tutorial">https://www.javatpoint.com/computer-network-tutorial</a>                                                 |
| 3                    | <a href="https://www.youtube.com/watch?v=4D55Cmj2t-A">https://www.youtube.com/watch?v=4D55Cmj2t-A</a>                                                                   |
| 4                    | <a href="https://www.youtube.com/watch?v=ET2W8DyA7zI">https://www.youtube.com/watch?v=ET2W8DyA7zI</a>                                                                   |

**MOOCs:**

| Resources No. | Website address                                        |
|---------------|--------------------------------------------------------|
| 1             | NPTEL/ Swayam                                          |
| 2             | <a href="http://www.edx.com">www.edx.com</a>           |
| 3             | <a href="http://www.coursera.com">www.coursera.com</a> |

| Programme:BCA CBCS –Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                                                                                                                                                   |                 |                  |                      |                 |                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------|----------------------|-----------------|-----------------------|
| Semester                                                                                                                                                                                                                                                                                                                                                                         | Course Code     | Course Title     |                      |                 |                       |
| II                                                                                                                                                                                                                                                                                                                                                                               | 402             | Advanced Java    |                      |                 |                       |
|                                                                                                                                                                                                                                                                                                                                                                                  | Prepared by     | Dr.Suvarna Patil |                      |                 |                       |
| Type                                                                                                                                                                                                                                                                                                                                                                             | Credits         | Evaluation       | Marks                |                 |                       |
| DSC                                                                                                                                                                                                                                                                                                                                                                              | 3               | UE:IE            | 60:40                |                 |                       |
| Course Objectives:                                                                                                                                                                                                                                                                                                                                                               |                 |                  |                      |                 |                       |
| <b>Course Objectives</b> <ul style="list-style-type: none"><li>To learn implementation of Thread</li><li>To understand collection classes and interfaces.</li><li>To understand working socket and using it for simple communication</li><li>To acquire knowledge about handling databases using Java.</li><li>To study web components for developing web applications</li></ul> |                 |                  |                      |                 |                       |
| Course Outcomes :                                                                                                                                                                                                                                                                                                                                                                |                 |                  |                      |                 |                       |
| CO1: Understand the concept of Concurrent Programming, Network programming, JDBC, Servlet and JSP<br>CO2: Apply the concept to write simple socket programs, server side validation<br>CO3: Create and deploy a web application using Servlet and Java Server Pages<br>CO4: Demonstrate the data retrieval from Database using JDBC                                              |                 |                  |                      |                 |                       |
| Content                                                                                                                                                                                                                                                                                                                                                                          | Sessions (Hrs ) | COs Number       | Teaching Methodology | Cognition Level | Evaluation Tools      |
| <b>Concurrent Programming</b><br>Concept of threads, lifecycle of threads, creating threads, Thread class, Runnable interface, Thread synchronization, inter thread communication – wait(), notify(), notifyAll() methods                                                                                                                                                        | 8               | CO 1             | Lecture with Ppts    | Understand      | Quiz<br>Short Answers |
| <b>Java Collections and Utility Classes</b><br><br>Introductions to generics: generic types and methods<br><br>Collection Basics- A Collection Hierarchy, Using ArrayList and Vector, LinkedList, making use of Iterator to access collection elements                                                                                                                           | 8               | CO 2             | Lecture with Ppts    | Understand      | Quiz<br>Short Answers |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |     |                    |            |                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|--------------------|------------|---------------------------|
| <b>Java Network Programming</b><br><br>The java.net package, Connection oriented transmission – Stream, Socket Class, Creating a Socket to a remote host on a port, (creating TCP client and server), Simple Socket Program Example                                                                                                                                                                                                                                                                                 | 10 | CO2 | Lecture with PPTs  | Understand | Quiz<br><br>Short Answers |
| <b>Java Database Connectivity</b><br><br>The role of JDBC, jdbcconfiguration, Types of drivers, Connectivity with database, JDBC Statements –Statement, PreparedStatement, CallableStatement, Scrollable and updatable result sets, Metadata – DatabaseMetadata, ResultSetMetadata                                                                                                                                                                                                                                  | 10 | CO3 | Lectures with PPTs | Understand | Quiz<br><br>Short Answers |
| <b>Java Servlet</b><br><br>Introduction to Servlets and Hierarchy of Servlets , Life cycle of a servlet, Tomcat configuration, Handling get and post request (HTTP), Handling a data from HTML to a servlet, Session tracking – Cookies and Http Session<br><br><b>Java Server Pages</b><br><br>Simple JSP program, Life cycle of a JSP, Implicit Objects, Scripting elements – Declarations, Expressions, Scriptlets, Comments JSP Directives<br><br>Page Directive, include directive, Mixing Scriptlets and HTML | 9  | CO4 | Lecture            | Create     | Quiz<br><br>Short Answers |

### Reference Books

| Sr. No. | Name of the Author                | Title of the Book                  | Publisher Company      |
|---------|-----------------------------------|------------------------------------|------------------------|
| 1       | Cay S. Horstmann                  | Core Java Volume I - Fundamentals  | PHI                    |
| 2       | Herbert Schildt                   | The Complete Reference             | McGraw-Hill Education, |
| 3       | Cay S. Horstmann                  | Core Java Volume II – Fundamentals | Prentice Hall          |
| 4       | Steven Holzner                    | Java 2 Programming                 | DreamTech Press        |
| 5       | Cay S. Horstmann and Gary Cornell | Core Java-Volume-2                 | Sun Core Series        |

### Online Resources

| Online Resources No. | Web site address                                                              |
|----------------------|-------------------------------------------------------------------------------|
| 1                    | <a href="https://www.tutorialspoint.com/">https://www.tutorialspoint.com/</a> |
| 2                    | <a href="https://www.javatpoint.com/">https://www.javatpoint.com/</a>         |
| 3                    | <a href="https://www.w3schools.in/">https://www.w3schools.in/</a>             |

### MOOCs:

| Resources No. | Web site address |
|---------------|------------------|
| 1             | NPTEL            |
| 2             | Swayam           |
| 3             | edx.com          |
| 4             | coursera.com     |

| Programme: BCA CBCS– Revised Syllabus w.e.f.-Year 2022 –2023                                                                                                                                                                                                                                                                                 |             |                                       |       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                     | Course Code | CourseTitle                           |       |
| IV                                                                                                                                                                                                                                                                                                                                           | 403         | Advanced HTML with JavaScript and CSS |       |
|                                                                                                                                                                                                                                                                                                                                              | Prepared by | Dr. Swati Desai                       |       |
| Type of Course                                                                                                                                                                                                                                                                                                                               | Credits     | Evaluation                            | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                          | 3           | UE(60)+IE(40)                         | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                    |             |                                       |       |
| Objectives :                                                                                                                                                                                                                                                                                                                                 |             |                                       |       |
| <ul style="list-style-type: none"> <li>To learn Web Supporting Technologies and develop website</li> </ul>                                                                                                                                                                                                                                   |             |                                       |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                      |             |                                       |       |
| <b>CO1:</b> To remember basic concepts of Web Supporting Technologies.<br><b>CO2:</b> To understand syntaxes of HTML, HTML5, CSS and JavaScript<br><b>CO3:</b> To design web pages by applying HTML, HTML5, CSS and JavaScript.<br><b>CO4:</b> To analyze and solve real life problem using web supporting technology given in the syllabus. |             |                                       |       |

| Unit | Content                                                                                                                                                                                                                                                                                                                                                              | Sessions (Hrs) | COs Number   | Teaching Methodology   | Cognition Level | Evaluation Tools                                                       |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|------------------------|-----------------|------------------------------------------------------------------------|
| 1    | <b>Overview of HTML</b> ,<br>concept of Tag, types of HTML tags, structure of HTML program, Text Formatting Through HTML: Emphasizing Material in a Web Page, Using Image tag, attributes of Image tag, Lists: Using unordered, ordered, definition lists,<br><br>Handling Tables: To define header rows & data rows, use of caption tag, changing height & width of | 10             | CO1,<br>CO2, | Explanation, Demo, PPT | understand      | Q-A in class, Quiz, theory assignment, Lab assignments, Mid Term Exam, |

|   |                                                                                                                                                                                                                                                                                                                                                               |    |                     |                        |          |                                                                                    |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---------------------|------------------------|----------|------------------------------------------------------------------------------------|
|   | table, BGcolor, Handling Tables: cell padding, cell spacing, colspan, row span, handling table data, images in table, Frames: Introduction To frames, using frames & framesets, named frames, Concept of hyperlink, types of hyperlinks, linking to the beginning of document, linking to a particular location in a document, image as hyperlinks            |    |                     |                        |          |                                                                                    |
| 2 | <b>HTML5</b><br>Introduction to HTML5, Features of HTML5, Elements of HTML5, HTML Media and Graphics                                                                                                                                                                                                                                                          | 7  | CO1, CO2            | Explanation Demo, PPT, | Remember | Q-A in class, Quiz, theory assignment, Lab assignments                             |
| 3 | <b>Cascading Style Sheets:</b><br><br>Introducing CSS, Types of style sheets: inline, embedded and external style sheets, working with CSS properties: text properties, color and background properties, border and shading, box and block properties, positioning with CSS, various types of CSS selectors, Using class and span tag, External style sheets, | 7  | CO1, CO2, CO3       | Explanation, Demo, PPT | design   | Q-A in class, Quiz, theory assignment, Lab assignments, Case based example solving |
| 4 | <b>Introduction to JavaScript (Client-Side Scripting):</b><br><br>Introduction to scripting, overview of Java Script, advantages, client-side java Script, capturing user input, writing JavaScript into HTML, Advantages and limitations of JavaScript,<br><br><b>JavaScript Basics:</b> Data types, literals, variables and                                 | 12 | CO1, CO2, CO3, CO4, | Explanation, Demo, PPT | analyse  | Mini projects, team work,                                                          |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |                    |                        |        |                                                                                 |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------------------|------------------------|--------|---------------------------------------------------------------------------------|
|   | <p>operators, Java Script arrays, dense array, operators, expressions,</p> <p><b>JavaScript Programming Constructs:</b> Assignment, data declaration, if, switch, while, for, do while, label, break, continue, function call, return, with, delete, method of invocation</p> <p><b>Dialog boxes</b> -Alert dialog box, prompt dialog box, confirm dialog box, window objects</p> <p><b>JavaScript Functions-</b> Types of functions in Java Script- Built in functions, User defined functions, function declaration, passing parameters, variable scope, return values, recursive functions</p> <p>Arrays- Introduction to arrays, arrays with methods</p> |   |                    |                        |        |                                                                                 |
| 5 | <p><b>Forms:</b></p> <p>Interactive web pages concepts, difference between static &amp; dynamic web pages, Concept of form, how form works, Different elements - text, password, button, submit, reset, checkbox, Radio, Text Area, select &amp; option, properties of form elements, form object's Method, Otherbuilt-in Object: String object, math object, date object, Regular Expressions, Form validation</p> <p><b>JavaScript Events:</b></p>                                                                                                                                                                                                         | 9 | CO1, CO2, CO3, CO4 | Explanation, Demo, PPT | Create | <p>Case Presentation</p> <p>Activity</p> <p>End Term: Theory</p> <p>Applied</p> |

|  |                                                                                                                                                                                                                  |  |  |  |  |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  | What is an Event? Onclick Event Type, onsubmit Event Type, onmouseover and onmouseout, onchange, onload, onkeydown, working with DOM, Concept of Cookies and sessions, when and how to use cookies and sessions, |  |  |  |  |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|

### Reference Books

| Sr.No. | NameoftheAuthor | TitleoftheBook        | Year Edition                 | Publisher Company |
|--------|-----------------|-----------------------|------------------------------|-------------------|
| 1      | Dromey          | How to solve computer | 2015,3 <sup>rd</sup> edition | PHI Publication   |
| 2      | P. K. Sinha     | Computer Fundamentals | 12 <sup>th</sup> edition     | PBP Publication   |
| 3      | V. Rajaraman    | Computer Fundamentals | 6TH EDN. 2014                | PHI Publication   |

### Online Resources

| OnlineResourcesNo. | Websiteaddress                                         |
|--------------------|--------------------------------------------------------|
| 1                  | <a href="http://www.edx.com">www. edx.com</a>          |
| 2                  | <a href="http://www.coursera.com">www.coursera.com</a> |

### MOOCs:

| ResourcesNo. | Websiteaddress |
|--------------|----------------|
| 1            | Alisons        |
| 2            | Swayam         |

| Programme:BCA CBCS– RevisedSyllabusw.e.f.-Year2022 –2023                                                                                                                                                                                                                                                                                                                                                                                                                                |             |                         |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CourseCode  | Course Title            |       |
| IV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 404         | Optimization Techniques |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Prepared by | Dr.A.B.Nadaf            |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Credits     | Evaluation              | Marks |
| MDC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3           | IE 40 + UA(60)          | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                         |       |
| <ul style="list-style-type: none"> <li>To make students to get familiar with basic concepts of Optimization Techniques</li> <li>To impart knowledge of the Linear Programming,Transportation model &amp; Assignment model</li> <li>To apply CPM and PERT techniques in Project Management.</li> </ul>                                                                                                                                                                                   |             |                         |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                         |       |
| <p>After completing the course, the students shall be able to</p> <p><b>CO1:</b> Understand the basic concepts of Optimization Techniques.</p> <p><b>CO2:</b> Design the optimal problem solving techniques using Linear Programming Problem.</p> <p><b>CO3:</b> Understand the concept of transportation and Assignment problem.</p> <p><b>CO4:</b> Design Solution by using Network Theory .</p> <p><b>CO5:</b> Design the Decision Table and Decision Tree for the given problem</p> |             |                         |       |

| Unit | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Hrs | COs No | Teaching Methodology                          | Cognition Level | Evaluation Tools |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------|-----------------------------------------------|-----------------|------------------|
| 1    | Origin of Optimization Techniques,History ,Methodology, different phases, Characteristics, Scope ,Applications of Optimization Techniques, Limitations of Optimization Techniques Introduction and requirement of LP, Assumption and Formulation of LP, General Statement of LP, Solution of LP by using Graphical Method(Maximization & Minimization), Special cases in Graphical Method-<br>i)Alternative solution<br>ii)Unbounded Solution<br>iii)Infeasible solution | 08  | CO1    | Power point Presentations, Classroom Sessions | Understand      | Assignment Quiz  |

|   |                                                                                                                                                                                                                                                                                                                                                                                                      |    |          |                                               |                    |                                                                    |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----------|-----------------------------------------------|--------------------|--------------------------------------------------------------------|
| 2 | Linear Programming formulation of Transportation Problem, General Procedure to solve Transportation Problem, Methods for finding Initial Feasible Solution-i)North -West Corner Method ii)Least CostMethod iii)Vogel's Aproximation Method, Final Transportation cost using MODI Method. Special Cases :i)Unbalanced problem ii)Mutiple Optimum Solution iii)Prohibited Routes iv)Case of Degeneracy | 12 | CO2, CO3 | Power point Presentations, Classroom Sessions | Remember           | Case Study Discussion ,<br><br>Class Test'<br><br>Class Assignment |
| 3 | Introduction, Hungerain Method to solve Assignment problem, Special cases- i)Unbalanced Problem ii)Alternate Solution iii)Prohibited Assignment iv)Maximization Problems                                                                                                                                                                                                                             | 8  | CO3      | Power point Presentations, Classroom Sessions | Understand , apply | Case Study, Question and Answer,                                   |
| 4 | Terms used in Network Analysis, Rules for Network construction, Drawing network diagrams, Backward Pass Calculation, Forward Pass Calculation, Critical Pass Method, Time estimates for critical path, PERT, Types of Float (Theoretical point of view only) , Probability of completion of project                                                                                                  | 8  | CO4      | Power point Presentations, Classroom Sessions | create             | Case Study,                                                        |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |     |                                               |        |            |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|-----------------------------------------------|--------|------------|
| 5 | Elements of Decision making problem, Decision making under risk-i)Expected Monetary value criterion ii)Expected value with perfect information iii)Expected Value of perfect information (E.V.P.I.)iv)Expected Opportunity Loss Decision Making under uncertainty-i)Maximax (gain) or Minimin (loss) criterion ii)Maximin criterion iii)Hurwicz Alpha criterion iv)Laplace criterion v)Minimax Regret criterion Decision Tree -simple Examples | 9 | CO5 | Power point Presentations, Classroom Sessions | create | assignment |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|-----------------------------------------------|--------|------------|

### Reference Books

| Sr.No. | Name of the Author                   | Title of the Book                                        | Year Edition | Publisher Company                   |
|--------|--------------------------------------|----------------------------------------------------------|--------------|-------------------------------------|
| 1      | J.K. Sharma                          | Operations Research                                      | 2016         | Laxmi Publications                  |
| 2      | Kanti Swaroop, P.K. Gupta, Man Mohan | Operations Research                                      | 2019         | Paperback                           |
| 3      | R. Panneerselvam                     | Operations Research : Introduction to Management Science | 2006         | 2019 Prentice Hall of India Pvt Ltd |
| 4.     | S. Kalavathy                         | Operations Research                                      | 2006         | Vikas Publishing House              |

### Online Resources:

| OnlineResourcesNo. | Website address                                                                                                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                  | <a href="https://www.youtube.com/watch?v=knZrhVkZ71Q&amp;list=PLU6SqYcYsfLyEPjMPHT_1ZhTRrnXA55R">https://www.youtube.com/watch?v=knZrhVkZ71Q&amp;list=PLU6SqYcYsfLyEPjMPHT_1ZhTRrnXA55R</a>                           |
| 2                  | <a href="https://www.youtube.com/watch?v=9vJx6tZgVQs&amp;list=PLU6SqYcYsfLyEPjMPHT_1ZhTRrnXA55R&amp;index=14">https://www.youtube.com/watch?v=9vJx6tZgVQs&amp;list=PLU6SqYcYsfLyEPjMPHT_1ZhTRrnXA55R&amp;index=14</a> |
| 3                  | <a href="https://www.youtube.com/watch?v=ydvnVw80I_8">https://www.youtube.com/watch?v=ydvnVw80I_8</a>                                                                                                                 |

|   |                                                                                                                                                                                                                       |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | <a href="https://www.youtube.com/watch?v=oBPIVV6AiPQ&amp;list=PLEjRWorvdxL6LnWXJxnFB_9DXHhUxJ3dk&amp;index=2">https://www.youtube.com/watch?v=oBPIVV6AiPQ&amp;list=PLEjRWorvdxL6LnWXJxnFB_9DXHhUxJ3dk&amp;index=2</a> |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### MOOCs:

| ResourcesNo. | Website address                                                                                                                                                                                 |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1            | <a href="https://www.youtube.com/watch?v=BDBhpxRzImI&amp;list=PLWoXNEI-KK1mCv_EL4OdF_-6FXryaZ11N">https://www.youtube.com/watch?v=BDBhpxRzImI&amp;list=PLWoXNEI-KK1mCv_EL4OdF_-6FXryaZ11N</a>   |
| 2            | <a href="https://www.youtube.com/watch?v=66aKgySf9vo&amp;list=PLLy_2iUCG87Bq8RGM TdeFZiB-87V4i9p1">https://www.youtube.com/watch?v=66aKgySf9vo&amp;list=PLLy_2iUCG87Bq8RGM TdeFZiB-87V4i9p1</a> |
| 3            | <a href="https://www.youtube.com/watch?v=a2QgdDk4Xjw&amp;list=PLjc8ejfjpgTf0LaDEHgL B3gCHZYcNtsoX">https://www.youtube.com/watch?v=a2QgdDk4Xjw&amp;list=PLjc8ejfjpgTf0LaDEHgL B3gCHZYcNtsoX</a> |

| Programme: BCA CBCS– Revised Syllabus w.e.f.-Year2022 –2023                                                                                                                                                                                                                                                                                                                                          |             |                      |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                             | Course Code | Course Title         |       |
| IV                                                                                                                                                                                                                                                                                                                                                                                                   | 405         | Lab on Advanced JAVA |       |
|                                                                                                                                                                                                                                                                                                                                                                                                      | Prepared by | Dr.Rahul Jadhav      |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                 | Credits     | Evaluation           | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                  | 2           | IA(40) + UA(60)      | 100   |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                              |             |                      |       |
| After completing the course the students shall be able to:<br><b>CO1:</b> Write Java code by making use of thread<br><b>CO2:</b> Construct a web application using Servlet and Java Server Pages<br><b>CO3:</b> Implement server-side validations with session<br><b>CO4:</b> Retrieve data effectively from database using JDBC<br><b>CO5:</b> Develop and deploy web-based enterprise applications |             |                      |       |

| Unit | Content                                                                                                                                                                                                                                                                                                                       | Sessions (Hrs ) | COs Number | Teaching Methodology | Cognition Level | Evaluation Tools |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|----------------------|-----------------|------------------|
| 1    | Write a program to demonstrate Multi-threading using Thread Class.<br>Write java program to implement Runnable interface<br>Write java program for demonstrating concept of Thread synchronization. Write java code for implementing the following Inter-thread communication methods:<br>using wait(), notify(), notifyAll() | 6               | CO1        | Lab Demonstration    | Applying        | Output / answer  |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |            |                   |          |                 |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------|-------------------|----------|-----------------|
| 2 | Develop java programs to implement Simple generic class and methods<br>Write java programs to demonstrate concept of ArrayList, Vector and LinkedList.<br>Write java code to implement Iterator to access collection elements.<br>Write java programs to demonstrate concept of HashSet, LinkedHashSet and TreeSet.                                                                                                                          | 6 | CO1        | Lab Demonstration | Applying | Output / answer |
| 3 | Implement jdbc connectivity to insert records and delete records into a table. Implement jdbc connectivity to demonstrate PreparedStatement.<br>Write java code to demonstrate stored procedures with Callable Statement.<br>Write java code to implement concept of Scrollable and updatable result sets. Write java code to Making use of Database Metadata and ResultSetMetadata                                                          | 6 | CO4        | Lab Demonstration | Applying | Output / answer |
| 4 | Write a servlet program to create a simple servlet and test it. Write a servlet program to read the client request parameters. Implement a Servlet to generate Multiplication Table for a Number Entered in Html Page.                                                                                                                                                                                                                       | 6 | CO2<br>CO3 | Lab Demonstration | Applying | Output / answer |
| 5 | Develop an application/s to demonstrate all the core tags available in JSP (Declaration, Expression, Directive and Scriptlet Tag) Develop a JSP Application to accept Details from user and store it into the database table. Develop a JSP Application to Authenticate User login as per registration details. If login success the forward user to Index Page otherwise show login failure Message. Write a web based student registration | 6 | CO4<br>CO5 | Lab Demonstration | Applying | Output / answer |

|  |                                                                                                                                                                                                                  |  |  |  |  |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  | application where students can register online with their enrolment number. The registered students should be able to log on to the site after getting registered. You are required to use JSP, Servlet and JDBC |  |  |  |  |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|

#### Reference Books:

| Sr.No. | Name of the Author                | Title of the Book           | Year                    | Publisher Company |
|--------|-----------------------------------|-----------------------------|-------------------------|-------------------|
| 1      | Herbert Schildt                   | The Complete Reference JAVA | 7 <sup>th</sup> Edition | McGraw-Hill       |
| 2      | Cay S. Horstmann and Gary Cornell | Core Java Volume-I          | 8 <sup>th</sup> Edition | Sun Core Series   |
| 3      | Bruce Eckel                       | Thinking In Java            | 4 <sup>th</sup> Edition | Printice Hall     |

#### Online Resources:

| Online Resources No. | Website address                                                                                       |
|----------------------|-------------------------------------------------------------------------------------------------------|
| 1                    | 1 <a href="https://docs.oracle.com/javase/tutorial/">https://docs.oracle.com/javase/tutorial/</a>     |
| 2                    | 2 <a href="https://www.javatpoint.com/java-tutorial">https://www.javatpoint.com/java-tutorial</a>     |
| 3                    | 3 <a href="https://www.programiz.com/java-programming">https://www.programiz.com/java-programming</a> |

#### MOOCs:

| Resources No. | Website address                                        |
|---------------|--------------------------------------------------------|
| 1             | NPTEL/ Swayam                                          |
| 2             | <a href="http://www.edx.com">www.edx.com</a>           |
| 3             | <a href="http://www.coursera.com">www.coursera.com</a> |

| Programme: BCA CBCS– Revised Syllabus w.e.f.-Year 2022 –2023                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |                                                |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Course Code | Course Title                                   |       |
| IV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 406         | Lab on HTML, JavaScript, and CSS & Project - I |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Prepared by | Dr. Ayesha Mujawar                             |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Credits     | Evaluation                                     | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2           | IE(40) + UA(60)                                | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |                                                |       |
| <p>To make students to:</p> <ul style="list-style-type: none"> <li>To teach the basic internet concepts and train them to develop internet applications.</li> <li>An overview of the HTML5 specification</li> <li>Practical knowledge to implement new HTML5 elements and attributes.</li> </ul> <p>Overview of JavaScript</p>                                                                                                                                                                                     |             |                                                |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                                |       |
| <p>After completing the course<br/>After completing the course the students shall be able to</p> <p><b>CO1:</b> To design simple web pages using HTML.<br/> <b>CO2:</b> To design web pages using text formatting, list, image tags in HTML.<br/> <b>CO3:</b> To apply various CSS styles to design pages.<br/> <b>CO4:</b> To apply various programming constructs and event handling mechanism using JavaScript for designing web pages.<br/> <b>CO5:</b> To develop minor project individually or in group.</p> |             |                                                |       |

| Unit                      | Content                                                                                                                                                                                                                                                           | Sessions | CO Number | Teaching Methodology | Cognitive Level | Evaluation Tools |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|----------------------|-----------------|------------------|
| <b>Basics of Internet</b> | 1. Design A webpage which has student's biodata with proper formatting and having student name as title.<br>2. Design a website for PNG jewellers, having images of different types of jewellerys which are linked with the pages giving details about the items. | 5        | CO1       | Live Demo            | Create          | Quiz             |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |     |           |        |      |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|-----------|--------|------|
| <b>Introduction to HTML</b>   | <ol style="list-style-type: none"> <li>Design a website for a class which shows student's listlinked with their biodata pages.</li> <li>Design a web page to display the following output. <ul style="list-style-type: none"> <li>List of subjects <ul style="list-style-type: none"> <li>Semester III <ul style="list-style-type: none"> <li>C++</li> <li>Dot.Net</li> </ul> </li> <li>Semester III <ul style="list-style-type: none"> <li>Java</li> <li>Industrial Projects</li> </ul> </li> </ul> </li> <li>Internet Programming <ul style="list-style-type: none"> <li>HTML</li> <li>VBScript</li> <li>Java Script</li> </ul> </li> </ul> </li> <li>Design a website for the college which lists all the faculties (ordered lists), courses (definition lists) every course explains details (fees, duration, intake capacity) as unordered list.</li> <li>Create a form having textboxes, radio buttons and check boxes and reset button. On clicking the reset button, the entire form should be reset.</li> </ol> | 5 | CO2 | Live Demo | Create | Quiz |
| <b>Cascading Style Sheets</b> | <ol style="list-style-type: none"> <li>Design a Style sheet to give following effects.<br/>The first latter of the paragraph should have 150% font size.<br/>The first line of the paragraph should have purple as background color and white as the fore color.</li> <li>Design a website for a college showing features of the university, college and list of different courses running in the institute. Course names have links with the pages having</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 6 | CO3 | Live Demo | Create | Quiz |

|                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |            |           |        |      |
|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------|-----------|--------|------|
|                                                                                                | <p>details of the courses having similar design using stylesheets.</p> <p>3. Design a CSS (inline) that displays the regular text at the center with green as background color and white as fore color and should be bold, using class.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |            |           |        |      |
| <b>Introduction to JavaScript (Client-Side Scripting)</b><br><br><b>Functions &amp; Arrays</b> | <p>1. Design a form using HTML that accepts information about your qualification, extracurricular activities, skill sets, achievements, hobbies, and expectation for a particular job.</p> <p>2. Write a JavaScript code which contains “show” button. When user clicks on show button, first 10 terms of Fibonacci series will be displayed in text box on another HTML page. This page contains button “back”. With this button user can come back to original page.</p> <p>3. Design a website which accepts a number from user and performs the selected operation (even/odd, prime/not prime, positive/negative).</p> <p>4. Design a webpage which provides calculator facilities.</p> <p>5. Write JavaScript to display table of numbers 2-10 (use form and form elements)</p> | 7 | CO4        | Live Demo | Create | Quiz |
| <b>Forms And Object Event Handling</b>                                                         | <p>1. Design a webpage which accepts users information with validations (name, std code (should not exceed 4 digits), landline number (no. of digits should be between</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 7 | CO4<br>CO5 | Live Demo | Create | Quiz |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |  |  |  |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  | <p>5 to 7), mobile number(exactly 10 digits), email (should have @ and.)).</p> <p>2. Develop a HTML form which accepts mathematical expression in one textbox and display its result in another textbox after clicking on a button showing mathematical operations.</p> <p>3. Create a HTML form that has a number of textboxes. When the form runs in the browser fill the textboxes with data. Write the JavaScript code which verifies that all textboxes have been filled. If the textbox has been left empty, popup an Alert indicating which textbox has been left empty. When alert's OK button is clicked on, set focus to that specific textbox.</p> <p>4. Design webpage which accepts no of lines and prints it in the form of triangular shaped pyramid.</p> <p>5. Accept data of a student wants to appear for entrance (name, marks at matriculation, higher secondary and graduation). Ask student to select the course he wants to take admission. If the student scores above 55 at matriculation, above 60 at higher secondary and graduation then he is eligible for any course. If he</p> |  |  |  |  |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  |  |  |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  | <p>has science degree or maths at 11th and 12th, then only he is eligible for MCA. Design the form accordingly. Give the according message.</p> <p>6. Create a form having textboxes, radio button and check boxes and reset button. On clicking the reset button, the entire form should be reset.</p> <p>7. Accept login name and password from user and display biodata of the corresponding user.</p> <p>8. Design a page for a user to create his login by accepting desired login name, password and confirm the password.</p> |  |  |  |  |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|

#### Reference Books:

| Sr.No. | Name of the Author                | Title of the Book                                                                      | Year | Publisher Company                  |
|--------|-----------------------------------|----------------------------------------------------------------------------------------|------|------------------------------------|
| 1      | Ivan Bayross                      | Web Enabled Commercial Application Development Using HTML, DHTML, JavaScript, Perl CGI | 2006 | BPB Publications                   |
| 2      | Thomas Powell                     | Web Design The complete Reference                                                      | 2004 | Tata McGrawHill                    |
| 3      | Thomas Powell and Fritz Schneider | JavaScript 2.0: The Complete Reference, Second Edition                                 | 2004 | McGraw-Hill Education; 2nd edition |

#### Online Resources:

| Online Resources No. | Website address                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.w3schools.com/html">https://www.w3schools.com/html</a>                     |
| 2                    | <a href="https://www.javatpoint.com/html-tutorial">https://www.javatpoint.com/html-tutorial</a> |
| 3                    | <a href="https://www.geeksforgeeks.org/html/">https://www.geeksforgeeks.org/html/</a>           |

**MOOCs:**

| Resources No. | Website address                                        |
|---------------|--------------------------------------------------------|
| 1             | NPTEL / Swayam                                         |
| 2             | <a href="http://www.edx.com">www.edx.com</a>           |
| 3             | <a href="http://www.coursera.com">www.coursera.com</a> |

| Programme: BCA CBCS – Revised Syllabus w.e.f – 2022-2023                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                       |                 |            |                                                       |                               |                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|-------------------------------------------------------|-------------------------------|------------------------------------------------------------------------|
| Semester                                                                                                                                                                                                                                                                                                                                                       | Course Code                                                                                                                                                                                                                                                                                                                                           | Course Title    |            |                                                       |                               |                                                                        |
| IV                                                                                                                                                                                                                                                                                                                                                             | 407                                                                                                                                                                                                                                                                                                                                                   | Cyber security  |            |                                                       |                               |                                                                        |
|                                                                                                                                                                                                                                                                                                                                                                | Prepared by                                                                                                                                                                                                                                                                                                                                           | Dr.Shabnam Mane |            |                                                       |                               |                                                                        |
| Type                                                                                                                                                                                                                                                                                                                                                           | Credits                                                                                                                                                                                                                                                                                                                                               | Evaluation      | Marks      |                                                       |                               |                                                                        |
| SEC                                                                                                                                                                                                                                                                                                                                                            | 2                                                                                                                                                                                                                                                                                                                                                     | IA              | 50         |                                                       |                               |                                                                        |
| Course Objectives: (CO)                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                       |                 |            |                                                       |                               |                                                                        |
| 1. To Understand the cyber security threat landscape.<br>2. To Develop a deeper understanding and familiarity with various types of cyberattacks, cyber crimes, vulnerabilities and remedies thereto.<br>3. To learn and apply existing legal framework and laws on cyber security                                                                             |                                                                                                                                                                                                                                                                                                                                                       |                 |            |                                                       |                               |                                                                        |
| Course Outcomes:                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                       |                 |            |                                                       |                               |                                                                        |
| The students will be in a position<br>CO1: Evaluate and communicate the human role in security systems with an emphasis on ethics, social engineering vulnerabilities and training.<br>CO2:Increase awareness about cyber-attack vectors and safety against cyber-frauds.<br>CO3:Take measures for self-cyber-protection as well as societal cyber-protection. |                                                                                                                                                                                                                                                                                                                                                       |                 |            |                                                       |                               |                                                                        |
| Unit                                                                                                                                                                                                                                                                                                                                                           | Contents                                                                                                                                                                                                                                                                                                                                              | Sessions (Hrs.) | COs Number | Teaching Methodology                                  | Cognition Level               | Evaluation Tools                                                       |
| 1. Introduction to Cyber security                                                                                                                                                                                                                                                                                                                              | Defining Cyberspace and Overview of Computer and Web-technology, Architecture of cyberspace, Communication and web technology, Internet, World wide web, Advent of internet, Internet infrastructure for data transfer and governance, Internet society, Regulation of cyberspace, Concept of cyber security, Issues and challenges of cyber security | 12              | CO 1       | Lecture with practical questions based on Cases Study | Understand, Analyze           | End Term: Short case and situation based questions / Applied Questions |
| 2. Cyber crime                                                                                                                                                                                                                                                                                                                                                 | Classification of cyber crimes, Common cyber crimes- cyber crime targeting computers and mobiles, cyber crime against women and                                                                                                                                                                                                                       | 08              | CO 2       | Lecture with practical questions                      | Understand, Analyze, Evaluate | End Term: Short case and situation                                     |

|              |                                                                                                                                                                                                                |    |      |                                                       |                               |                                                                        |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------|-------------------------------------------------------|-------------------------------|------------------------------------------------------------------------|
|              | children, financial frauds, social engineering attacks, malware and ransomware attacks, zero day and zero click attacks                                                                                        |    |      | based on Cases Study                                  |                               | based questions / Applied Questions                                    |
| 3. Cyber law | Remedial and mitigation measures, Legal perspective of cyber crime, IT Act 2000 and its amendments, Cyber crime and offences, Organizations dealing with Cyber crime and Cyber security in India, Case studies | 10 | CO 3 | Lecture with practical questions based on Cases Study | Understand, Analyze, Evaluate | End Term: Short case and situation based questions / Applied Questions |

### Reference Books

| Sr. No. | Name of the Author             | Title of the Book                         | Year Addition       | Publisher Company     |
|---------|--------------------------------|-------------------------------------------|---------------------|-----------------------|
| 1       | R. C Mishra                    | Cyber Crime Impact in the New Millennium  | 2010                | Author Press. Edition |
| 2       | SumitBelapure and Nina Godbole | Computer Forensics and Legal Perspectives | First Edition, 2011 | Wiley India Pvt. Ltd  |

### Online Resources

| MOOCS | Website address  |
|-------|------------------|
| 1     | NPTEL/ Swayam    |
| 2     | www.edx.com      |
| 3     | www.coursera.com |

| Programme:BCA CBCS –Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                        |             |                              |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------|-------|
| Semester                                                                                                                                                                                                                              | Course Code | Course Title                 |       |
| IV                                                                                                                                                                                                                                    | 408         | Mathematical Aptitude        |       |
|                                                                                                                                                                                                                                       | Prepared by | Dr. Dhanashree Sahasrabuddhe |       |
| Type                                                                                                                                                                                                                                  | Credits     | Evaluation                   | Marks |
| AEC                                                                                                                                                                                                                                   | 2           | IA                           | 50    |
| <b>Course Objectives:</b>                                                                                                                                                                                                             |             |                              |       |
| <ul style="list-style-type: none"> <li>To develop mathematical and logical thinking</li> <li>To prepare base for various aptitude tests being conducted by companies</li> <li>To develop their ability to draw conclusions</li> </ul> |             |                              |       |
| <b>Course Outcomes :</b>                                                                                                                                                                                                              |             |                              |       |
| <b>CO1:</b> To Learn various reasoning techniques.<br><b>CO2:</b> To Apply reasoning techniques to solve real time problems<br><b>CO3:</b> To analyse the given problem with the view of development of an efficient solution         |             |                              |       |

| Unit | Content                                                                                                                                                                                                                                                                                                                                                                                     | Sess<br>ions<br>(Hrs<br>) | COs Number       | Teaching<br>Methodolog<br>y | Cognitio<br>nLevel      | Evaluatio<br>nTools |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|------------------|-----------------------------|-------------------------|---------------------|
| 1    | <b>Numerical Reasoning</b> <ul style="list-style-type: none"> <li>Problems on Numbers like divisibility tests, basic arithmetic operations</li> <li>LCM (Least Common Multiplier), HCF (Highest Common Factor)</li> <li>Profit and Loss</li> <li>Partnership</li> <li>Speed and Distance</li> <li>Simple and Compound Interest</li> <li>Problems on ages</li> <li>Simplification</li> </ul> | 10                        | CO 1,<br>CO2,CO3 | Lecture with<br>Quiz        | Understand<br>and Apply | Quiz                |

|   |                                                                                                                                                                                          |    |                |                   |                      |      |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----------------|-------------------|----------------------|------|
|   |                                                                                                                                                                                          |    |                |                   |                      |      |
| 2 | <b>Logical Reasoning</b> <ul style="list-style-type: none"> <li>• Series</li> <li>• Directions</li> <li>• Blood Relations</li> <li>• Seating Arrangements</li> <li>• Calendar</li> </ul> | 10 | CO 1, CO2, CO3 | Lecture with Quiz | Understand and Apply | Quiz |
| 3 | <b>Mathematical Aptitude</b> <ul style="list-style-type: none"> <li>• Permutations and combinations</li> <li>• Mensuration</li> <li>• Set Theory</li> </ul>                              | 10 | CO 1, CO2, CO3 | Lecture with Quiz | Understand and Apply | Quiz |

#### Reference Books :

| Sr. No. | Name of the Author | Title of the Book     | Year Edition | Publisher Company |
|---------|--------------------|-----------------------|--------------|-------------------|
| 1       | R.S. Aggarwal      | Quantitative Aptitude | 2016         | S.Chand           |

#### MOOCs:

| Resources No. | Web site address                                                                                                                                                                                                                                                    |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1             | <a href="https://www.mygreatlearning.com/academy/learn-for-free/courses/crash-course-on-quantitative-aptitude-and-logical-reasoning">https://www.mygreatlearning.com/academy/learn-for-free/courses/crash-course-on-quantitative-aptitude-and-logical-reasoning</a> |
| 2             | <a href="https://www.geeksforgeeks.org/quantitative-aptitude-course-free-online/">https://www.geeksforgeeks.org/quantitative-aptitude-course-free-online/</a>                                                                                                       |

| <b>Programme: BCA CBCS Revised Syllabus w.e.f.-Year 2022–2023</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                    |                     |              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------|--------------|
| <b>Semester</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Course Code</b> | <b>Course Title</b> |              |
| V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 501                | Python Programming  |              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Prepared by        | Dr.M.K.Patil        |              |
| <b>Type</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Credits</b>     | <b>Evaluation</b>   | <b>Marks</b> |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3                  | UE: IE              | 60:40        |
| <b>Course Objectives</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                     |              |
| <ul style="list-style-type: none"> <li>• A Python programming course is designed to equip students with a comprehensive understanding of the language and its application.</li> <li>• Starting with an introduction to Python's history and community, the course guides students through setting up their development environment and mastering fundamental syntax and data types.</li> <li>• Students learn control flow structures, functions, and modules, progressing to file handling, object-oriented programming (OOP) principles, and data structures.</li> <li>• The curriculum includes essential skills such as error handling, debugging, and the use of popular libraries and frameworks.</li> <li>• Emphasis is placed on best practices, code style, collaborative development using version control (e.g., Git), testing, and debugging techniques.</li> <li>• Overall, the objectives aim to empower students with a well-rounded skill set for effective Python programming and application development.</li> </ul> |                    |                     |              |
| <b>Course Outcome</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |                     |              |
| <p><b>CO1:</b> Using some motivating examples to remember and quickly builds up basic concepts such as conditionals, loops, functions, lists, strings and tuples.</p> <p><b>CO2:</b> Students will get acquainted built in data structures in python, understand features and programming constructs of python language. During this course, they will understand main control structures of procedural programming languages.</p> <p><b>CO3:</b> They will make of function to reduce problem into small modules, To familiarize with exceptions and mechanism to handle it, make use of python to read and write data into files</p> <p><b>CO4:</b> Analyzing the different problems based on CSV files</p> <p><b>CO5:</b> Ability to choose appropriate data dictionary for problem solving</p> <p><b>CO6:</b> Design and create their own programs for solving a real life problem</p>                                                                                                                                             |                    |                     |              |

| <b>Unit</b>                    | <b>Contents</b>                                                                                          | <b>Sessions (Hrs.)</b> | <b>COs Number</b> | <b>Teaching Methodology</b>            | <b>Cognition Level</b>               | <b>Evaluation Tools</b> |
|--------------------------------|----------------------------------------------------------------------------------------------------------|------------------------|-------------------|----------------------------------------|--------------------------------------|-------------------------|
| <b>Introduction to Python:</b> | History of Python, Unique features of Python, Python Identifiers, Keywords and Indentation, Comments and | 5                      | CO1, CO2, CO3     | Classroom Teaching, ICT-based teaching | Remembering, Understanding, Applying | Assignments, Quizzes    |

|                                                                                |                                                                                                                                                                                                                                                                                                                      |   |                      |                                                 |                                                    |                    |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------------|-------------------------------------------------|----------------------------------------------------|--------------------|
|                                                                                | document<br>interlude in<br>Python, Getting<br>User Input<br>Python, Data<br>Types,<br>variables,<br>Python Core<br>objects and<br>Functions<br>Number and<br>Maths                                                                                                                                                  |   |                      |                                                 |                                                    |                    |
| <b>Statements<br/>and<br/>Control<br/>Structures:</b>                          | Assignment<br>statement,<br>import<br>statement, print<br>statement, if:<br>elif: else:<br>statement,<br>for: statement.,<br>while:<br>statement.,<br>continue and<br>break<br>statements, try:<br>except<br>statement., raise<br>statement., with<br>statement, del,<br>case statement                              | 5 | CO1,<br>CO2,<br>CO3  | Classroom<br>Teaching,<br>ICT-based<br>teaching | Rememberin<br>g,<br>Understandin<br>g,<br>Applying | Lab<br>Assignments |
| <b>List,<br/>Ranges &amp;<br/>Tuples &amp;<br/>Dictionarie<br/>s in Python</b> | Introduction,<br>Lists in Python,<br>Understanding<br>Iterators,<br>Generators,<br>Comprehensions<br>and Lambda<br>Expressions,<br>Generators and<br>Yield Next and<br>Ranges,<br>Understanding<br>and using<br>Ranges,<br>Ordered Sets<br>with tuples,<br>Introduction to<br>Python<br>Dictionaries,<br>Python Sets | 4 | CO1,<br>CO2,<br>CO3  | Classroom<br>Teaching,<br>ICT-based<br>teaching | Rememberin<br>g,<br>Understandin<br>g,<br>Applying | Lab<br>Assignments |
| <b>Functions,<br/>Modules,<br/>Packages,<br/>and</b>                           | The def<br>statement<br>Returning<br>values,                                                                                                                                                                                                                                                                         | 5 | CO1,<br>CO2,<br>CO3, | Classroom<br>Teaching,<br>ICT-based<br>teaching | Rememberin<br>g,<br>Understandin<br>g,             | Lab<br>Assignments |

|                                                            |                                                                                                                                                                                                                                                           |   |                         |                                            |                                                             |                                                              |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------------------|--------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------|
| <b>Debugging Functions:</b>                                | Parameters, Arguments, Local variables, Other things to know about functions, Global variables and the global statement, Doc strings for functions, Decorators for functions, lambda Iterators and generators, Modules, Doc strings for modules, Packages |   | CO5                     |                                            | Applying, Evaluating                                        |                                                              |
| <b>Python Object Oriented</b>                              | Overview of OOP, Creating Classes and Objects, Accessing attributes Built-In Class Attributes, Destroying Objects                                                                                                                                         | 4 | CO1, CO2                | Classroom Teaching, ICT-based teaching     | Remembering, Understanding                                  | Lab Assignments                                              |
| <b>Python Exceptions Handling</b>                          | What is Exception? Handling an exception try....except...else Try-finally clause Argument to fan Exception. Python Standard Exceptions Raising and exceptions, User-Defined Exceptions                                                                    | 6 | CO1, CO2, CO3, CO5      | Classroom Teaching, ICT-based teaching     | Remembering, Understanding, Applying, Evaluating            | Lab Assignments                                              |
| <b>Input and Output in Python &amp; Built in Functions</b> | File Objects, creating a file object, reading File contents, writing data into file, reading and writing CSV files, using with clause, Using                                                                                                              | 6 | CO1, CO2, CO3, CO5, CO6 | Project-based teaching, ICT-based teaching | Remembering, Understanding, Applying, Analyzing, Evaluating | Lab Assignments, Live case study from the website Kaggle.com |

|  |                                          |  |  |  |  |  |
|--|------------------------------------------|--|--|--|--|--|
|  | Exception handling with file operations, |  |  |  |  |  |
|--|------------------------------------------|--|--|--|--|--|

**Textbook:**

1. Artificial Intelligence by Elaine Rich and Kevin Knight, Tata McGraw Hill
2. Understanding Machine Learning. Shai Shalev-Shwartz and Shai Ben-David. Cambridge University Press.
3. Artificial Neural Network, B. Yegnanarayana, PHI, 2005
- Tom Mitchell, "Machine Learning", McGraw Hill, 1997
2. E. Alpaydin, "Introduction to Machine Learning", PHI, 2005.

**Reference Book:**

1. Christopher M. Bishop. Pattern Recognition and Machine Learning (Springer)
2. Introduction to Artificial Intelligence and Expert Systems by Dan W. Patterson, Prentice Hall of India
3. Andrew Ng, Machine learning yearning, <https://www.deeplearning.ai/machine-learning-yearning/>
4. Aurolien Geron," Hands-On Machine Learning with Scikit-Learn and TensorFlow, Shroff/O'Reilly",2017

5. Andreas Muller and Sarah Guido," Introduction to Machine Learning with Python: A Guide for Data Scientists", Shroff/O'Reilly, 2016

| Programme:BCA(CBCS)– Revised Syllabus w.e.f.-Year 2022–2023                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |                              |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Course Code | Course Title                 |       |
| V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 502         | Dot Net Programming using C# |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Prepared by | Mr.Alok Shah                 |       |
| Type of Course                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Credits     | Evaluation                   | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3           | UE(60)+IE(40)                | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |                              |       |
| Objectives: <ul style="list-style-type: none"> <li>To introduce .Net framework.</li> <li>To introduce C# as OOP language.</li> <li>To understand Event driven programming in C#.</li> <li>To understand working with windows forms.</li> </ul>                                                                                                                                                                                                                                                                     |             |                              |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                              |       |
| After completing the course the students shall be able to<br><br><b>CO1:</b> Understand .NET Framework, its runtime environment and application development IDE of Visual Studio.<br><b>CO2:</b> Understand the concept of object oriented for making programs.<br><b>CO3:</b> Implement C# language constructs in the form of stand-alone console and window form applications.<br><b>CO4:</b> Understand database concepts in ADO.NET and apply the knowledge to implement distributed data-driven applications. |             |                              |       |

| Unit                                                       | Content                                                                                                                                                                                                                                                                                                                                                                                    | Sessions | COs Number | Teaching Methodology   | Cognitive Level      | Evaluation Tools                                               |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|------------------------|----------------------|----------------------------------------------------------------|
| <b>Introduction to Dot.Net Framework :</b>                 | <b>History and Overview of Dot.Net framework</b><br>Framework Components and Versions<br><b>Introduction to C # :</b><br>C# Language, C# Language elements , Data types -Reference Type and Value Type, Boxing and Unboxing , Enum and Constant, Operators Control Statements, Working with Arrays and Strings , Pass by value and by reference, outparameters, Variable length parameter. | 7        | CO 1       | Lecture with Ppts Quiz | Understand           | Quiz<br>End Term<br>Internals:Short Answers and Practical Test |
| <b>Implementation of Object Oriented Concept using C#:</b> | Object oriented concepts ,Working with Indexer and Properties ,Constructor & Destructor , Working with "static" Members,                                                                                                                                                                                                                                                                   | 7        | CO 2       | Lecture with Ppts      | Understand and Apply | End Term: Applied Questions and Practical Test                 |

|                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |      |                    |                      |                                                |
|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------|--------------------|----------------------|------------------------------------------------|
|                                                                      | Inheritance & Polymorphism, Types of Inheritance, Constructor in Inheritance, Interface Implementation, Operator and method Overloading, and overriding, - Static and Dynamic Binding and Virtual, methods, Abstract Class, sealed keyword                                                                                                                                                                                                                                                       |   |      |                    |                      |                                                |
| <b>Exception Handling:</b>                                           | What is Exception, Rules for Handling Exception, Exception classes and its important properties, Understanding & using try, catch keywords, Throwing exceptions, Importance of finally block, Writing Custom Exception Classes.<br><b>Using I/O Class: Streams</b><br><b>Class:</b> Text Stream and Binary Stream, System.IO and Base classes of Stream., Console I/O Streams, Working with File System -File, FileInfo, Directory, DirectoryInfo classes                                        | 7 | CO 2 | Lecture with PPTs  | Understand and Apply | End Term: Applied Questions and Practical Test |
| <b>Delegates &amp; Events:</b><br><b>Introduction of Delegation:</b> | Types of delegate, Anonymous Methods, What is Events?, Multicast Events, Lambda Expression.<br><b>Collections and Generics:</b> Importance of IList and IDictionary, Collection classes: ArrayList, Hashtable, stack, queue, Writing custom generic classes, Working with Generic Collection Classes.<br><b>Multithreading:</b> Multithreading Fundamentals, Thread Class, Creating and Managing Threads, Threads Priority, Thread Synchronization, Suspending, Resuming and Terminating threads | 8 | CO3  | Lectures with PPTs | Understand and Apply | End Term: Applied Questions and Practical Test |
| <b>WinForms</b>                                                      | : Introduction, Controls: Common control Group, Data, control Group, Dialog control Group, Container control Group, Menus and Context Menus: Menu Strip, Toolbar Strip., SDI and MDI Applications, Developing Custom, Composite and                                                                                                                                                                                                                                                              | 8 | CO4  | Lecture With PPTs  | Understand and Apply | End Term: Applied Questions and Practical Test |

|                 |                                                                                                                                                                                                                                                                                                                     |   |     |                    |                      |                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|--------------------|----------------------|-----------------------------------|
|                 | Extended Controls, WPF, Developing WPF application                                                                                                                                                                                                                                                                  |   |     |                    |                      |                                   |
| <b>ADO.net:</b> | Evolution of ADO.NET , Connected and Disconnect Classes, Establishing Connection with Database, Executing simple Insert, Update and Delete , Statements, DataReader and DataAdapter, What is DataSet?, Advantages of DataSet, Working with DataRelation Prepared Statements, Stored Procedures, Master Detail Form. | 8 | CO4 | Lectures with PPTs | Understand and Apply | Activity End Term: Theory Applied |

#### Reference Books:

| Sr.No. | NameoftheAuthor    | TitleoftheBook                  | Year Edition | Publisher Company |
|--------|--------------------|---------------------------------|--------------|-------------------|
| 1      | Jesse Liberty      | Programming C#                  |              | O'Reilly Press    |
| 2      | Robinson et al     | Professional C#"-               |              | Wrox Press, 2002  |
| 3      | Herbert Schildt    | The Complete Reference: C#"-    |              | Tata McGraw Hill  |
| 4      | Jerk               | The Complete Reference: Ado.Net |              | Tata McGraw Hill  |
| 5      | Deilte             | C# for programmer               |              | Pearson           |
| 6      | hilyard and teiler | C# cookbook                     |              | Orelly            |

#### Online Resources

| OnlineResourcesNo. | Websiteaddress                                                                                                |
|--------------------|---------------------------------------------------------------------------------------------------------------|
| 1                  | <a href="https://www.w3schools.com/cs/index.php">https://www.w3schools.com/cs/index.php</a>                   |
| 2                  | <a href="https://www.tutorialspoint.com/csharp/index.htm">https://www.tutorialspoint.com/csharp/index.htm</a> |
| 3                  | <a href="https://www.youtube.com/watch?v=GhQdlIFylQ8">https://www.youtube.com/watch?v=GhQdlIFylQ8</a>         |

#### MOOCs:

| ResourcesNo. | Websiteaddress |
|--------------|----------------|
| 1            | Alisons        |
| 2            | Swayam         |

| Programme: BCA-CBCS–RevisedSyllabusw.e.f.-Year2022–2023                                                                                                                                                                                                                                                                                                                                                                                                            |             |                              |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Course Code | Course Title                 |       |
| V                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 503         | Entrepreneurship Development |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Prepared by | Mr.Akhilesh Jadhav           |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Credits     | Evaluation                   | Marks |
| MDC                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3           | UE:IE                        | 60:40 |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |                              |       |
| <ul style="list-style-type: none"> <li>To understand the concept of entrepreneurship.</li> <li>To create interest amongst the students to think of becoming entrepreneurs.</li> <li>To provide ways and means to start an enterprise.</li> </ul>                                                                                                                                                                                                                   |             |                              |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                              |       |
| <p>At the end of this course, student should be able to understand</p> <p><b>CO1:</b> Study meaning of Entrepreneur and entrepreneurship.</p> <p><b>CO2:</b> Understand Role of Entrepreneurship in Economic Development, Concept of Intellectual property rights and Financial Sources</p> <p><b>CO3:</b> Identify Business Opportunity</p> <p><b>CO4:</b> Study Importance of Business plan and Support Agencies</p> <p><b>CO5:</b> Create new Business plan</p> |             |                              |       |

| Unit | Content                                                                                                                                                                                                                                                               | Sessions (Hrs) | COs Number | Teaching Methodology | CognitionLevel | EvaluationTools       |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|----------------------|----------------|-----------------------|
| 1    | <b>Introduction to Entrepreneurship :</b><br>Concept and definition of an entrepreneur,types of entrepreneurs, Qualities of good Entrepreneur, Growth of Entrepreneurship in India, role of Entrepreneurship in Economic Development, Women Entrepreneurship in India | 8              | CO1, CO2   | Lecture with Ppts    | Understand     | Quiz<br>Short Answers |
| 2    | <b>Business Opportunity Identification :</b>                                                                                                                                                                                                                          | 8              | CO 3       | Lecture with Ppts    | Understand     | Quiz<br>Short Answers |

|   |                                                                                                                                                                                                                                       |   |           |                                 |            |                          |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------|---------------------------------|------------|--------------------------|
|   | Process of searching business ideas, Need of market assessment prior to finalise the product or services, Sources of market information, Environmental analysis, Government's initiatives in entrepreneurship , selection of business |   |           |                                 |            |                          |
| 3 | <b>Business Plan Preparation :</b><br>Meaning of Business plan, Significance and Contents of a Business Plan, developing Business Plan, Presenting Business Plan, Elevator Pitch                                                      | 8 | CO4, CO 5 | Lecture with Ppts<br>Case Study | Create     | Quiz<br>Short<br>Answers |
| 4 | <b>Availability of Financial Sources and Assistance:</b><br>Types of Finance, Sources of Finance, Venture Capital, Start-up and Make-in-India program, MUDRA                                                                          | 6 | CO2       | Lecture with Ppts               | Understand | Quiz<br>Short<br>Answers |
| 5 | <b>Support Agencies for Entrepreneurs :</b><br>Support to Entrepreneurs by DIC, SIDBI, SIDCO, SSIB, NSIC, SISI, Other Institutions etc.<br>Entrepreneurship promotion by Government through various schemes                           | 8 | CO4       | Lecture with Ppts               | Understand | Quiz<br>Short<br>Answers |
| 6 | <b>Entrepreneurial Motivation and Development :</b><br>Factors in motivating entrepreneurs, Basic course contents of EDP , Evaluation of EDP, Organizations involved in EDP, Basics of Intellectual property rights                   | 7 | CO2       | Lecture with Ppts               | Understand | Quiz<br>Short<br>Answers |

### Reference Books

| Sr.No. | Name of the Author | Title of the Book                                      | Year Edition             | Publisher Company                   |
|--------|--------------------|--------------------------------------------------------|--------------------------|-------------------------------------|
| 1      | Vasant Desai       | Dynamics of Entrepreneurial Development and Management | 2001, Millennium edition | Himalaya Publication                |
| 2      | Jasmer Singh Saini | Entrepreneurship Development                           | 2003,                    | Deep and Deep Publications Pvt. Ltd |

|   |                                                      |                                                |                                 |                                     |
|---|------------------------------------------------------|------------------------------------------------|---------------------------------|-------------------------------------|
| 3 | B.S Bhatia and G. S.Batra                            | Entrepreneurship and Small Business Management | 2003                            | Deep and Deep Publications Pvt. Ltd |
| 4 | Dr. Sudhir Sharma<br>Balraj Singh<br>Sandeep Singhal | Entrepreneurship Development                   | 1 <sup>st</sup> Edition<br>2003 | Wisdom Publications                 |
| 5 | Mary Coulter                                         | Entrepreneurship I in Action                   | 2 <sup>nd</sup> Edition<br>2005 | Prentice Hall of India Pvt. Ltd     |

### Online Resources

| OnlineResourcesNo. | Websiteaddress                                                                                                                                                          |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                  | <a href="https://www.vedantu.com/commerce/entrepreneurship-development-process">https://www.vedantu.com/commerce/entrepreneurship-development-process</a>               |
| 2                  | <a href="http://www.startupindia.gov.in">www.startupindia.gov.in</a>                                                                                                    |
| 3                  | <a href="https://www.simplynotes.in/e-notes/mbabba/entrepreneurship-development/">https://www.simplynotes.in/e-notes/mbabba/entrepreneurship-development/</a>           |
| 4                  | <a href="https://www.scribd.com/document/554249314/Entrepreneurship-development-notes">https://www.scribd.com/document/554249314/Entrepreneurship-development-notes</a> |

### MOOCs:

| ResourcesNo. | Websiteaddress |
|--------------|----------------|
| 1            | Udemy          |
| 2            | Vedantu        |

| <b>Programme: BCA CBCS Revised Syllabus w.e.f.-Year 2022–2023</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                    |                      |              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----------------------|--------------|
| <b>Semester</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Course Code</b> | <b>Course Title</b>  |              |
| V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 505                | <b>LAB ON Python</b> |              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Prepared by</b> | <b>Dr.M.K.Patil</b>  |              |
| <b>Type</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Credits</b>     | <b>Evaluation</b>    | <b>Marks</b> |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2                  | UE: IE               | 60:40        |
| <b>Course Objectives</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                    |                      |              |
| <ul style="list-style-type: none"> <li>To reinforce theoretical knowledge gained in the classroom through hands-on, practical exercises.</li> <li>To include honing skills in basic syntax and data types, mastering controlflow structures, and gaining practical experience in functions, modules, and file handling.</li> <li>To focus on applying object-oriented programming (OOP) principles and manipulating data structures effectively.</li> <li>To provide a platform for students to develop proficiency in error handling and debugging techniques, fostering an understanding of best practices and coding standards.</li> <li>To reinforce practical problem-solving abilities, preparing students for real-world Python programming challenges.</li> </ul>                                                               |                    |                      |              |
| <b>Course Outcome</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                    |                      |              |
| <p><b>CO1:</b> Using some motivating examples to remember and quickly build up basic concepts such as conditionals, loops, functions, lists, strings, and tuples.</p> <p><b>CO2:</b> By remembering students, the basic concepts students will understand the concepts of searching and sorting algorithms, dynamic programming, and backtracking, as well as topics such as exception handling and using files</p> <p><b>CO3:</b> Students will Have thorough knowledge of data structures and will be able to design &amp; and develop programs for solving problems</p> <p><b>CO4:</b> Analyzing the different problems based on CSV files</p> <p><b>CO5:</b> Ability to choose an appropriate data dictionary for problem-solving</p> <p><b>CO6:</b> Design and create their own data structure for solving a real-life problem</p> |                    |                      |              |

| <b>Unit</b>                    | <b>Contents</b>                                                                                                                                                                                         | <b>Sessions (Hrs.)</b> | <b>COs Number</b> | <b>Teaching Methodology</b>            | <b>Cognition Level</b>               | <b>Evaluation Tools</b> |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------|----------------------------------------|--------------------------------------|-------------------------|
| <b>Introduction to Python:</b> | Installation of Python IDE, understanding various platforms for Python (google collaborator, Jupitar Notebook) <ul style="list-style-type: none"> <li>Basic program to understand Data Types</li> </ul> | 4                      | CO1, CO2, CO3     | Classroom Teaching, ICT-based teaching | Remembering, Understanding, Applying | Assignments, Quizzes    |

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |                              |                                        |                                                                       |                 |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------------|----------------------------------------|-----------------------------------------------------------------------|-----------------|
|                                            | <ul style="list-style-type: none"> <li>▪ creating variables, accepting input variables from user, and printing their datatype</li> <li>□ Mathematical functions (apply various operations on data +, -, /, *)</li> </ul>                                                                                                                                                                                                                                                                                                                  |   |                              |                                        |                                                                       |                 |
| <b>Statements and Control Structures :</b> | <ul style="list-style-type: none"> <li>▪ Python Program to Check if a Number is Positive, Negative or Zero</li> <li>▪ Python Program to Check if a Number is Odd or Even</li> <li>▪ Python Program to Check Leap Year</li> <li>▪ Python Program to Check Prime Number</li> <li>▪ Python Program to Print all Prime Numbers in an Interval</li> <li>▪ Python Program to Find the Factorial of a Number</li> <li>▪ Python Program to Display the Multiplication Table</li> <li>▪ Python Program to Print the Fibonacci sequence.</li> </ul> | 4 | CO1, CO2, CO3, CO4, CO5, CO6 | Classroom Teaching, ICT-based teaching | Remembering, Understanding, Applying, Analyzing, Evaluating, Creating | Lab Assignments |

|                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |                              |                                        |                                                                       |                 |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------------|----------------------------------------|-----------------------------------------------------------------------|-----------------|
|                                                               | <ul style="list-style-type: none"> <li>Python Program to Check Armstrong Number</li> <li>Python Program to Find Armstrong Number in an Interval</li> <li>Python Program to Find the Sum of Natural Numbers</li> </ul>                                                                                                                                                                                                                                                                                                               |   |                              |                                        |                                                                       |                 |
| <b>List, Ranges &amp; Tuples &amp; Dictionaries in Python</b> | <ul style="list-style-type: none"> <li>Operations on Strings, Lists, tuples and arrays</li> <li>Creating lists/tuples/arrays and accessing list elements using index</li> <li>Access the list/tuple element using –ve index.</li> <li>Extract specific elements from list/tuple/array</li> <li>Use Len(), del(), remove() and range functions on list/tuple</li> <li>Applying different searching and sorting algorithm on data (list)</li> <li>Create Dictionaries with key, value pair, and access various elements of</li> </ul> | 4 | CO1, CO2, CO3, CO4, CO5, CO6 | Classroom Teaching, ICT-based teaching | Remembering, Understanding, Applying, Analyzing, Evaluating, Creating | Lab Assignments |

|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |                                             |                                                 |                                                                                      |                    |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------|--------------------|
|                                                                                   | Dictionaries,<br>Various<br>operations<br>using<br>Dictionaries.<br>Usage of map,<br>filter functions<br>on list                                                                                                                                                                                                                                                                                                                                                                                                  |   |                                             |                                                 |                                                                                      |                    |
| <b>Functions,<br/>Modules,<br/>Packages,<br/>and<br/>Debugging<br/>Functions:</b> | <ul style="list-style-type: none"> <li>• Python Program to Find LCM</li> <li>• Python Program to Find HCF</li> <li>• Python Program to Convert Decimal to Binary, Octal and Hexadecimal</li> <li>• Python Program To Find ASCII value of a character</li> <li>• Python Program to Make a Simple Calculator</li> <li>• Python Program to Display Calendar</li> <li>• Python Program to Display Fibonacci Sequence Using Recursion</li> <li>• Python Program to Find Factorial of Number Using Recursion</li> </ul> | 4 | CO1,<br>CO2,<br>CO3,<br>CO4,<br>CO5,<br>CO6 | Classroom<br>Teaching,<br>ICT-based<br>teaching | Remembering,<br>Understanding,<br>Applying,<br>Analyzing,<br>Evaluating,<br>Creating | Lab<br>Assignments |
| <b>Python<br/>Object<br/>Oriented</b>                                             | Python Program to Get the ClassName of an Instance                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4 | CO1,<br>CO2,<br>CO3,<br>CO4,<br>CO5,<br>CO6 | Classroom<br>Teaching,<br>ICT-based<br>teaching | Remembering,<br>Understanding,<br>Applying,<br>Analyzing,<br>Evaluating,             | Lab<br>Assignments |

|                                                            |                                                                                                                                                                                                                                     |   |                              |                                            |                                                                       |                                                              |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------------|--------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------|
|                                                            | Python Program to Differentiate Between type() and is instance()                                                                                                                                                                    |   |                              |                                            | Creating                                                              |                                                              |
| <b>Python Exceptions Handling</b>                          | Exception handling routines programs                                                                                                                                                                                                | 4 | CO1, CO2, CO3, CO5, CO6      | Classroom Teaching, ICT-based teaching     | Remembering, Understanding, Applying, Evaluating, Creating            | Lab Assignments                                              |
| <b>Input and Output in Python &amp; built in functions</b> | <ul style="list-style-type: none"> <li>• Read, write, search operations on File data structure</li> <li>• Write Programs based on exception handling</li> <li>• Write program for various operations on string variables</li> </ul> | 6 | CO1, CO2, CO3, CO4, CO5, CO6 | Project-based teaching, ICT-based teaching | Remembering, Understanding, Applying, Analyzing, Evaluating, Creating | Lab Assignments, Live case study from the website Kaggle.com |

### Reference Book:

1. Introduction To Computation And Programming Using Python: With Application To Understanding Data, John V. Guttag
2. Think Python, By Allen B. Downey, O'reilly
3. Introducing Python: Modern Computing In Simple Packages By Bill Lubanovic
4. Python Programming: An Introduction To Computer Science By John Zelle
5. Core Python Programming, Dr. R. Nageshwara Rao, Dreamtech
6. Introduction to Computer Science using Python, Charles Dierbach, Wiley

| Programme:BCACBCS– Revised Syllabusw.e.f.- Year 2022–2023                                                                                                                                                                         |             |                       |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------|-------|
| Semester                                                                                                                                                                                                                          | Course Code | CourseTitle           |       |
| V                                                                                                                                                                                                                                 | 506         | Lab on Dot Net and C# |       |
|                                                                                                                                                                                                                                   | Prepared by | Mr.Alok Shah          |       |
| Type ofCourse                                                                                                                                                                                                                     | Credits     | Evaluation            | Marks |
| DSC                                                                                                                                                                                                                               | 2           | UE(60)+IE(40)         | 100   |
| <b>CourseObjectives:</b>                                                                                                                                                                                                          |             |                       |       |
| Objectives: <ul style="list-style-type: none"> <li>To learn basic C#.NET basic programming framework and designing.</li> <li>To learn and develop different C#.NET programs like classes, threads and delegations etc.</li> </ul> |             |                       |       |
| <b>CourseOutcomes:</b>                                                                                                                                                                                                            |             |                       |       |
| <b>CO1:</b> Display proficiency in C# by buildingst and-alone applications in the .NET framework using C#.                                                                                                                        |             |                       |       |
| <b>CO2:</b> Create distributed data-driven applications using the.NET Framework,C#,SQLServer and ADO.NET.                                                                                                                         |             |                       |       |
| <b>CO3:</b> Create Windows-based distributed applications using C#, SQL Server and ADO.NET                                                                                                                                        |             |                       |       |

| Unit | Content | Sessions | COs Number | Teaching Methodology | CognitionLevel | EvaluationTools |
|------|---------|----------|------------|----------------------|----------------|-----------------|
|------|---------|----------|------------|----------------------|----------------|-----------------|

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |      |                        |            |                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------|------------------------|------------|----------------------------------------------------------------------|
| <b>Basic Console Applications</b> | <ul style="list-style-type: none"> <li>• Write a C# Program to design simple calculator</li> <li>• Write a console application that obtain four int values from the user and displays the product.</li> <li>• If you have two integers stored in variables var1 and var2,what Boolean test can you perform to see if one or the other (but not both) is greater than10?</li> <li>• Write an application that receives the following informationfrom a set of students:Student Id:<br/>Student Name: Course Name: Date of Birth:<br/>The application should also display the</li> </ul> | 5 | CO 1 | Lecture with Ppts Quiz | Understand | Quiz<br>End Term<br>Internals:Short Answers<br>and Practical<br>Test |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------|------------------------|------------|----------------------------------------------------------------------|

|                      |                                                                                                                                                                                                                                                                                                                                                                                                      |   |          |                   |                      |                                                |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------|-------------------|----------------------|------------------------------------------------|
|                      | <p>information of all the students once the data is entered. Implement this using an Array of Structures.</p> <ul style="list-style-type: none"> <li>• Write a C# Program to Get a Number and Display the Number with its Reverse</li> <li>• Write a Program in C# to demonstrate Command line arguments processing.</li> <li>• Write a Program in C# to demonstrate boxing and Unboxing.</li> </ul> |   |          |                   |                      |                                                |
| <b>Date and Time</b> | <ul style="list-style-type: none"> <li>• Write a C# Program to Display the Date in Various Formats</li> <li>• Write a C# Program to Check Whether the Entered Year is a Leap Year or Not.</li> <li>• Write a C# Program to find difference between Two Dates</li> </ul>                                                                                                                              | 5 | CO 1     | Lecture with PPTs | Understand and Apply | End Term: Applied Questions and Practical Test |
| <b>Classes</b>       | <p>Write a program to demonstrate abstract class and abstract methods in C#.</p> <ul style="list-style-type: none"> <li>• Find the sum of all the elements present in a jagged array of 3 inner arrays.</li> <li>• Write a program to demonstrate Operator overloading.</li> <li>• Demonstrate arrays of interface types (for runtime polymorphism) with a C# program.</li> </ul>                    | 5 | CO 1     | Lecture with PPTs | Understand and Apply | End Term: Applied Questions and Practical Test |
| <b>ADO.NET</b>       | <p>Consider the Database STUDENT consisting of following tables:</p> <ul style="list-style-type: none"> <li>• Course (C_ID: int, C_Name: string)</li> <li>• Student (RollNo: int, S_Name: string, Address: string, C_ID: int, Admissiyear: int)</li> </ul>                                                                                                                                           | 5 | CO2, CO3 | Lecture with PPTs | Understand and Apply | End Term: Applied Questions and Practical Test |

|                            |                                                                                                                                                                                                                                                                                                                                 |   |      |                   |                      |                                                |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------|-------------------|----------------------|------------------------------------------------|
|                            | Develop suitable windows application using C#.NET having following options:<br>1. Entering new course details.<br>2. Entering new student details.<br>3. Display the details of students (in a Grid) who belong to a particular course.<br>4. Display the details of the students who have taken admission in a particular year |   |      |                   |                      |                                                |
| <b>EXCEPTION HANDLING</b>  | <ul style="list-style-type: none"> <li>write a program in C# to demonstrate error handling</li> </ul>                                                                                                                                                                                                                           | 5 | CO 2 | Lecture with PPTs | Understand and Apply | End Term: Applied Questions and Practical Test |
| <b>EVENTS AND DELEGATE</b> | <ul style="list-style-type: none"> <li>To develop a C# program to implement threading concepts.</li> <li>To develop a C# program to implement the following concepts:<br/>(a) Delegates<br/>(b) Events</li> </ul>                                                                                                               | 5 | CO 2 | Lecture with PPTs | Understand and Apply | End Term: Applied Questions and Practical Test |

### Reference Books

| Sr.No. | Name of the Author | Title of the Book               | Year Edition | Publisher Company |
|--------|--------------------|---------------------------------|--------------|-------------------|
| 1      | Jesse Liberty      | Programming C#                  |              | O'Reilly Press    |
| 2      | Robinson et al     | Professional C#                 |              | Wrox Press, 2002  |
| 3      | Herbert Schildt    | The Complete Reference: C#      |              | Tata McGraw Hill  |
| 4      | Jerk               | The Complete Reference: ADO.NET |              | Tata McGraw Hill  |
| 5      | Deitel             | C# for programmer               |              | Pearson           |
| 6      | Hilyard and Teiler | C# cookbook                     |              | O'Reilly          |

## Online Resources

| OnlineResource<br>sNo. | Websiteaddress                                                                                                                                                                                                              |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                      | <a href="https://www.w3resource.com/csharp-exercises/">https://www.w3resource.com/csharp-exercises/</a>                                                                                                                     |
| 2                      | <a href="https://home.cs.colorado.edu/~kena/classes/5448/f11/presentation-materials/csharp_dotnet_adnanreza.pdf">https://home.cs.colorado.edu/~kena/classes/5448/f11/presentation-materials/csharp_dotnet_adnanreza.pdf</a> |
| 3                      | <a href="https://www.w3resource.com/csharp-exercises/">https://www.w3resource.com/csharp-exercises/</a>                                                                                                                     |

## MOOCs:

| ResourcesNo. | Websiteaddress |
|--------------|----------------|
| 1            | Alisons        |
| 2            | Swayam         |

| Programme:BCA CBCS –Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                                                                                                                 |             |                             |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                       | Course Code | Course Title                |       |
| V                                                                                                                                                                                                                                                                                                                                              | 507         | IT Based Aptitude           |       |
|                                                                                                                                                                                                                                                                                                                                                | Prepared by | Dr.Dhanashree Sahasrabuddhe |       |
| Type                                                                                                                                                                                                                                                                                                                                           | Credits     | Evaluation                  | Marks |
| AEC                                                                                                                                                                                                                                                                                                                                            | 2           | IA                          | 50    |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                      |             |                             |       |
| <ul style="list-style-type: none"> <li>To develop skills in understanding various constructs in basic programming</li> <li>To learn applications of different types of algorithms</li> <li>To develop skills in writing SQL queries</li> <li>To learn applications of OOP concepts</li> <li>To prepare for IT company aptitude test</li> </ul> |             |                             |       |
| <b>Course Outcomes :</b>                                                                                                                                                                                                                                                                                                                       |             |                             |       |
| CO1: Applying and testing algorithms to various computing problems                                                                                                                                                                                                                                                                             |             |                             |       |
| CO2: Calculating efficiency of algorithms                                                                                                                                                                                                                                                                                                      |             |                             |       |
| CO3: Develop programming skills                                                                                                                                                                                                                                                                                                                |             |                             |       |

| Unit | Content                                                                                                                                                                                                                                                                   | Sessions (Hrs) | COs Number     | Teaching Methodology      | Cognition Level                     | Evaluation Tools |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------|---------------------------|-------------------------------------|------------------|
| 1    | <b>Algorithms and their complexity</b> -<br>Types of algorithms, efficiency of algorithms (complexity of algorithms), sorting and searching algorithms and their complexities.                                                                                            | 10             | CO 1, CO2, CO3 | Lecture with Ppts<br>Quiz | Understand, Apply, Evaluate, Create | Quiz             |
| 2    | <b>Programming with ‘c’ and Data Structures</b><br>Aptitude questions in ‘c’ with reference to datatypes, operators, different programming constructs, arrays, pointers.<br>Aptitude questions on Linear and non-linear Data structures with reference to representation, | 10             | CO 1, CO2, CO3 | Lecture with Ppts<br>Quiz | Understand, Apply, Evaluate, Create | Quiz             |

|   |                                                                                                                                                                                                                                                                                                                                                 |    |                |                        |                                     |      |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----------------|------------------------|-------------------------------------|------|
|   | characteristics, traversing algorithms                                                                                                                                                                                                                                                                                                          |    |                |                        |                                     |      |
| 3 | <b>Object Oriented Programming Concepts-</b><br>Aptitude on OOP with reference to Data Binding, data hiding, data abstraction, data encapsulation, class, object, inheritance, polymorphism, message passing<br><b>SQL –</b><br>Aptitude on SQL with reference to Usage, Types of commands, Select query and various options used with ‘select’ | 10 | CO 1, CO2, CO3 | Lecture with Ppts Quiz | Understand, Apply, Evaluate, Create | Quiz |

#### Reference Books :

| Sr. No. | Name of the Author | Title of the Book                                                   | Year Edition                | Publisher Company       |
|---------|--------------------|---------------------------------------------------------------------|-----------------------------|-------------------------|
| 1       | S. Sridhar         | Design and Analysis of Algorithms                                   | 15/12/2014<br>First Edition | Oxford University Press |
| 2       | Yashvant Kanetkar  | Let us c                                                            | 19 <sup>th</sup> Edition    | BPB Publication         |
| 3       | Ivan Bayross       | SQL, PL/SQL the Programming Language of Oracle                      | 4 <sup>th</sup> Edition     | BPB Publication         |
| 4       | Rakesh Singh       | OOP Concepts<br>Booster : Take Your Coding Skills to the Next Level | 25 Nov. 2019                | Notion Press            |

| Programme:BCA CBCS –Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                                                                                                                                                                                                                               |             |                 |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Course Code | Course Title    |       |
| V                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 508         | Human Rights    |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Prepared by | Dr.Deepali Gala |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Credits     | Evaluation      | Marks |
| VBC                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2           | IE              | 50    |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |                 |       |
| <ul style="list-style-type: none"> <li>• Foundational Understanding of Human Rights</li> <li>• Proficiency in Interpreting Human Rights Instruments</li> <li>• Critical Analysis of Judicial Activism and Human Rights</li> </ul>                                                                                                                                                                                                                            |             |                 |       |
| <b>Course Outcomes :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |                 |       |
| <p>After completing this course, the student will be able to :</p> <p><b>CO1:</b> Students will acquire a solid understanding of the foundational principles, meaning, and scope of human rights</p> <p><b>CO2:</b> Gain proficiency in interpreting and applying human rights instruments.</p> <p><b>CO3:</b> Develop critical thinking skills to analyze instances of judicial activism and understand its implications for human rights jurisprudence</p> |             |                 |       |

| Unit | Content                                                                                                                                                                                                                                                                                                                                                                            | Sessions (Hrs) | COs Number | Teaching Methodology                 | Cognition Level | Evaluation Tools                     |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|--------------------------------------|-----------------|--------------------------------------|
| 1    | <p><b>Chapter 1:</b> Concept and Development of Human Rights</p> <p><b>Meaning and Scope of Human Rights:</b> Define and explore the fundamental concept of human rights and their scope.</p> <p><b>Development of Human Rights:</b> Trace the historical development of human rights and highlight key milestones.</p> <p><b>Universal Declaration of Human Rights (UDHR)</b></p> | 10             | CO1        | As per individual faculty discretion | Remember        | As per individual faculty discretion |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |     |                                      |            |                                      |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|--------------------------------------|------------|--------------------------------------|
|   | <p><b>1948:</b> Discuss the significance and provisions of the UDHR, a landmark document in the field of human rights.</p> <p><b>International Covenant on Civil and Political Rights (ICCPR) 1996:</b> Examine the provisions and implications of this international covenant.</p> <p><b>International Covenant on Economic, Social and Cultural Rights (ICESCR) 1966:</b> Explore the content and importance of ICESCR.</p>                                                                                            |    |     |                                      |            |                                      |
| 2 | <p><b>Chapter 2: Human Rights in India</b></p> <p><b>Protection of Human Rights Act, 1993:</b> Analyze the key features and provisions of this legislation.</p> <p><b>Third Generation Human Rights (Group Rights) and Fourth Generation Human Rights (Right to Development and Environmental Rights):</b> Explore emerging categories of human rights, emphasizing group rights, right to development, and environmental rights.</p> <p><b>Convention on the Elimination of All Forms of Discrimination Against</b></p> | 10 | CO2 | As per individual faculty discretion | Understand | As per individual faculty discretion |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |     |                                 |         |                                      |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|---------------------------------|---------|--------------------------------------|
|   | <p><b>Women (CEDAW):</b> Discuss the international convention focused on women's rights.</p> <p><b>Convention on the Rights of the Child:</b> Examine the international convention addressing the rights of children.</p>                                                                                                                                                                                                                                                                                                                                                                                        |    |     |                                 |         |                                      |
| 3 | <p><b>Chapter 3: Enforcement of Human Rights</b></p> <p><b>National Human Rights Commission (NHRC):</b> Analyze the role, functions, and significance of the NHRC in India.</p> <p><b>State Human Rights Commission:</b> Explore the functions and role of State Human Rights Commissions in India.</p> <p><b>Judicial Activism and Human Rights:</b> Discuss instances of judicial activism in upholding human rights and the impact on legal interpretation.</p> <p><b>Human Rights Courts in India:</b> Examine the establishment and functioning of specialized courts dedicated to human rights issues.</p> | 10 | CO3 | Lecture with PPTs<br>Case Study | Analyse | As per individual faculty discretion |

**Reference Books**

| <b>Sr.No.</b> | <b>NameoftheAuthor</b> | <b>TitleoftheBook</b>    | <b>Year<br/>Edition</b> | <b>Publisher<br/>Company</b> |
|---------------|------------------------|--------------------------|-------------------------|------------------------------|
| 1             | Charles R. Beitz       | The Idea of Human Rights | 2009                    | Oxford                       |
| 2             | Amartya Sen            | The Argumentative Indian | 2006                    | Penguin                      |

**Online Resources**

| <b>Online Resources No.</b> | <b>Website address</b>                                        |
|-----------------------------|---------------------------------------------------------------|
| 1                           | <a href="https://www.who.int/">https://www.who.int/</a>       |
| 2                           | <a href="https://www.icrc.org/en">https://www.icrc.org/en</a> |

**MOOCs:**

| <b>Resources No.</b> | <b>Website address</b> |
|----------------------|------------------------|
| 1                    | Alisons                |
| 2                    | Swayam                 |

| Programme:BCA CBCS –Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                                                                                                                                                                                                                                      |             |                                  |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Course Code | Course Title                     |       |
| VI                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 601         | Data Warehousing And Data Mining |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Prepared by | Dr.Rajendra Pujari               |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Credits     | Evaluation                       | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3           | UE:IE                            | 60:40 |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |                                  |       |
| <ul style="list-style-type: none"> <li>To identify the scope and essentiality of Data Warehousing and Mining.</li> <li>To analyze data, choose relevant models and algorithms for respective applications.</li> <li>To study spatial and web data mining.</li> <li>To develop research interest towards advances in data mining.</li> </ul>                                                                                                                         |             |                                  |       |
| <b>Course Outcomes :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                  |       |
| <p>After completing this course, the student will be able to :</p> <p><b>CO1 :</b> Identify the need for data warehousing</p> <p><b>CO2 :</b> Understand the data warehousing architecture and understand various of Data warehouse.</p> <p><b>CO3:</b> Familiar with basic concepts of data mining</p> <p><b>CO4 :</b> Applying knowledge using association rule mining algorithms, classification techniques and prediction methods in real life applications</p> |             |                                  |       |

| Unit | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Sessi<br>ons<br>(Hrs<br>) | COs<br>Nu<br>m<br>ber | Teachin<br>g<br>Method<br>ology | Cognit<br>i on<br>Level | Evaluat<br>ion<br>Tools           |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------------------|---------------------------------|-------------------------|-----------------------------------|
| 1    | <b>Introduction to Data warehousing:</b><br>Data Warehousing, Difference between operational database system and data warehouse, Data Warehouse Users, Benefits of Data Warehousing, Metadata, Classification of Metadata, and Importance of Metadata. Data Marts, Reasons for creating Data Marts, Building Data Marts: Top down Approach & Bottom up Approach, Data Warehouse Architecture, Two Tier Architecture, Three Tier Architecture. Data Warehouse Schema, Star, Snow Flake & Fact Constellation Schema. OLAP, Need for OLAP, OLAP Operations, OLAP Models. | 8                         | CO 1                  | Lecture with Ppts Quiz          | Underst and             | End Term Internals :Short Answers |

|   |                                                                                                                                                                                                                                                          |   |      |                       |                 |                                                       |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------|-----------------------|-----------------|-------------------------------------------------------|
| 2 | <b>Data Preprocessing:</b><br>Need, Objectives and Techniques, Descriptive data summarization, Data Cleaning, Data Integration, Data Transformation, Data Reduction.                                                                                     | 8 | CO 1 | Lecture with Ppts     | Apply (Analyse) | End Term Internals :Short Answers                     |
| 3 | <b>Introduction to Data Mining:</b><br>Introduction, Need for Data Mining, KDD Process, Data Mining Architecture, Data Mining Functionalities, Data Mining Task Primitives, Integration of a Data Mining System with a Database or Data Warehouse System | 8 | CO 3 | Lecture with PPTs     | Analyse         | End Term Internals :Short Answers , Viva              |
| 4 | <b>Mining Frequent Items and Associations:</b><br>Frequent Item Set, Closed Item Set, Association Rule Mining, Market Basket Analysis, Classification of Association Rules, Apriori Algorithm                                                            | 6 | CO1  | Lectures with PPTs    | Evaluate        | End Term Internals :Short Answers , Practice examples |
| 5 | <b>Classification and Prediction:</b><br>Classification & Prediction, Issues regarding classification & Prediction, Comparing Classification Methods, Classification by Decision Tree Induction                                                          | 7 | CO2  | Lecture Case Activity | Create          | End Term Internals :Short Answers                     |
| 6 | <b>Clustering:</b><br>Introduction, Cluster Analysis, Need, Categorization of Major clustering methods. Types of Data in Cluster Analysis, Partitioning Methods: K-Means Method, K-Medoids Method, Applications of data mining in various sectors        | 8 | CO4  | Lectures with PPTs    | Evaluate        | End Term Internals :Short Answers                     |

**Reference Books**

| <b>Sr. No.</b> | <b>Name of the Author</b>       | <b>Title of the Book</b>                                      | <b>Year Edition</b> | <b>Publisher Company</b> |
|----------------|---------------------------------|---------------------------------------------------------------|---------------------|--------------------------|
| 1              | Jiawei Han and Micheline Kamber | Data Mining Concepts and Techniques                           | 2012                | ELSEVIER                 |
| 2              | M.Humphires, M.Hawkins          | Data Warehousing: Architecture and Implementation             | 2008                | Pearson Education        |
| 3              | Kargupta, Joshi                 | Data Mining: Next Generation Challenges and Future Directions | 2004                | Prentice Hall of India   |
| 4              | Margaret H.Dunham               | Data mining Introductory and advanced Topics                  | 20018               | Pearson Education        |

**MOOCs:**

| <b>Resources No.</b> | <b>Web site address</b>                                |
|----------------------|--------------------------------------------------------|
| 1                    | NPTEL / Swayam                                         |
| 2                    | <a href="http://www.edx.com">www. edx.com</a>          |
| 3                    | <a href="http://www.coursera.com">www.coursera.com</a> |

| Programme:BCA CBCS –Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                                                                                                                    |             |                      |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                          | Course Code | Course Title         |       |
| VI                                                                                                                                                                                                                                                                                                                                                | 602         | Web Programming(PHP) |       |
|                                                                                                                                                                                                                                                                                                                                                   | Prepared by | Dr.Suvarna Patil     |       |
| Type                                                                                                                                                                                                                                                                                                                                              | Credits     | Evaluation           | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                               | 3           | UE:IE                | 60:40 |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                         |             |                      |       |
| <ul style="list-style-type: none"> <li>To get knowledge of dynamic web site development</li> <li>To make students able to design, develop the various types of web based applications.</li> <li>To get student familiar with various functionality of PHP</li> </ul>                                                                              |             |                      |       |
| <b>Course Outcomes :</b>                                                                                                                                                                                                                                                                                                                          |             |                      |       |
| <p><b>CO1:</b> To study the basic of PHP language as control structures , array , function, strings and file handling</p> <p><b>CO2:</b> To understand the concept of cookie and session</p> <p><b>CO3:</b> To under the MYSQL components, and Database connectivity</p> <p><b>CO4:</b> To create website with implementation of all concepts</p> |             |                      |       |

| Unit | Content                                                                                                                                                                                                                                                                  | Sessions (Hrs ) | COs Number | Teaching Methodology | CognitionLevel | EvaluationTools          |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|----------------------|----------------|--------------------------|
| 1    | <b>Introduction To PHP:</b><br>Installing and configuring PHP, Building blocks of PHP:PHP tags, variables, data types, operators, expressions, constants, Control Structures: conditional statements, loops, switch statement                                            | 9               | CO 1       | Lecture with Ppts    | Understand     | Quiz<br>Short<br>Answers |
| 2    | <b>Working With Functions And Arrays:</b><br>Working with functions:<br>What is a function? Function declaration and definition, Calling function, user defined functions, variable scope, working with arrays:<br>Creating, sorting and reordering arrays, PHP classes. | 9               | CO 1       | Lecture with Ppts    | Understand     | Quiz<br>Short<br>Answers |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |          |                   |                    |                    |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------|-------------------|--------------------|--------------------|
| 3 | <b>String Manipulation:</b><br>Working with strings, dates and time: Formatting, investigating and manipulating strings with PHP, using date and time functions in PHP, working with forms: Creating a simple input form.<br>File Handling: Saving data, storing and retrieving Bob's order, processing files, opening file, writing to a file, closing a file, reading from a file, uses other useful file functions             | 9 | CO 1     | Lecture with Ppts | Understand , Apply | Quiz Short Answers |
| 4 | <b>Working With CookiesAnd Sessions :</b><br>Working with cookies: Introducing cookies, setting and deleting cookies with PHP<br><br>Working with session: starting a session, working with session variables, passing session IDs in the query string, destroying sessions and unsetting variables, using sessions                                                                                                               | 9 | CO2      | Lecture with Ppts | Understand , Apply | Quiz Short Answers |
| 5 | <b>MYSQL :</b><br>Creating web database: Using MySQL monitor, logging into MySQL, creating databases and users, setting users and privileges, column data types<br><br><b>Working with MySQL database:</b> Inserting data into database, retrieving data from the database, retrieving data with specific criteria, retrieving data from multiple tables, retrieving data in particular order, grouping and aggregate data, using | 9 | CO3, CO4 | Lecture with Ppts | Create, Apply      | Quiz Short Answers |

|  |                                                                                              |  |  |  |  |  |
|--|----------------------------------------------------------------------------------------------|--|--|--|--|--|
|  | sub queries, updating records, deleting records from databases, dropping table and database. |  |  |  |  |  |
|--|----------------------------------------------------------------------------------------------|--|--|--|--|--|

### Reference Books

| Sr. No. | Name of the Author | Title of the Book                    | Year Edition             | Publisher Company   |
|---------|--------------------|--------------------------------------|--------------------------|---------------------|
| 1       | Welling Thomson    | PHP and MySQL Web Development        | Fourth Edition           | Pearson Publication |
| 2       | Julie C. Meloni    | Teach Yourself PHP, MySQL and Apache | 12 <sup>th</sup> edition | Pearson Publication |

### Online Resources

| Online Resources No. | Web site address                                                                                        |
|----------------------|---------------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.tutorialspoint.com/php/index.htm">https://www.tutorialspoint.com/php/index.htm</a> |
| 2                    | <a href="https://www.w3schools.com/php/">https://www.w3schools.com/php/</a>                             |
| 3                    | <a href="https://www.javatpoint.com/php-tutorial">https://www.javatpoint.com/php-tutorial</a>           |

### MOOCs:

| Resources No. | Web site address                                       |
|---------------|--------------------------------------------------------|
| 1             | NPTEL / Swayam                                         |
| 2             | <a href="http://www.edx.com">www.edx.com</a>           |
| 3             | <a href="http://www.coursera.com">www.coursera.com</a> |

| Programme: BCA CBCS–Revised Syllabus w.e.f.-Year2022–2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |                             |       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | CourseCode  | CourseTitle                 |       |
| VI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 603         | Software Project Management |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Prepared by | Mr.B.D.Patil                |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Credits     | Evaluation                  | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3           | UE:IE                       | 60:40 |
| CourseObjectives:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |                             |       |
| <ul style="list-style-type: none"> <li>To provide basic project management skills with a strong emphasis on issues</li> <li>To understand problems associated with delivering successful IT projects</li> <li>To understand of the particular issues encountered in handling IT projects</li> <li>To offer students methods, techniques to manage IT projects</li> <li>To provide 'hands-on' experience in dealing with IT projects</li> </ul>                                                                                                                                                                                                     |             |                             |       |
| Course Outcomes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |                             |       |
| <p><b>CO1:</b> Remember basic concept of software, types, SDLC, Process models</p> <p><b>CO2:</b> By remembering basic concept of software student will understand concept of project management formulation, project management</p> <p><b>CO3:</b> Student will have thorough knowledge of software project management life cycle and apply up to real life project</p> <p><b>CO4:</b> Student will acquire a good knowledge of software project management, PMBOK, accurate software estimation, risk and software quality.</p> <p><b>CO5:</b> Student will have ability to make estimation and planning and scheduling of real life project</p> |             |                             |       |

| Unit | Content                                                                                                                                                                                                                                                                                                                                                          | Sess<br>ions<br>(Hrs<br>) | COs<br>Number | Teaching<br>Methodolo<br>gy | Cognitio<br>n Level | Evaluatio<br>nTools                                |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------|-----------------------------|---------------------|----------------------------------------------------|
| 1    | <b>Introduction to project management:</b><br>Project, project management, Importance, characteristics of project how software projects are diff. than other projects, Problems with software projects, Phases: Initiation phase, planning phase, execution phase, monitoring and controlling phase, and closing phase. All parties involved in project, Role of | 5                         | CO 1          | Lecture with Ppts<br>Quiz   | Understand          | Quiz<br>End Term<br>Internals:<br>Short<br>Answers |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |      |                                                           |                 |                                                                                               |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------|-----------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------------------|
|   | Project Manager, Project management framework, Software tool for project management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |      |                                                           |                 |                                                                                               |
| 2 | <b>Project planning:</b><br>Integration management: What is integration management, plan development and execution, What is scope management, methods for selecting project, scope statement, Work Breakdown Structure, main steps in Project planning: identify project scope and objective, identify project infrastructure, analyze project characteristics, identify project products and activities, estimate effort for each activity, identify risk activity, allocate resources, review plan, execute plan. Use of software (Microsoft Project) to assist in project planning activities. | 10 | CO 2 | Lecture with Ppts<br>Case Study<br>Microsoft Project Demo | Apply (Analyse) | Case Study ,<br>Business cases<br>End Term:<br>Applied Questions                              |
| 3 | <b>Project scheduling:</b><br>Time management: importance of Project schedules, schedules and activities, sequencing and scheduling activities, Network Planning models, duration estimation and schedule development, Critical path analysis, PERT, Use of software( Microsoft project) to assist in project scheduling.                                                                                                                                                                                                                                                                         | 10 | CO 3 | Lecture with PPTs<br>Case Study<br>Microsoft Project Demo | Analyse         | Case Study with<br>Presentations<br>End Term Exams:<br>Case based Questions/Applied Questions |
| 4 | <b>Project cost management:</b><br>Importance and principles of project cost management, Resource planning, Attributes to be considered in cost estimation, factors affecting the cost, various costs involved in it. Traditional method: Estimation by analogy, Expert judgment, Parkinson, price to win, top down,                                                                                                                                                                                                                                                                              | 10 | CO 4 | Lectures with PPTs<br><br>Group Activity<br>Video Cases   | Evaluate        | Group Activity<br><br>End Term Exam: Short business cases and situation based questions       |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |      |                        |                    |                                                                  |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------|------------------------|--------------------|------------------------------------------------------------------|
|   | bottom up. COCOMO Model, Function point analysis, Function point analysis, Cost control, Use of software( Microsoft project) to assist in cost management.                                                                                                                                                                                                                                                                                                                                              |    |      |                        |                    |                                                                  |
| 5 | <b>Project quality management and Project Risk Management:</b><br>Quality of information technology project, Stages of software quality management, PMBOK, Quality standards, Tools and techniques for quality control.<br><b>Project risk management:</b><br>The importance, Top risk in projects, Common sources of risk in IT projects, elements in risk mgt., Risk identification, Risk quantification, Risk response development and control, using software to assist in project risk management. | 10 | CO 5 | Lecture, Case Activity | Analyze / Evaluate | Case Presentation Activity<br>End Term: Theory Applied Questions |

### Reference Books

| Sr.No. | Name of the Author            | Title of the Book                                  | Publisher Company        |
|--------|-------------------------------|----------------------------------------------------|--------------------------|
| 1      | Kathy schwalbe,               | Information Technology Project Management: THOMSON | course Technology, 2003. |
| 2      | Bob Hughes and Mike Cottrell, | Software project management Third edition          | Tata McGraw-Hill         |
| 3      | Microsoft project Tool.       | Software Requirement:                              | Microsoft project Tool.  |

**Online Resources:**

| Online ResourcesNo | Websiteaddress                                                                        |
|--------------------|---------------------------------------------------------------------------------------|
| 1                  | <a href="https://onlinecourses.swayam2.ac.in">https://onlinecourses.swayam2.ac.in</a> |
| 2                  | <a href="https://www.coursera.org/courses">https://www.coursera.org/courses</a>       |
| 3                  | <a href="https://www.udemy.com/courses">https://www.udemy.com/courses</a>             |
| 4                  | <a href="https://www.edx.org">https://www.edx.org</a>                                 |
| 5                  | NPTEL / Swayam                                                                        |
| 6                  | <a href="https://www.classcentral.com">https://www.classcentral.com</a>               |

| Programme:BCA CBCS –Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                                                                                                  |             |                                     |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                        | Course Code | Course Title                        |       |
| VI                                                                                                                                                                                                                                                                                                                              | 605         | Lab on Web programming with Project |       |
|                                                                                                                                                                                                                                                                                                                                 | Prepared by | Dr.Suvarna Patil                    |       |
| Type                                                                                                                                                                                                                                                                                                                            | Credits     | Evaluation                          | Marks |
| DSC                                                                                                                                                                                                                                                                                                                             | 2           | UE:IE                               | 60:40 |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                       |             |                                     |       |
| To make students to:                                                                                                                                                                                                                                                                                                            |             |                                     |       |
| <ul style="list-style-type: none"> <li>To get knowledge of dynamic web site development</li> <li>To make students able to design, develop the various types of web based applications.</li> <li>To get student familiar with various functionality of PHP</li> </ul>                                                            |             |                                     |       |
| <b>Course Outcomes :</b>                                                                                                                                                                                                                                                                                                        |             |                                     |       |
| <p><b>CO1:</b> To apply concept of array, looping, function,, file handling, string function</p> <p><b>CO2:</b> To create form with basic functionality</p> <p><b>CO3:</b> To create Database and Tables with SQL and create Database connectivity</p> <p><b>CO4:</b> To create website with implementation of all concepts</p> |             |                                     |       |

| Unit | Content                                                                                                                                                                                                                                                              | Sessions (Hrs) | COs Number | Teaching Methodology | Cognition Level | Evaluation Tools |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|----------------------|-----------------|------------------|
| 1    | <p>Write a Program for finding the biggest number in an array without using any array functions.</p> <p>Write a program to square of a number.</p> <p>Write a program to print Factorial of any number.</p> <p>Write a program in PHP to print Fibonacci series.</p> | 5              | CO 1       | Practical Demo       | Create          | Quiz             |



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |         |                |        |      |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------|----------------|--------|------|
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |         |                |        |      |
|  | <ul style="list-style-type: none"> <li>• Write a PHP program to create and manage a database using SQL commands.</li> <li>• Write a PHP program to create and validate a email id.</li> <li>• Using PHP and SQL, create and validate a sample login form.</li> <li>• Write a PHP script to accept personal details of student (rno, name, class) on first page. On second page accept marks of six subjects (out of 100). On third page print marklist (rno, name, class, marks, total, percentage)</li> <li>• Write a PHP file that will output a form containing 2 fields: username and password. Upon submission of the form, the code should check against the database to see whether the username-password pair was correct. If so, display a welcome message. If not, display the message "Invalid username or password" followed by the same login form.</li> </ul> | 6 | CO2,CO3 | Practical Demo | Create | Quiz |
|  | Write a PHP file that can be added to other PHP files using                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 5 | CO4     | Practical Demo | Create | Quiz |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |  |  |  |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  | <p>the include or require functions. This file should:</p> <ol style="list-style-type: none"> <li>Make a connection to a MySQL database, and log in with valid credentials. The connection resource should be stored in a variable with an appropriate name.</li> <li>Create a database TEST if it does not exist.</li> <li>Select the TEST database.</li> <li>Create a table USER exerciseusers if it does not exist with the following field <ol style="list-style-type: none"> <li>USERNAME VARCHAR(100), PASSWORD_HASH CHAR(40), PHONE VARCHAR(10)</li> </ol> </li> <li>The USERNAME field should be designated as UNIQUE.</li> <li>If any of these operations cause an error, stop execution and print the error message</li> </ol> <p>Design a web page that accepts inputs(username and password) and authenticate the username and password from a given database using PHP.</p> |  |  |  |  |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|

### Reference Books

| Sr. No. | Name of the Author | Title of the Book                    | Year Edition             | Publisher Company   |
|---------|--------------------|--------------------------------------|--------------------------|---------------------|
| 1       | Welling Thomson    | PHP and MySQL Web Development        | Fourth Edition           | Pearson Publication |
| 2       | Julie C. Meloni    | Teach Yourself PHP, MySQL and Apache | 12 <sup>th</sup> edition | Pearson Publication |

### Online Resources

| Online Resources No. | Web site address                                                                                        |
|----------------------|---------------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.tutorialspoint.com/php/index.htm">https://www.tutorialspoint.com/php/index.htm</a> |
| 2                    | <a href="https://www.w3schools.com/php/">https://www.w3schools.com/php/</a>                             |
| 3                    | <a href="https://www.javatpoint.com/php-tutorial">https://www.javatpoint.com/php-tutorial</a>           |

**MOOCs:**

| <b>Resources<br/>No.</b> | <b>Web site address</b> |
|--------------------------|-------------------------|
| 1                        | NPTEL                   |
| 2                        | Swayam                  |
| 3                        | edx.com                 |
| 4                        | coursera.com            |

| Programme: BCA CBCS – Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                           |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Course Code | Course Title              |       |
| VI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 606         | Lab on Data Visualization |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Prepared by | Prof. Niket Tajane        |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Credits     | Evaluation                | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2           | UE:IE                     | 60:40 |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |                           |       |
| <ul style="list-style-type: none"> <li>Introduce the basic concepts of Statistics and Data Visualization techniques.</li> <li>Explore the types of data visualization by using small as well as large datasets.</li> <li>To present the result using various visualization techniques by using Python.</li> </ul>                                                                                                                                                                              |             |                           |       |
| <b>Course Outcomes :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |                           |       |
| <p>After completing the course, the students shall be able to:</p> <p><b>CO1</b> :To comprehend how Statistics techniques are used.</p> <p><b>CO2</b> :To comprehend how data visualization techniques are used.</p> <p><b>CO3</b> :To apply different forms of visual encoding and data visualization.</p> <p><b>CO4</b> :Students can demonstrate various methods of data visualization to present the relevant analysis's outcome by using python programming after solving case study.</p> |             |                           |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |                           |       |

| Unit | Content                                                                                                                                                                                                                                                                                                         | Sessions (Hrs) | COs Number | Teaching Methodology                      | Cognition Level         | Evaluation Tools |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|-------------------------------------------|-------------------------|------------------|
| 1    | <p><b>Basic statistical operations</b></p> <p>Apply basic statistical operations on a dataset. For example - compute the mean, median, mode, range, quartiles, and variance for one or more attributes.</p> <p>a. Create a dataframe for students' information such as name, graduation percentage and age.</p> | 5              | CO 1       | Lab Demonstration / Practical Assignments | Understand and Applying | Short answer     |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |           |                                           |                         |              |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------|-------------------------------------------|-------------------------|--------------|
|   | <p>Display average age of students, average of graduation percentage. And, also describe all basic statistics of data. (Hint: use describe()).</p> <p>b. Download iris dataset file. Read this csv file using read_csv() function. Take samples from entire dataset. Display maximum and minimum values of all numeric attributes.</p> <p>c. Continue with above dataset, find number of records for each distinct value of class attribute. Consider entire dataset and not the samples.</p> <p>Display column-wise mean, and median for iris dataset from (Hint: Use mean() and median() functions of pandas dataframe)</p> |   |           |                                           |                         |              |
| 2 | <p><b>Introduction to Python</b></p> <p>a. Download the heights and weights dataset and load the dataset from a given csv file into a dataframe. Print the first, last 10 rows and random 20 rows. (<a href="https://www.kaggle.com/burnoutminer/heightsand-weights-dataset">https://www.kaggle.com/burnoutminer/heightsand-weights-dataset</a>)</p> <p>b. Write a Python program to find the shape, size, datatypes of the dataframe object.</p>                                                                                                                                                                             | 5 | CO 2, CO3 | Lab Demonstration / Practical Assignments | Understand and Applying | Short answer |

|   | <p>c. Write a Python program to view basic statistical details of the data.</p> <p>d. Write a Python program to get the number of observations, missing values and nan values.</p> <p>e. Write a Python program to add a column to the dataframe “BMI” which is calculated as : weight/height<sup>2</sup></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |            |        |     |        |      |      |      |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |      |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |      |                                           |                         |              |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------|--------|-----|--------|------|------|------|---|-------|------|--------|----|-----|--------|---|---|-------|------|------|----|-----|--------|---|---|-------|------|------|----|-----|--------|---|---|-------|------|------|----|-----|--------|---|---|-------|------|--------|----|-----|--------|---|---|-------|------|------|----|-----|--------|---|---|------|------|------|----|-----|--------|---|---|-------|------|------|----|-----|--------|---|---|-------|------|------|----|-----|--------|---|---|-------|------|------|----|-----|--------|---|---|------|-------------------------------------------|-------------------------|--------------|
| 3 | <p><b>Basic of Data Visualization</b></p> <p>Create the Data set as per the following table in .csv file and draw the following types of Graph / Charts. Use Matplotlib Library.</p> <table><thead><tr><th></th><th>total_bill</th><th>tip</th><th>sex</th><th>smoker</th><th>day</th><th>time</th><th>size</th></tr></thead><tbody><tr><td>0</td><td>16.99</td><td>1.01</td><td>Female</td><td>No</td><td>Sun</td><td>Dinner</td><td>2</td></tr><tr><td>1</td><td>10.34</td><td>1.66</td><td>Male</td><td>No</td><td>Sun</td><td>Dinner</td><td>3</td></tr><tr><td>2</td><td>21.01</td><td>3.50</td><td>Male</td><td>No</td><td>Sun</td><td>Dinner</td><td>3</td></tr><tr><td>3</td><td>23.68</td><td>3.31</td><td>Male</td><td>No</td><td>Sun</td><td>Dinner</td><td>2</td></tr><tr><td>4</td><td>24.59</td><td>3.61</td><td>Female</td><td>No</td><td>Sun</td><td>Dinner</td><td>4</td></tr><tr><td>5</td><td>25.29</td><td>4.71</td><td>Male</td><td>No</td><td>Sun</td><td>Dinner</td><td>4</td></tr><tr><td>6</td><td>8.77</td><td>2.00</td><td>Male</td><td>No</td><td>Sun</td><td>Dinner</td><td>2</td></tr><tr><td>7</td><td>26.88</td><td>3.12</td><td>Male</td><td>No</td><td>Sun</td><td>Dinner</td><td>4</td></tr><tr><td>8</td><td>15.04</td><td>1.96</td><td>Male</td><td>No</td><td>Sun</td><td>Dinner</td><td>2</td></tr><tr><td>9</td><td>14.78</td><td>3.23</td><td>Male</td><td>No</td><td>Sun</td><td>Dinner</td><td>2</td></tr></tbody></table> <p>a. Scatter Plot<br/>b. Line Chart<br/>c. Bar Chart</p> <p>Histogram</p> |      | total_bill | tip    | sex | smoker | day  | time | size | 0 | 16.99 | 1.01 | Female | No | Sun | Dinner | 2 | 1 | 10.34 | 1.66 | Male | No | Sun | Dinner | 3 | 2 | 21.01 | 3.50 | Male | No | Sun | Dinner | 3 | 3 | 23.68 | 3.31 | Male | No | Sun | Dinner | 2 | 4 | 24.59 | 3.61 | Female | No | Sun | Dinner | 4 | 5 | 25.29 | 4.71 | Male | No | Sun | Dinner | 4 | 6 | 8.77 | 2.00 | Male | No | Sun | Dinner | 2 | 7 | 26.88 | 3.12 | Male | No | Sun | Dinner | 4 | 8 | 15.04 | 1.96 | Male | No | Sun | Dinner | 2 | 9 | 14.78 | 3.23 | Male | No | Sun | Dinner | 2 | 8 | CO 3 | Lab Demonstration / Practical Assignments | Understand and Applying | Short answer |
|   | total_bill                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | tip  | sex        | smoker | day | time   | size |      |      |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |      |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |      |                                           |                         |              |
| 0 | 16.99                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1.01 | Female     | No     | Sun | Dinner | 2    |      |      |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |      |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |      |                                           |                         |              |
| 1 | 10.34                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1.66 | Male       | No     | Sun | Dinner | 3    |      |      |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |      |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |      |                                           |                         |              |
| 2 | 21.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3.50 | Male       | No     | Sun | Dinner | 3    |      |      |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |      |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |      |                                           |                         |              |
| 3 | 23.68                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3.31 | Male       | No     | Sun | Dinner | 2    |      |      |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |      |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |      |                                           |                         |              |
| 4 | 24.59                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3.61 | Female     | No     | Sun | Dinner | 4    |      |      |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |      |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |      |                                           |                         |              |
| 5 | 25.29                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4.71 | Male       | No     | Sun | Dinner | 4    |      |      |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |      |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |      |                                           |                         |              |
| 6 | 8.77                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2.00 | Male       | No     | Sun | Dinner | 2    |      |      |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |      |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |      |                                           |                         |              |
| 7 | 26.88                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3.12 | Male       | No     | Sun | Dinner | 4    |      |      |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |      |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |      |                                           |                         |              |
| 8 | 15.04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1.96 | Male       | No     | Sun | Dinner | 2    |      |      |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |      |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |      |                                           |                         |              |
| 9 | 14.78                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3.23 | Male       | No     | Sun | Dinner | 2    |      |      |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |        |    |     |        |   |   |       |      |      |    |     |        |   |   |      |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |       |      |      |    |     |        |   |   |      |                                           |                         |              |

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|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---------------------|-------------------------------------------------------------------------------|-------------------------------|-----------------|
| 4 | <p><b>Case Study on Data Visualization</b></p> <p>Student must use Iris flower data set for LabAssignments. The Iris flower data set or Fisher's Iris data set is a multivariate data set introduced by the British statistician and biologist Ronald Fisher in his 1936 paper.</p>  <p>The data set consists of 50 samples from each of three species of Iris (Iris setosa, Iris virginica and Iris versicolor). Four features were measured from each sample: the length and the width of the sepals and petals, in centimeters. Based on the combination of these four features, Fisher developed a linear discriminant model to distinguish the species from each other.</p> <p>The downloadable dataset (.csv format) can be found at:</p> | 12 | CO2,<br>CO3,<br>CO4 | Lab<br>Demonstration /<br>Practical<br>Assignments / Case<br>Study<br>Solving | Understand<br>and<br>Applying | Short<br>answer |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---------------------|-------------------------------------------------------------------------------|-------------------------------|-----------------|

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  |  |  |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  | <p><a href="https://archive.ics.uci.edu/ml/datasets/iris">https://archive.ics.uci.edu/ml/datasets/iris</a></p> <ol style="list-style-type: none"> <li>Generate a random array of 50 integers and display them using a line chart, scatter plot, histogram and box plot. Apply appropriate color, labels and styling options.</li> <li>Add two outliers to the above data and display the box plot.</li> <li>Create two lists, one representing subject names and the other representing marks obtained in those subjects. Display the data in a pie chart and bar chart.</li> <li>Write a Python program to create a Bar plot to get the frequency of the three species of the Iris data.</li> <li>Write a Python program to create a Pie plot to get the frequency of the three species of the Iris data.</li> <li>Write a Python program to create a histogram of the three species of the Iris data.</li> <li>Write a Python program to create a graph to find relationship between the petal length and petal width.</li> <li>Write a Python program to draw scatter plots to</li> </ol> |  |  |  |  |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|

|  |                                                                                                                                                                                                                                |  |  |  |  |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  | <p>compare two features of the iris dataset.</p> <p>Write a Python program to create box plots to see how each feature i.e. Sepal Length, Sepal Width, Petal Length, Petal Width are distributed across the three species.</p> |  |  |  |  |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|

### Reference Books

| Sr. No. | Name of the Author                            | Title of the Book                  | Year Edition            | Publisher Company         |
|---------|-----------------------------------------------|------------------------------------|-------------------------|---------------------------|
| 1       | Vijay Kotu and Bela Deshpande                 | Data Science Concepts and Practice | 2 <sup>nd</sup> Edition | Morgan Kaufmann Publisher |
| 2       | Field Cady                                    | The Data Science Handbook          | 1 <sup>st</sup> Edition | John Wiley & Sons         |
| 3       | Chun-houh Chen, Wolfgang Härdle, Antony Unwin | Handbook of Data Visualization     | 1 <sup>st</sup> Edition | Springer                  |

### Online Resources

| Online Resources No. | Web site address                                                                                                                                |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                    | ( <a href="https://www.kaggle.com/burnoutminer/heightsand-weights-dataset">https://www.kaggle.com/burnoutminer/heightsand-weights-dataset</a> ) |
| 2                    | <a href="https://archive.ics.uci.edu/ml/datasets/iris">https://archive.ics.uci.edu/ml/datasets/iris</a>                                         |

### MOOCs:

| Resources No. | Web site address                                       |
|---------------|--------------------------------------------------------|
| 1             | NPTEL/ Swayam                                          |
| 2             | <a href="http://www.edx.com">www.edx.com</a>           |
| 3             | <a href="http://www.coursera.com">www.coursera.com</a> |

| Programme: BCA CBCS – Revised Syllabus w.e.f – 2022-2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                              |                       |                   |                                                        |                                                  |                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------|--------------------------------------------------------|--------------------------------------------------|------------------------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Course Code                                                                                                                                                                                                                                                                                                                                                  | Course Title          |                   |                                                        |                                                  |                        |
| VI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 607                                                                                                                                                                                                                                                                                                                                                          | Digital Marketing     |                   |                                                        |                                                  |                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Prepared by                                                                                                                                                                                                                                                                                                                                                  | Dr.Pratap Desai       |                   |                                                        |                                                  |                        |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Credits                                                                                                                                                                                                                                                                                                                                                      | Evaluation            | Marks             |                                                        |                                                  |                        |
| SEC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 02                                                                                                                                                                                                                                                                                                                                                           | IE                    | 50                |                                                        |                                                  |                        |
| Course Objectives:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                              |                       |                   |                                                        |                                                  |                        |
| <div>1. Gain a comprehensive understanding of the core concepts and channels of digital marketing and its strategic significance in contemporary business.</div> <div>2. Develop practical skills in search engine optimization (SEO) to optimize websites, conduct keyword research, and implement on-page and off-page strategies.</div> <div>3. Learn to create and implement engaging social media marketing strategies, including content creation, audience engagement, and effective use of social media advertising.</div> <div>4. Acquire proficiency in using digital marketing analytics tools to interpret data, measure campaign success, and make data-driven decisions for optimization.</div> <div>5. Learn the planning and execution of digital advertising campaigns across platforms like Google Ads and Facebook Ads,</div> |                                                                                                                                                                                                                                                                                                                                                              |                       |                   |                                                        |                                                  |                        |
| Course Learning Outcomes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                              |                       |                   |                                                        |                                                  |                        |
| <b>CO 1.</b> Students will demonstrate a comprehensive understanding of the fundamental concepts, principles, and components of digital marketing,.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                              |                       |                   |                                                        |                                                  |                        |
| <b>CO 2.</b> Students will develop the ability to analyze and interpret digital marketing data using analytical tools and metrics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                              |                       |                   |                                                        |                                                  |                        |
| <b>CO 3.</b> Students will be proficient in developing and executing content marketing strategies and will demonstrate the skills needed to create compelling and relevant content for various digital platforms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                              |                       |                   |                                                        |                                                  |                        |
| <b>CO 4.</b> Students will acquire expertise in utilizing social media platforms for marketing purposes and will create and execute social media campaigns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                              |                       |                   |                                                        |                                                  |                        |
| <b>CO 5.</b> Students will be capable of planning, designing, and executing digital advertising campaigns across various channels such as Google Ads, Facebook Ads, and other display networks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                              |                       |                   |                                                        |                                                  |                        |
| Unit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Contents                                                                                                                                                                                                                                                                                                                                                     | Sessi<br>ons<br>(Hrs) | COs<br>Nu<br>mber | Teaching<br>Methodology                                | Cognition<br>Level                               | Evaluation<br>Tools    |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Digital Marketing Fundamentals</b><br>1.Definition and scope of digital marketing<br>2.Historical perspective and evolution3.Impact on traditional marketing 4.Major digital marketing channels (SEO, SEM, SMM, Email Marketing)<br>5.Comparative analysis of channels<br>6.Case studies of successful digital campaigns<br>7. Digital Marketing Strategy | 9                     | CO1,<br>CO2       | Lectures,<br>Experts<br>form<br>Industry<br>Case study | Understan<br>ding<br>Remember<br>ing<br>Planning | Quiz<br><br>Class test |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |            |                                                                                           |                                        |                                                                                    |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------------|-------------------------------------------------------------------------------------------|----------------------------------------|------------------------------------------------------------------------------------|
|   | 8. Setting objectives and goals<br>9. Target audience identification<br>10. Developing a digital marketing plan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |            |                                                                                           |                                        |                                                                                    |
| 2 | <b>Search Engine Optimization (SEO) and Search Engine Marketing (SEM)</b><br>1. Understanding search engines and algorithms<br>2. On-page and off-page optimization techniques<br>3. SEO best practices<br>4. Using tools like Google Analytics and Search Console<br>5. Keyword research and analysis<br>6. Monitoring website performance<br>7. Overview of search engine marketing<br>8. Basics of pay-per-click advertising<br>9. Campaign setup and management<br>Keyword selection and bidding strategies<br>10. Ad copywriting and design<br>11. Budgeting and ROI measurement                                                             | 10 | CO3        | Lectures<br>Case Studies<br>Group Discussion<br>DM Plan Development                       | Understanding<br>Implying<br>Analysing | Class Test<br>Online Quiz<br>Group Discussion                                      |
| 3 | <b>Social Media Marketing (SMM), Email Marketing, and Analytics</b><br>1. Overview of major social media platforms<br>2. Building a social media strategy<br>3. Creating engaging content<br>4. Visual storytelling and multimedia strategies<br>5. Social media scheduling and management tools<br>6. Paid advertising on social platforms<br>Analytics and performance measurement<br>7. Email Marketing, Content Marketing, and Analytics<br>8. Building email lists and segmentation<br>9. Designing effective email campaigns<br>10. Automation and personalization<br>11. Content strategy and planning<br>12. Measurement and optimization | 12 | CO4<br>CO5 | Lectures<br>Case studies<br>Presentation<br>Evaluation<br>Field Visits<br>Content Writing | Creating<br>Evaluating                 | Online Tests<br>Internship<br>Dummy Campaigns<br>Peer Review<br>Digital Assessment |

|                                                                                                                                                               |  |  |  |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
| 13. Performance Measurement<br>Importance of data-driven decision-making<br>14. Key metrics in digital marketing<br>Analyzing and interpreting analytics data |  |  |  |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|

#### Reference Books:

| Sr. No. | Name of the Author                                 | Title of the Book                | Year Edition:           | Publisher Company       |
|---------|----------------------------------------------------|----------------------------------|-------------------------|-------------------------|
| 1       | Mathur, Vibha, Arora, Saloni                       | Digital Marketing                |                         | PHI Learning            |
| 2       | Vandana Ahuja                                      | Digital Marketing                | 1 <sup>st</sup> Edition | Oxford University Press |
| 3       | Dr Tanvi Gupta<br>Dr Smita Mishra Ms Kaushi Katyal | A text book on Digital Marketing | 2nd Edition             | Puffins Publishers      |
| 4       | Seema Gupta                                        | Digital marketing                | 3 <sup>rd</sup> Edition | Mc Graw Hill            |

#### Online Resources:

| Online Resources | Website address                                                                                                                                                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                | <a href="https://india.oup.com/product/digital-marketing-2e-9789354972478?">https://india.oup.com/product/digital-marketing-2e-9789354972478?</a>                                           |
| 2                | <a href="https://kamarajcollege.ac.in/wp-content/uploads/Core-14-Digital-Marketing.pdf">https://kamarajcollege.ac.in/wp-content/uploads/Core-14-Digital-Marketing.pdf</a>                   |
| 3                | <a href="https://tech-vismera.myinstamojo.com/product/2868999/digital-marketing-study-material/">https://tech-vismera.myinstamojo.com/product/2868999/digital-marketing-study-material/</a> |

#### MOOC:

| MOOCS | Website address                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <a href="https://ugcmoocs.inflibnet.ac.in/index.php/courses/view_ug/269">https://ugcmoocs.inflibnet.ac.in/index.php/courses/view_ug/269</a>                               |
| 2     | <a href="https://soravjain.com/digital-marketing-course-for-free/">https://soravjain.com/digital-marketing-course-for-free/</a>                                           |
| 3     | <a href="https://www.socialbeat.in/top-free-digital-marketing-courses-online-in-india/">https://www.socialbeat.in/top-free-digital-marketing-courses-online-in-india/</a> |
| 4     | <a href="https://onlinecourses.swayam2.ac.in/ugc19_hs26/preview">https://onlinecourses.swayam2.ac.in/ugc19_hs26/preview</a>                                               |

| Programme:BCA CBCS – Revised Syllabus w.e.f. - Year 2022 – 23                                                                                                                                                                                                                                                                                                                  |             |                |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                       | CourseCode  | Course         |       |
| VI                                                                                                                                                                                                                                                                                                                                                                             | 608         | Indian Culture |       |
|                                                                                                                                                                                                                                                                                                                                                                                | Prepared by | Dr Mona Sinha  |       |
| Type                                                                                                                                                                                                                                                                                                                                                                           | Credits     | Evaluation     | Marks |
| AEC                                                                                                                                                                                                                                                                                                                                                                            | 2           | IE             | 100   |
| <b>CourseObjectives:</b>                                                                                                                                                                                                                                                                                                                                                       |             |                |       |
| <ul style="list-style-type: none"> <li>To acquaint students with the Cultural History of India</li> <li>To Understand various phases in and the process of the evolution of Indian Culture</li> <li>Review of the Theoretical framework of the evolution of Indian Languages and literature</li> </ul>                                                                         |             |                |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                        |             |                |       |
| <p><b>CO1:</b> Understand various phases in the evolution of Indian Culture and appreciate the glorious Past and achievements of Indians.</p> <p><b>CO2:</b> Students should know about Vedas, Indian philosophy and Religion.</p> <p><b>CO3:</b> It allows students from a variety of disciplines to gain a comprehensive grasp of the core principles of Indian Culture.</p> |             |                |       |

| Unit | Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Sessions (Hrs) | COs Number | Teaching Methodology                         | Cognition Level | Evaluation Tools |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|----------------------------------------------|-----------------|------------------|
| 1    | <b>Indian Culture: An Introduction</b> 1. Characteristics of Indian culture, Significance of Geography on Indian Culture.<br>2. Society in India through ages- Ancient period- Varna and Jati, family and marriage in India, position of women in ancient India, Contemporary period; caste system and communalism.<br>3. Religion and Philosophy in India: Ancient Period: Pre-Vedic and Vedic Religion, Buddhism and Jainism, Indian philosophy – Vedanta and Mimamsa school of Philosophy. | 8              | CO1        | lectures, ppts                               | Understand      | quiz             |
| 2    | <b>Indian Literature</b> 1. Short History of the Sanskrit literature: The Vedas, The Brahmanas and Upanishads & Sutras, Epics: Ramayana and Mahabharata & Puranas.                                                                                                                                                                                                                                                                                                                            | 8              | CO2        | lectures, ppts, stories, class presentations | Analyze         | class discussion |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |     |                                      |         |               |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|--------------------------------------|---------|---------------|
|   | 2. History of Buddhist and Jain Literature in Pali, Prakrit and Sanskrit, Sangama literature & Odia literature.                                                                                                                                                                                                                                                                                                                       |   |     |                                      |         |               |
| 3 | <b>A Brief History of Indian Arts and Architecture</b> 1. Indian Art & Architecture: Gandhara School and Mathura School of Art; Hindu Temple Architecture, Buddhist Architecture, Medieval Architecture and Colonial Architecture.<br><b>2.Performing Arts:</b> Divisions of Indian classical music: Hindustani and Carnatic, Dances of India: Various Dance forms: Classical and Regional, Rise of modern theatre and Indian cinema. | 6 | CO3 | Lectures with PPTs<br>Flip Classroom | Analyze | class test    |
| 4 | Field Visit is Compulsory to one of the Following Sites. This should be followed by the submission of the Report on the given assignment.<br><b>Field Visits</b><br>1. 'Discovery of India' Exhibition at Nehru Centre, Worli, Mumbai<br>2. Deccan College PGRI Deemed University Museum, Pune.<br>3. Chatrapati Shivaji Maharaj Vastu Sangrahalaya, Mumbai.                                                                          | 8 | CO3 | Demo                                 | Analyze | Presentat ion |

#### Reference Books:

| Sr. No. | Name of the Author                          | Title of the Book                       | Year Edition | Publisher Company |
|---------|---------------------------------------------|-----------------------------------------|--------------|-------------------|
| 1       | J.L.Mehta, Sarita mehta                     | History of Ancient India                | 2012         |                   |
| 2       | Shastri K. A. Nilakanth                     | History of India Part I – Ancient India |              |                   |
| 3       | R.C.Majumdar, H.C. Raychaudhari, Kalikinkar | An Advanced History of India            | 2020         |                   |

|   |               |                                                            |      |  |
|---|---------------|------------------------------------------------------------|------|--|
| 4 | Kosambi D. D. | The culture and civilization of ancient India              | 1975 |  |
| 5 | Kosambi D. D. | An introduction to study of Indian History 1975            | 1975 |  |
| 6 | Sharma R. S.  | Aspect of political ideas and institution in ancient India | 1959 |  |

**Online Resources:**

| Online ResourcesNo | Website address                                                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                  | <a href="https://www.researchgate.net/publication/339726396_A_Brief_History_of_India">https://www.researchgate.net/publication/339726396_A_Brief_History_of_India</a> |
| 2                  | <a href="https://www.pdfdrive.com/indian-history-books.html">https://www.pdfdrive.com/indian-history-books.html</a>                                                   |

**MOOCs:**

| Resources No | Web site address                                                                                        |
|--------------|---------------------------------------------------------------------------------------------------------|
| 1            | <a href="https://www.edx.org/course/natural-disasters">https://www.edx.org/course/natural-disasters</a> |
| 2            | <a href="https://swayam.gov.in/">https://swayam.gov.in/</a>                                             |
| 3            | <a href="https://www.coursera.org/">https://www.coursera.org/</a>                                       |
| 4            | <a href="https://nptel.ac.in/">https://nptel.ac.in/</a>                                                 |

| <b>Programme: BCA CBCS Revised Syllabus w.e.f.-Year 2022–2023</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                    |                                              |              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----------------------------------------------|--------------|
| <b>Semester</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Course Code</b> | <b>Course Title</b>                          |              |
| VII                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 701                | Artificial Intelligence and Machine Learning |              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Prepared by        | Dr.M.K.Patil                                 |              |
| <b>Type</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Credits</b>     | <b>Evaluation</b>                            | <b>Marks</b> |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3                  | UE: IE                                       | 60:40        |
| <b>Course Objectives</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                    |                                              |              |
| <ul style="list-style-type: none"> <li>The aim of the Artificial Intelligence &amp; Machine Learning course is to prepare students for a career in computer science &amp; and engineering where knowledge of AI &amp; ML techniques leads to the advancement of research and technology.</li> </ul>                                                                                                                                                                                                                                                                                                                         |                    |                                              |              |
| <b>Course Outcome</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                    |                                              |              |
| <p><b>CO1:</b> Demonstrate a fundamental understanding of artificial intelligence (AI) and expert systems.</p> <p><b>CO2:</b> Apply basic principles of AI in solutions that require problem-solving, inference, perception, knowledge representation, and learning.</p> <p><b>CO3:</b> Demonstrate proficiency in applying scientific methods to models of machine learning.</p> <p><b>CO4:</b> Discuss the basics of ANN and different optimization techniques.</p> <p><b>CO5:</b> Design and Concrete implementations of various machine learning algorithms to solve a given problem using languages such as Python</p> |                    |                                              |              |

| <b>Unit</b>                           | <b>Contents</b>                                                                                                                                                                                                                                                              | <b>Sessions (Hrs.)</b> | <b>COs Number</b> | <b>Teaching Methodology</b>                   | <b>Cognition Level</b>      | <b>Evaluation Tools</b>                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------|-----------------------------------------------|-----------------------------|-------------------------------------------------|
| <b>Overview and Search Techniques</b> | Introduction to AI, Problem Solving, State space search, Blind search: Depth-first search, Breadth-first search, Informed Search: Heuristic function, Hill climbing search, best-first search, A* & AO* Search, Constraint satisfaction, Mini-Max search, Alpha-beta pruning | 10                     | CO1<br>CO2        | Classroom Teaching, Presentations, Video Demo | Understand, apply (Analyze) | Quiz<br>End Term<br>Internals:<br>Short Answers |

|                                      |                                                                                                                                                                                                                                                                                                              |    |     |                                               |                     |                                         |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|-----------------------------------------------|---------------------|-----------------------------------------|
| <b>Knowledge Representation (KR)</b> | Introduction to KR, Predicate logic, Inference rule & and theorem proving, forward chaining, backward chaining, resolution; Propositional knowledge, Rule-Based Systems, Forward reasoning: Conflict resolution, backward reasoning: Structured KR: Semantic Net, slots, inheritance, Conceptual Dependency. | 10 | CO2 | Classroom Teaching, Presentations, Case study | Apply (Analyze )    | Case Study, End Term: Applied Questions |
| <b>Handling uncertainty</b>          | Source of uncertainty, Probabilistic                                                                                                                                                                                                                                                                         | 6  | CO3 | Application Demo, Use of Theorem              | Understand, Analyze | Case Study with Presentations End-Term  |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |            |                             |                 |                                                              |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------------|-----------------------------|-----------------|--------------------------------------------------------------|
|                         | inference, Bayes' theorem, Limitation of naïve Bayesian system, Bayesian Belief Network (BBN)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |            |                             |                 | Exams: Case-based Questions/Applied Questions                |
| <b>Machine Learning</b> | <p>Machine learning, Terminologies, Challenges in ML, Application of ML.</p> <p>Types of machine learning: supervised, unsupervised, semi-supervised learning.</p> <p>Decision Trees and Issues in Decision Tree, Clustering (K-means, Hierarchical, etc), Dimensionality reduction</p> <p>Linear Regression (with one variable and multiple variables)</p> <p>Discriminative Models: Least Square Regression,</p> <p>Gradient Descent Algorithm, Univariate and Multivariate Linear Regression, Prediction Model, probabilistic interpretation, Regularization, Logistic regression, multi-class classification, Support Vector Machines- Large margin classifiers, Nonlinear SVM</p> | 10 | CO3<br>CO5 | Practical Demo using Python | Apply, Evaluate | <p>Group Activity</p> <p>End Term Exam:<br/>Lab Exercise</p> |

|                                   |                                                                                                                      |   |     |                                               |                  |                                                        |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------|---|-----|-----------------------------------------------|------------------|--------------------------------------------------------|
| <b>Artificial Neural Networks</b> | Introduction, Artificial Neurons, Perceptron, Multilayer Networks, Back-propagation Rule back-propagation Algorithm- | 9 | CO4 | Classroom Teaching, Presentations, Video Demo | Create, Evaluate | Case Presentation Activity<br>End Term: Theory Applied |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------|---|-----|-----------------------------------------------|------------------|--------------------------------------------------------|

### Reference Book:

| Sr. No. | Name of the Author                     | Title of the Book                                                         | Publisher Company                                                                                                         |
|---------|----------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 1       | Elaine Rich and Kevin Knight           | Artificial Intelligence                                                   | Tata McGraw Hill                                                                                                          |
| 2       | Shai Shalev-Shwartz and Shai Ben-David | Understanding Machine Learning                                            | Cambridge University Press                                                                                                |
| 3       | B. Yegnanarayana                       | Artificial Neural Network                                                 | CRC Press, Taylor& Francis group,2010                                                                                     |
| 4       | Tom Mitchell                           | Machine Learning                                                          | Tata Mc Graw Hill edition,2010                                                                                            |
| 5       | E. Alpaydin                            | Introduction to Machine Learning”,                                        | PHI, 2005.                                                                                                                |
| 6       | Christopher M. Bishop                  | Pattern Recognition and Machine Learning (Springer)                       |                                                                                                                           |
| 7       | Dan W. Patterson,                      | Introduction to Artificial Intelligence and Expert Systems                | Prentice Hall of India                                                                                                    |
| 8       | Andrew Ng                              | Machine learning yearning                                                 | <a href="https://www.deeplearning.ai/machine-learning-yearning">https://www.deeplearning.ai/machine-learning-yearning</a> |
| 9       | Aurolien Geron                         | Hands-On Machine Learning with Scikit-Learn and TensorFlow                | Shroff/O’Reilly”, 2017                                                                                                    |
| 10      | Andreas Muller and Sarah Guido         | Introduction to Machine Learning with Python: A Guide for Data Scientists | Shroff/O’Reilly, 2016                                                                                                     |

| Programme: BCA CBCS –Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                                                                                                                                                                                                 |             |                                   |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                        | Course Code | Course Title                      |       |
| VII                                                                                                                                                                                                                                                                                                                                                                                                                             | 702         | Object Oriented Analysis & Design |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                 | Prepared by | Dr.Swati Desai                    |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                            | Credits     | Evaluation                        | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                             | 3           | UE:IE                             | 60:40 |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                       |             |                                   |       |
| <b>Course Objective :</b> <ol style="list-style-type: none"> <li>1. To understand system development through object oriented techniques.</li> <li>2. Students should be able to apply object oriented concepts and UML diagrams to the defined problem.</li> <li>3. Students should be able to understand requirements of the user.</li> <li>4. Students should be able to evaluate design of the existing software.</li> </ol> |             |                                   |       |
| <b>Course Outcomes :</b>                                                                                                                                                                                                                                                                                                                                                                                                        |             |                                   |       |
| At the end of course students will know – <p><b>CO1:</b> Various steps carried out in development of software.<br/> <b>CO2:</b> Object oriented concepts and UML diagrams to the defined problem<br/> <b>CO3:</b> How to analyze requirements of the user and convert to functionalities of the software.<br/> <b>CO4:</b> How to design their own software.</p>                                                                |             |                                   |       |

| Unit | Contents                                                                                                                                                                                                                                                                                                                                                  | Sessions (Hrs) | COs Number | Teaching Methodology               | Cognition Level | Evaluation Tools |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|------------------------------------|-----------------|------------------|
| 1    | Object Oriented Concepts, Modeling and UML:<br><br>What is Object Orientation : (Introduction to class, object, inheritance, polymorphism),<br>Model : Introduction of Modeling, Object Oriented Modeling , Object oriented system development:<br>Function/data methods, Object oriented analysis, Object oriented construction, Object oriented testing | 6              | CO1        | Lecture with Ppts, Q/A, Discussion | Understanding   | Assignment       |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |     |                                       |                                     |                                                                                |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|---------------------------------------|-------------------------------------|--------------------------------------------------------------------------------|
| 2 | <p>Iterative Development and UML:</p> <p>Understanding requirements, Rational Unified process &amp; RUP Phases – Inception, Elaboration, Construction, Transition</p> <p>UML : Designing Tool for OOAD : Introduction to UML, Overview of UML, Conceptual Model of UML, Diagrams in UML, Advantages of UML</p> <p>Behavioral Modeling</p> <p>Use Case Diagram : Realization of Use Cases, Finding Actors, Defining Relations among Use case, Writing Use Cases, Activity Diagram</p> | 8 | CO2 | Lecture with Ppts, Demo, Lab Sessions | Understanding, Analyzing & creating | <p>Theory&amp; Practical assignments/scenario to design</p> <p>Use of Tool</p> |
| 3 | <p>Basic and Advanced Structural Modeling</p> <p>Class Diagram : Identifying the elements of an object model, Identifying classes and objects, Specifying the attributes, Defining operations, Finalizing the object definition, Advanced class Modelling, Interface, Types and Roles</p> <p>Diagrams Based on Classes : State Chart Diagram, Package Diagram, Object Diagram</p>                                                                                                    | 8 | CO2 | Lecture with Ppts, Demo               | Understanding, Analyzing & creating | <p>Theory &amp; Practical assignments</p> <p>Use of Tool</p>                   |
| 4 | <p>Interaction Modelling :</p> <p>Introduction to Interaction Diagrams, Need of Interaction Diagrams, Interaction Diagrams, Collaboration Diagram, Sequence Diagram</p>                                                                                                                                                                                                                                                                                                              | 7 | CO2 | Lecture with Ppts, Demo               | Understanding, Analyzing & creating | <p>Theory &amp; Practical assignments</p> <p>Use of Tool</p>                   |
| 5 | <p>Architectural Modeling</p> <p>Component Diagram: Need of Component Diagram, Realization of Components, Relating Components.</p> <p>Deployment Diagram : Purpose of deployment diagram, Architecture of System, Different Architectures used for System, Representing Architecture using Deployment Diagram</p>                                                                                                                                                                    | 7 | CO4 | Lecture with Ppts, Demo               | Understanding, Analyzing & creating | <p>Theory &amp; Practical assignments</p> <p>Use of Tool</p>                   |
| 6 | <p>Object Oriented Programming Styles</p> <p>Object Oriented Style with reference to Reusability and</p>                                                                                                                                                                                                                                                                                                                                                                             | 9 | CO3 |                                       | Understanding, Analyzing &          | <p>Theory &amp; Practical assignments</p>                                      |

|  |                                                                                                                                                                                                                            |  |  |  |          |                |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|----------|----------------|
|  | Extensibility, Robustness, 3<br>Programming in the Large,<br>Discussion on case Studies e.g.<br>Library Management System,<br>Hospital Management System, .<br>Online Shopping, Nukari.com<br>website, Matrimonial website |  |  |  | creating | Use of<br>Tool |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|----------|----------------|

## Reference Books

| Sr. No. | Name of the Author                                 | Title of the Book                                                      | Year Edition | Publisher Company                  |
|---------|----------------------------------------------------|------------------------------------------------------------------------|--------------|------------------------------------|
| 1       | Grady Booch, James<br>Raumbaugh, Ivar<br>Jacobson. | The Unified Modeling<br>Language User Guide                            | -            | Addison-<br>Wesley<br>professional |
| 2       | Ivar Jacobson                                      | Object Oriented<br>Software Engineering<br>Use case driven<br>approach | -            | Pearson                            |
| 3       | Martin Fowler                                      | UML Distilled                                                          | -            | Addison-<br>Wesley<br>Professional |

## Online Resources

| Online Resources No. | Web site address                                                                                                                                                  |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.tutorialspoint.com">https://www.tutorialspoint.com</a>                                                                                       |
| 2                    | <a href="https://www.javatpoint.com/uml">https://www.javatpoint.com/uml</a>                                                                                       |
| 3                    | <a href="https://www.guru99.com/uml-tutorial.html">https://www.guru99.com/uml-tutorial.html</a>                                                                   |
| 4                    | <a href="https://www.geeksforgeeks.org/unified-modeling-language-uml-introduction/">https://www.geeksforgeeks.org/unified-modeling-language-uml-introduction/</a> |

## MOOCs:

| Resources No. | Web site address |
|---------------|------------------|
| 1             | Swayam           |
| 2             | NPTEL            |

| Programme: BCA CBCS –Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                             |              |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|--------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Course Code                                 | Course Title |       |
| VII                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 704-Mobile Application Development with Lab |              |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Prepared by Dr. Rahul Jadhav                |              |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Credits                                     | Evaluation   | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2                                           | IE           | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                             |              |       |
| <ul style="list-style-type: none"><li>• To understand architecture of mobile application using Android</li><li>• To get acquainted with life cycle of android application and its component</li><li>• To develop proficiency in creating Mobile based applications using the JavaProgramming Language.</li><li>• To develop application using android with data handling(database access)</li></ul>                                                                                                                                                                                     |                                             |              |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                             |              |       |
| At the end of this course, student should be able to understand<br>CO1: State features of Android, components of android architecture and android application.<br>CO2: Describe components of android application along with life cycle of activity, intent, fragment etc.<br>CO3: Apply android knowledge to design and develop mobile applications<br>CO4: Analyse the use of Intent, Fragment, content providers and sensors.<br>CO5: Evaluate use of various component of android application.<br>CO6: Create and publish Android application using various component and database. |                                             |              |       |

| Unit | Contents                                                                                                                                                                                                                                                                 | Sessions (Hrs) | COs Number | Teaching Methodology | CognitionLevel | EvaluationTools |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|----------------------|----------------|-----------------|
| 1    | <b>Introduction to Android</b><br>Android OS, evolutionand advantages of android, Dalvik VirtualMachine, Features of Android, API Level Introduction, Linux Kernel, Libraries, Android Libraries, Android Application Framework, Introduction to Application components. | 4              | CO1 , CO2  | Lecture withPPT      | Understand     | Quiz            |

|   |                                                                                                                                                                                                                                              |   |     |                                        |                |      |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|----------------------------------------|----------------|------|
| 2 | Android Studio<br>Downloading and<br>installing Android<br>Studio, Android<br>StudioOverview,<br>Creating a first<br>project<br>(HelloWorld),<br>Understanding<br>Project internals and<br>configuration files.<br>Creating and<br>Launching | 4 | CO2 | Lecture<br>withPPT,<br>Hands<br>OnDemo | Understan<br>d | Quiz |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|----------------------------------------|----------------|------|

|   |                                                                                                                                                                                                                                                                                                                                                                                                           |   |           |                                 |                            |                                          |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------|---------------------------------|----------------------------|------------------------------------------|
|   | emulator(Android Virtual Device), Editing emulator settings, Running first android application on emulator<br><b>Practical:</b>                                                                                                                                                                                                                                                                           |   |           |                                 |                            |                                          |
| 3 | <b>Working with Activities and Layouts</b><br>Android Activities<br>Introduction, Life Cycle, Working with Activities, handling events, making use of resource files, concept of intents and using it to launch new activities.<br><br>UI Layouts, Types of Layout, Configuration of Layouts, View Identification, UI Controls, Event Handling, understanding and using fragments, Making use of adapters | 4 | CO 3      | Lecture with PPT, Hands On Demo | Analyze                    | Class Test, Lab assignment, MidTerm Exam |
| 4 | <b>Content Providers:</b><br>Working with Shared Preferences, storing and retrieving shared key-value pairs. Store data using SQLite database, Content Providers, Content Resolver, Loader                                                                                                                                                                                                                | 4 | CO3 , CO6 | Lecture with PPT, Hands On Demo | Evaluate , Create          | Lab Assignment                           |
| 5 | <b>Intents and Intent Filters</b><br>Understanding the Intents, Android Intent Messaging via Intent Objects, Intent Resolution, Intent Filters, Explicit Intents, Implicit Intents, Working with Intents, Using Intents with Activities, Android Services, Using Intents with Broadcast Receivers                                                                                                         | 4 | CO2 , CO4 | Lecture with PPT, Hands On Demo | Evaluate , analyze, Create | Lab Assignment                           |

|   |                                                                                                                                                                         |   |     |                               |                            |                                           |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|-------------------------------|----------------------------|-------------------------------------------|
| 6 | <b>Sensor, Location and Maps</b><br>Sensor Basic, Motion and Position<br>Sensors, Using Orientation and Accelerometer<br>sensors Using Location Based Services, Finding | 5 | CO5 | Lecture withPPT, Hands OnDemo | Evaluate , analyze, Create | Class test, End Term Exam, lab Assignment |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|-------------------------------|----------------------------|-------------------------------------------|

|   |                                                                                                                                                                                                                                                                     |   |     |                                 |                           |                             |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|---------------------------------|---------------------------|-----------------------------|
|   | current location and listening for changes in location , Proximity alerts, Working with Google Maps, Showing Google map in an Activity, Map Overlays, Itemized overlays, Geocoder, Displaying route on map                                                          |   |     |                                 |                           |                             |
| 7 | <b>Performance Improvement and Publishing</b><br>Performance Parameters, Profiling Tools, Rendering and Layout, Garbage Collection and Memory Leaks, Best Practices. Preparing for publishing ,Signing and preparing the graphics , ublishing to the Android Market | 5 | CO6 | Lecture with PPT, Hands On Demo | Evaluate, analyze, Create | End Term Exam: Mini Project |

### Practical:

Following is the sample practical assignments. Student has to identify the similar problems and solve during the practical sessions. Student has to develop minor project based on above syllabus.

### Sample questions for Practical

|    |                                                                                                                                                                                                                                                                                                                                                                              |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Create "hello world" application to display "hello world" in the middle of the screen in the emulator as well as android phone                                                                                                                                                                                                                                               |
| 2  | Create an android app to display various android lifecycle phases.                                                                                                                                                                                                                                                                                                           |
| 3  | Create an android app with first activity having edittext and send button. On click of send button, use explicit intent to send the text within edittext to a second activity and displayed within textview                                                                                                                                                                  |
| 4  | Create an android app with first activity having edittext and send button. On click of send button, use implicit intent that uses send action, and let user select app from app chooser and navigate to that application.                                                                                                                                                    |
| 5  | Create a calculator app that performs addition, subtraction, division and multilpication operation on numbers.                                                                                                                                                                                                                                                               |
| 6  | Create a spinner application with strings taken from resource directory res/values/strings.xml and on changing the spinner value, image will change. Image is saved in the drawable directory                                                                                                                                                                                |
| 7  | Create an app that uses radiobutton group which calculates discount on shopping bill amount. Use ediitext to enter bill amount and select one of three radio buttons to determine a discount for 10, 15, or 20 percent.the discount is calculated upon selection of one of the buttons and displayed in a textview control.                                                  |
| 8  | Create an app that uses radiobutton group of all courses in your college. On selecting one of the buttons, the TIC of that course should be displayed in a textview control at the bottom of the screen.                                                                                                                                                                     |
| 9  | Create an application that uses checkbox for construction of a shopping list so the user can check off items as they are picked up. The checked items should be displayed in a textview control.                                                                                                                                                                             |
| 10 | Create a login application to verify username and password. Create a registration page to register a user. On successful login, "welcome user" should appear as a pop-up message.                                                                                                                                                                                            |
| 11 | Create a login application to verify username and password. On successful login, redirect to another activity that has a textview to display "welcome user" with logout button. On click of logout button, a dialog should appear with ok and cancel buttons. On click of oK button, go back to the login activity and on click of cancel button, stay on the same activity. |

|    |                                                                                                                                                                                                                                 |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12 | Create a menu with 5 options. The selected option should appear in the textbox.                                                                                                                                                 |
| 13 | Use linear layout to create a simple application that will take the contents of a predefined textview and use a button to cause the application to take that text, convert it to uppercase, and display it in an edittext field |
| 14 | When working with edittext controls on the screen, create an application to respond to a particular keystroke rather than requiring the user to touch a button using keyevent.                                                  |
| 15 | Create an application that uses tablelayout with textview, edittext and buttons. Also, create ur own styles.xml file within res/values directory, to style text of textview control.                                            |
| 16 | Create an application to perform the operations of create, insert, delete, view and update, using sqlite database.                                                                                                              |
| 17 | Create an app to display 3 button controls vertically aligned. On selecting a button, the color of the screen will change.                                                                                                      |

### Reference Books

| Sr. No. | Name of the Author                 | Title of the Book                                                 | Year Edition            | Publisher Company           |
|---------|------------------------------------|-------------------------------------------------------------------|-------------------------|-----------------------------|
| 1       | Barry A. Burd                      | Android Application Development All-in-One For Dummies            | August 2015             | For Dummies                 |
| 2       | Bryan Sills, Brian Gardner, et al  | Android Programming: The Big Nerd Ranch Guide Programming Android | 5 <sup>th</sup> edition | Addison-Wesley Professional |
| 3.      | J F DiMarzio                       | Beginning Android Programming with Android Studio                 | 4th Edition 2016        | Wiley India Pvt Ltd         |
| 4.      | Dawn Griffiths and David Griffiths | Head First Android Development: A Brain-Friendly Guide            | 2nd Edition, 2017       | Shroff/O'Reilly             |

### MOOCs:

| Resources No. | Web site address                                                                                            |
|---------------|-------------------------------------------------------------------------------------------------------------|
| 1             | <a href="https://alison.com/">https://alison.com/</a>                                                       |
| 2             | <a href="https://nptel.ac.in/courses/106/106/106106147/">https://nptel.ac.in/courses/106/106/106106147/</a> |

| Programme: BCA CBCS –Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                     |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Course Code | Course Title        |       |
| VIII                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 801         | Cloud Computing     |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Prepared by | Dr. Mukund Kulkarni |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Credits     | Evaluation          | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3           | UE:IE               | 60:40 |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |                     |       |
| <ul style="list-style-type: none"> <li>Students will learn an overview of the field of Cloud Computing.</li> <li>Students will understand virtualization and its role in cloud computing.</li> <li>Students will gain hands-on experience solving relevant problems through projects that will utilize existing public cloud tools.</li> <li>Students will develop the skills needed to use cloud computing technique and will be able to create strategies for flexible and scalable cloud infrastructure.</li> </ul>                                          |             |                     |       |
| <b>Course Outcomes :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |                     |       |
| <p><b>CO1:</b> Define the key characteristics of cloud computing and recall different cloud service models (IaaS, PaaS, SaaS) and deployment models.</p> <p><b>CO2:</b> Explain the concept of virtualization and its role in cloud computing.</p> <p><b>CO3:</b> Apply security measures to address challenges in a cloud environment.</p> <p><b>CO4:</b> Analyze components of Infrastructure as a Service (IaaS) such as computing, storage, and networking.</p> <p><b>CO5:</b> Critically assess compliance and legal issues related to cloud security.</p> |             |                     |       |

| Unit | Content                                                                                                                                                                                                                            | Sessions (Hrs) | COs Number    | Teaching Methodology                    | Cognition Level           | Evaluation Tools                                   |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------|-----------------------------------------|---------------------------|----------------------------------------------------|
| 1    | <b>Cloud Computing Fundamentals</b><br>Definition of Cloud Computing, private, public and hybrid cloud. Cloud types; IaaS, PaaS, SaaS. Benefits and challenges of cloud computing, public Vs private clouds                        | 6              | CO 1          | Lecture with Ppts<br>Quiz               | Remembering               | Quiz<br><br>End Term<br>Internals:Short<br>Answers |
| 2    | <b>Virtualization And Cloud Computing</b><br>Role of virtualization in enabling the cloud; Business Agility: Benefits and challenges to Cloud architecture. Application availability, performance, security and disaster recovery; | 8              | CO 1,<br>CO 2 | Lecture with Ppts<br>Psychometric Tools | Understanding<br>Applying | Quiz<br>End Term:<br>Applied<br>Questions          |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |              |                                                         |                     |                                                                                  |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------------|---------------------------------------------------------|---------------------|----------------------------------------------------------------------------------|
|   | next generation Cloud Applications, Visualizing Virtualization, Managing Virtualization, Taking Virtualization into the Cloud                                                                                                                                                                                                                                                                                                                            |   |              |                                                         |                     |                                                                                  |
| 3 | <b>Service Oriented Architecture And The Cloud</b><br>Defining Service Oriented Architecture, Understanding the Coupling, Implementation of Service Oriented Architecture (SOA), Understanding Services in the Cloud, Serving the Business with SOA and Cloud Computing.                                                                                                                                                                                 | 8 | CO 3         | Lecture with PPTs<br>Case Study                         | Applying            | Presentations<br>End Term Exams:<br>Case based Questions/<br>Applied Questions   |
| 4 | <b>Deploying Web Services in the Cloud</b><br>Technologies and the processes required when deploying web services; Deploying a web service from inside and outside a cloud architecture, advantages and disadvantages.                                                                                                                                                                                                                                   | 5 | CO3<br>CO4   | Lectures with PPT, Tutorial and practical demonstration | Analyze             | Group Activity<br><br>End Term Exam:<br>Short case and situation based questions |
| 5 | <b>Management of Cloud Services</b><br>Reliability, availability, and security of services deployed from the cloud. Performance and scalability of services, tools and technologies used to manage cloud services deployment; Cloud Economics: Cloud Computing infrastructures available for implementing cloud-based services. Economics of choosing a Cloud platform for an organization, based on application requirements, economic constraints, and | 8 | CO3,<br>CO 4 | Lecture<br>Case Activity                                | Analyze<br>Evaluate | Case Presentation<br>Activity<br>End Term:<br>Theory Applied                     |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |              |                                      |                    |                                            |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------------|--------------------------------------|--------------------|--------------------------------------------|
|   | business needs (e.g Amazon, Microsoft and Google, Salesforce.com, Ubuntu and Redhat)                                                                                                                                                                                                                                                                                                                                                                                                           |    |              |                                      |                    |                                            |
| 6 | <b>Application Development &amp; Case Studies</b><br>Application Development Service creation environments to develop cloud-based applications. Development environments for service development; Amazon, Azure, Google App. Analysis of Case Studies when deciding to adopt cloud computing architecture. How to decide if the cloud is right for your requirements. Cloud based service, applications and development platform deployment so as to improve the total cost of ownership (TCO) | 10 | CO 4<br>CO 5 | Lectures with PPTs<br>Flip Classroom | Evaluate<br>Create | Activity<br>End Term:<br>Theory<br>Applied |

**Reference Book:**

| Sr. No. | Name of the Author                                     | Title of the Book                                       | Publisher Company                      |
|---------|--------------------------------------------------------|---------------------------------------------------------|----------------------------------------|
| 1       | Rajkumar Buyya, James Broberg and Andrzej M. Goscinski | Cloud Computing: Principles and Paradigms               | Wiley, 2011.                           |
| 2       | Kai Hwang, Geoffery C. Fox, Jack Elsevierm             | Distributed & Cloud computing                           | 2012                                   |
| 3       | John W. Rittinghouse, James E Ransome                  | Cloud Computing implementation management, and security | CRC Press, Taylor& Francis group, 2010 |

|   |                                                   |                                                        |                                |
|---|---------------------------------------------------|--------------------------------------------------------|--------------------------------|
| 4 | Anthony T. Velte, Toby J. Velte Robert Elsenpeter | Cloud Computing a practical approach                   | Tata Mc Graw Hill edition,2010 |
| 5 | George Reese,                                     | Cloud Application Architecture                         | Oreilly publishers             |
| 6 | David S. Linthicum,                               | Cloud computing and SOA convergence in your enterprise | Addison-Wesley                 |

| Programme: BCA CBCS– Revised Syllabus w.e.f.-Year 2022 –2023                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |                              |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Course Code | CourseTitle                  |       |
| VIII                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 802         | Enterprise Resource Planning |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Prepared by | Prof. Deelip Patil           |       |
| Type of Course                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Credits     | Evaluation                   | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3           | UE(60)+IE(40)                | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |                              |       |
| <p>Objectives:</p> <ul style="list-style-type: none"> <li>To provide students with a thorough understanding of the fundamental concepts, principles, and frameworks of Enterprise Resource Planning.</li> <li>To equip students with the skills necessary for successful ERP implementation in real-world organizational settings.</li> <li>To provide students with a detailed understanding of core ERP modules and related technologies that enhance organizational efficiency.</li> </ul> |             |                              |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |                              |       |
| <p>After completing the course, the students shall be able to</p> <p><b>CO1:</b> Understand concept, need and significance of ERP.</p> <p><b>CO2:</b> Apply different concept regarding ERP implementation.</p> <p><b>CO3:</b> Analyze ERP models and related technologies.</p> <p><b>CO4:</b> Describe popular products and future trends in ERP.</p>                                                                                                                                        |             |                              |       |

| Unit                                                     | Content                                                                                                                                                                                                                                                                     | Sessions | COs Number | Teaching Methodology | Cognition Level | Evaluation Tools      |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|----------------------|-----------------|-----------------------|
| <b>Foundations of Enterprise Resource Planning (ERP)</b> | Introduction to ERP, Definition and significance of ERP, Historical evolution of ERP systems, Core principles and concepts of ERP, Organizational Structure and Processes, ERP Components and Architecture, Benefits and Risks of ERP, Business Process Reengineering (BPR) | 9        | CO1        | Lecture              | Understand      | Quiz<br>Short Answers |

|                                                     |                                                                                                                                                                              |    |     |                    |          |                    |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|--------------------|----------|--------------------|
| <b>ERP Implementation Strategies and Life Cycle</b> | ERP Implementation challenges, ERP Implementation Strategies, Selection of ERP Subsystems, ERP Implementation Life Cycle, Vendor Selection and Role of Consultants           | 12 | CO2 | Lectures with PPTs | Apply    | Quiz Short Answers |
| <b>Core ERP Modules</b>                             | Financial and Accounting Module, Inventory and Supply Chain Module, Sales and Distribution Module, Production and Human Resource Module, Customer Relationship Module        | 12 | CO3 | Lectures with PPTs | Analyze  | Quiz Short Answers |
| <b>ERP Related Technologies</b>                     | Business Process Reengineering (BPR), Supply Chain Management (SCM), Customer Relationship Management (CRM), Management Information System (MIS), Role of MIS in ERP systems | 6  | CO3 | Lectures with PPTs | Remember | Quiz Short Answers |
| <b>Marketplace and Future Trends in ERP</b>         | ERP Market Dynamics, Overview of key ERP vendors: SAP AG, Oracle, JD Edwards, Emerging Trends in ERP, Critical Evaluation and Case Studies                                   | 6  | CO4 | Lectures with PPTs | Remember | Quiz Short Answers |

#### Reference Books:

| Sr. No. | Name of the Author                        | Title of the Book            | Year Edition | Publisher Company |
|---------|-------------------------------------------|------------------------------|--------------|-------------------|
| 1       | Alexis Leon                               | ERP Demystified              | 2008         | Tata McGraw Hill  |
| 2       | Vinod Kumar Grag and N.K. Venkitakrishnan | ERP- Concepts and Practice   | 2006         | PHI               |
| 3       | Sumner, M.                                | Enterprise Resource Planning | 2005         | Prentice Hall     |

#### Online Resources:

| Online Resources No. | Website address                                                                                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.tutorialspoint.com/management_concepts/enterprise_resource_planning.htm">https://www.tutorialspoint.com/management_concepts/enterprise_resource_planning.htm</a> |

**MOOCs:**

| Resources No. | Website address                                        |
|---------------|--------------------------------------------------------|
| 1             | NPTEL / Swayam                                         |
| 2             | <a href="http://www.edx.com">www.edx.com</a>           |
| 3             | <a href="http://www.coursera.com">www.coursera.com</a> |

| Programme :BCA CBCS– Revised Syllabus w.e.f.-Year2022 –2023                                                                                                                                                                                                                                                                                                                                                                                                   |             |                       |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Course Code | Course Title          |       |
| VIII                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 803         | Blockchain Technology |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Prepared by | Dr.Pratibha Jadhav    |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Credits     | Evaluation            | Marks |
| DSC                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3           | UA(60)+IE(40)         | 100   |
| <b>Pre-requisite:</b><br>Basic knowledge of cryptography, networking, distributed systems and expertise in object-oriented programming.                                                                                                                                                                                                                                                                                                                       |             |                       |       |
| <b>Course Objectives:</b><br>To make students to: <ul style="list-style-type: none"> <li>• Provide the overview of the structure and mechanisms of Blockchain.</li> <li>• Explain permissioned and decentralized Blockchain concepts.</li> <li>• Understand cryptocurrency transactions and mining Blockchain.</li> <li>• Understand the applications of Blockchain technology.</li> </ul>                                                                    |             |                       |       |
| <b>Course Outcomes:</b><br>After completing the course, the students shall be able to<br><b>CO1:</b> Understand Blockchain technologies and their components.<br><b>CO2:</b> Interpret the uses of cryptographic techniques in Blockchain.<br><b>CO3:</b> Understand and analyze the consensus mechanisms in Bitcoin.<br><b>CO4:</b> Understand and handles the smart contracts.<br><b>CO5:</b> Demonstrate the use of hyperledger fabric and its components. |             |                       |       |

| Unit                              | Content                                                                                                                                                                                                        | Sessions (in Hrs) | COs Number | Teaching Methodology | Cognition Level | Evaluation Tools      |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------|----------------------|-----------------|-----------------------|
| <b>Introduction to Blockchain</b> | Basics of blockchain, History, Uses of Blockchain, Structure of a block, Transactions, Public Ledger, Distributed Consensus. Peer to peer systems, centralized and decentralized systems, Types of blockchain. | 9                 | CO1        | Lecture with Ppt     | Understand      | Quiz<br>Short Answers |

|                                 |                                                                                                                                                                                                                                                                                            |    |     |                    |             |                               |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|--------------------|-------------|-------------------------------|
| <b>Cryptographic Primitives</b> | Basics of cryptography (Symmetric and Asymmetric) RSA algorithm<br>Cryptographic hash functions<br>– collision free, hiding, puzzle friendly (properties), Hash Chain, Hash tree-Merkle Tree, Public Key cryptography, Digital signatures.                                                 | 9  | CO2 | Lecture with Ppt   | Apply       | Quiz Short Answers            |
| <b>Bitcoin</b>                  | Basics (Structure of block, creation of coins), Double Spending, Script (FORTH), Mining Process, Objectives of consensus mechanisms, Consensus in Bitcoin – Proof of Work, Proof of Stake, Proof of Burn                                                                                   | 9  | CO3 | Lecture with Ppt   | Apply       | Quiz Short answers            |
| <b>Permissioned Blockchain</b>  | Smart Contracts, Distributed Consensus, Faults in DC, Algorithms – Paxos, RAFT, Byzantine Fault Tolerance                                                                                                                                                                                  | 8  | CO4 | Lecture with ppt   | Apply       | Quiz Short answers            |
| <b>Ethereum</b>                 | History, Architecture, Account Types, Gas, Transactions, Structure (Blocks, Transactions), Accounts, Ether, Gas, Ethereum Virtual Machine, Solidity.<br>Hyperledger Fabric: Features of hyperledger, Architecture, ordering service, Transaction Flow, Membership and Identity Management. | 10 | CO5 | Lectures with PPTs | Demonstrate | Quiz Short Answer, Case study |

#### Reference Books:

| Sr.No. | Name of the Author                                                                                                                                    | Title of the Book | Year | Publisher Company |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------|-------------------|
| 1      | Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, Steven Goldfeder, Bitcoin and Cryptocurrency Technologies, Princeton University Press |                   |      |                   |
| 2      | Don Tapscott, Alex Tapscott, Blockchain Revolution, ISBN No. 9781101980132                                                                            |                   |      |                   |

|   |                                                                                                                                                        |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | Mark Gates, Blockchain ultimate Guide to understanding Blockchain, Bitcoin, Cryptocurrencies, Smart Contracts and Future of money, Wise Fox Publishing |
| 4 | VikramDhillon, David Metcalf, Max Hooper, Blockchain Enabled Applications, Apress, ISBN No.13:978-1-4842-3081-7                                        |
| 5 | Melanie Swan, Blockchain Blueprint for a new economy, O'Reilly, First Edition, ISBN No.978-1-491-92049-7                                               |
| 6 | MayukhMukhopadhyay, Ethereum Smart Contract Development, Packt publishing, First Edition, ISBN No.978-1-78847-304-0                                    |
| 7 | Chris Dannen, Introducing Ethereum and Solidity, Apress, ISBN No.978-1-4842- 2535-6                                                                    |
| 8 | Nitin Gaur, Luc Desrosiers, Petr Novotny, Venkatraman Ramakrishna, Anthony O'Dowd, Salman A. Baset, Hands-On Blockchain with Hyperledger, Packt        |

### Online Resources:

| Online ResourcesNo. | Website address                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------|
| 1                   | <a href="https://blockexplorer.com/">https://blockexplorer.com/</a>                                           |
| 2                   | <a href="https://en.wikipedia.org/wiki/Digital_signature">https://en.wikipedia.org/wiki/Digital_signature</a> |
| 3                   | Public Key Cryptography - GlobalSign                                                                          |
| 4                   | What is Asymmetric Cryptography? Definition from SearchSecurity (techtargget.com)                             |
| 5                   | What is Blockchain Technology? A Step-by-Step Guide For Beginners (blockgeeks.com)                            |
| 6                   | The Truth About Blockchain (hbr.org)                                                                          |
| 7                   | How Does Ethereum Work? Understanding the Ethereum Network (coindesk.com)                                     |
| 8                   | A (Short) Guide to Blockchain Consensus Protocols - CoinDesk                                                  |
| 9                   | Hyperledger Fabric                                                                                            |
| 10                  | Proof of Work vs Proof of Stake: Basic Mining Guide - Blockgeeks                                              |

**MOOCs:**

| <b>Resources No.</b> | <b>Website address</b>                                 |
|----------------------|--------------------------------------------------------|
| 1                    | NPTEL / Swayam                                         |
| 2                    | <a href="http://www.edx.com">www.edx.com</a>           |
| 3                    | <a href="http://www.coursera.com">www.coursera.com</a> |

### Elective Group I – Data Analysis

| Programme: BCA CBCS– Revised Syllabus w.e.f.-Year 2022 –2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                          |                           |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Course Code              | CourseTitle               |       |
| V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Data Analysis<br>504-1-A | Data Analysis Using Excel |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Prepared by              | Dr.Kirti Mahajan          |       |
| Type of Course                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Credits                  | Evaluation                | Marks |
| DSE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3                        | 60(UE)+40(IA)             | 100   |
| <b>Course Objective:</b><br>To train the student for using the spreadsheet package MS-Excel for business applications.<br>To impart skill so finalizing data and presenting it using MS-Excel.                                                                                                                                                                                                                                                                                                                                                                                       |                          |                           |       |
| <b>Course Outcomes:</b><br>After completing the course the students shall be able to <ul style="list-style-type: none"> <li><b>CO1: Visualization:</b> Students will be able to create and manage a variety of charts and graphs in Excel, such as column, line, pie, and scatter plots, as well as work with multiple sheets and hyperlinks.</li> <li><b>CO2: Decision Making:</b> Students will develop the ability to analyze data to make informed decisions, including using functions like IF and SUMIF, and evaluating results to ensure accuracy and reliability.</li> </ul> |                          |                           |       |

| Unit | Content                                                                                                                                                                                                                                                                                                    | Sessions (Hrs) | COs Number | Teaching Methodology                          | Cognition Level | Evaluation Tools                                   |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|-----------------------------------------------|-----------------|----------------------------------------------------|
| 1    | <b>Introduction to Excel</b><br>MS excel screen elements – Tool bar, title bar, ribbon, formula bar, status bar. Moving around a Worksheet, entering and formatting (e.g. Number, Text, Date and Currency) data. Cell referencing (relative, absolute, mixed), using formulae, Use of Find, Replace, Goto. | 6              | CO 1       | Lecture with Ppts<br>Quiz<br>Excel assignment | Understand      | Quiz<br>End Term<br>Internals:<br>Short<br>Answers |
| 2    | <b>Working with Excel</b><br>Insert, delete-cells, rows, columns. Sorting (basic, custom), filtering, grouping, ungrouping data, dealing with subtotals and grand totals. Validating data,                                                                                                                 | 6              | CO 1       | Lecture with Ppts<br><br>Excel assignment     | Apply           | End Term:<br>Applied<br>Questions                  |

|   |                                                                                                                                                                                                                      |    |      |                                                          |            |                                            |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------|----------------------------------------------------------|------------|--------------------------------------------|
|   | protecting cells. Create, manage, and format pivot tables and pivot charts.                                                                                                                                          |    |      |                                                          |            |                                            |
| 3 | <b>Conditional Formatting</b><br>Once defined, it will automatically change the formats as per conditions user inputs. Work with functions to manipulate strings of text and data                                    | 6  | CO 1 | Lecture with PPTs<br>Excel assignment                    | Understand | End Term Exams:<br>Applied Questions       |
| 4 | <b>Commonly used functions</b><br>Sum, Max, Min, Average, Count, To day, Now, Datedif, Countif, CountA, CountBlank, Round, Roundup, RoundDown, ABS, Sign, Ceiling, Floor, Trim, Value, Clean, sqrt, if, sumif        | 8  | CO2  | Lectures with PPTs<br><br>Excel assignment               | Understand | End Term Exam:<br>Applied Questions        |
| 5 | <b>Data Viewing and Reviewing</b><br>Inserting comments, spell checks and changes to the worksheet data etc, Viewing data in different ways eg. Page break, normal etc                                               | 10 | CO2  | Lecture<br>Excel assignment                              | Create     | Activity<br>End Term:<br>Theory<br>Applied |
| 6 | <b>Creating and managing charts</b><br>Create and modify graphs/charts like Column, Line, Pie, Bar, Area, Scatter, 3D etc. Working with multiple sheets, hyper linking Work with sparklines. Perform Look UP tables. | 9  | CO2  | Lectures with PPTs<br>Flip Classroom<br>Excel assignment | Create     | Activity<br>End Term:<br>Theory<br>Applied |

**References (Books, Websites etc) :**

1. Albright: Data Analysis and Decision Making Using MS Excel
2. Stephen Nelson: Data Analysis For Dummies
3. Narayan Ash Sah: Data Analysis Using Microsoft Excel 1/e, Excel Books

| Programme: BCA CBCS– Revised Syllabus w.e.f.-Year 2022 –2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                          |                                   |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-----------------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Course Code              | CourseTitle                       |       |
| VI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Data Analysis<br>604-1-B | Data Analysis Using R Programming |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Prepared by              | Dr.Kirti Mahajan                  |       |
| Type of Course                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Credits                  | Evaluation                        | Marks |
| DSE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                        | 60(UE)+40(IA)                     | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                          |                                   |       |
| <b>Course Objective:</b><br>To teach the Beginners of R Programming of the master level. A variety of topics will be covered that are important for Data Analysis in order to prepare the students for real life prediction of data engineering.<br><br>To impart knowledge of the concepts related to Probability and Application on data sets. It also gives the idea how data is managed in various environments with emphasis on Predictions measures as implemented in data sets.                                                                                                                                                    |                          |                                   |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                          |                                   |       |
| After completing the course the students shall be able to<br><b>CO1 :</b> Apply Data Distribution Techniques: Students will gain knowledge about different types of data distributions including exponential, binomial, normal, and Poisson distributions. They will learn to generate random numbers and conduct Monte Carlo simulations.<br><br><b>CO2:</b> Apply Statistical Models: Students will learn to implement correlation and regression analysis, analysis of variance (ANOVA), and create complex data structures for statistical analysis. They will be able to summarize data and analyze case studies using R programming |                          |                                   |       |

| Unit | Content                                                                                                                                                                               | Sessions (Hrs) | COs Number | Teaching Methodology                               | Cognition Level | Evaluation Tools                                |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|----------------------------------------------------|-----------------|-------------------------------------------------|
| 1    | <b>Introduction of Probability</b><br>Concept, Types of Probability, Permutation and Combination concept ,Addition and Multiplication Theorem, Condition Probability, Bayes's Theorem | 8              | CO 1       | Lecture with Ppts<br>Quiz<br>Statistics assignment | Understand      | Quiz<br>End Term<br>Internals:<br>Short Answers |
| 2    | <b>Random Variable</b><br>Concept, Discrete and Continuous Random Variable, Probability density function,                                                                             | 8              | CO 1       | Lecture with Ppts<br>Statistics assignment         | Apply           | End Term:<br>Applied Questions                  |

|   |                                                                                                                                                                                                                                                                                                                                                                            |    |      |                                                                  |            |                                      |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------|------------------------------------------------------------------|------------|--------------------------------------|
|   | Mathematical Expectation and their Theorem                                                                                                                                                                                                                                                                                                                                 |    |      |                                                                  |            |                                      |
| 3 | <b>Data Distribution</b><br>Distribution, Types of Data distribution, Exponential distribution, Binomial distribution, Normal distribution, Poisson distribution, Random number generation, Monte Carlo Simulation.                                                                                                                                                        | 8  | CO 1 | Lecture with PPTs<br>Statistics assignment                       | Understand | End Term Exams: Applied Questions    |
| 4 | <b>Testing of Hypothesis</b><br>Procedure of Testing Hypothesis, Standard Error and Sampling distribution, Estimation, Student's t-distribution, Chi-Square test and goodness of fit, F-test and analysis of variance. Factor analysis.                                                                                                                                    | 10 | CO2  | Lectures with PPTs<br><br>Statistics assignment                  | Understand | End Term Exam: Applied Questions     |
| 5 | <b>Introduction to R programming language</b><br>Getting R, Managing R, Arithmetic and Matrix Operations, Introduction to Functions, Control Structures. Working with Objects and Data: Introduction to Objects, Manipulating Objects, Constructing Data Objects, types of Data items, Structure of Data items, Reading and Getting Data, Manipulating Data, Storing Data. | 6  | CO2  | Lecture<br>R programming assignment                              | Create     | Activity<br>End Term: Theory Applied |
| 6 | <b>Graphical Analysis using R</b><br>Basic Plotting, Manipulating the plotting window, Box Whisker Plots, Scatter Plots, Pair Plots, Pie Charts, Bar Charts.<br><b>Advanced R</b><br>Statistical models in R, Correlation and regression analysis, Analysis of Variance (ANOVA), creating data for complex analysis, Summarizing data, and case studies                    | 5  | CO2  | Lectures with PPTs<br>Flip Classroom<br>R programming assignment | Create     | Activity<br>End Term: Theory Applied |

#### Text Books

"Fundamentals of Statistics" Seven Edition By S.C.Gupta

#### References (Books, Websites etc) :

1. "Fundamentals of Statistics" Seven Edition By S.C.Gupta
2. "R Programming Fundamentals by KaelenMedeiras

3. “ Reinforcement Learning e-book.

4. Learning R Programming Guide on line

**Suggested MOOC :**

Please refer these websites for MOOCS:

NPTEL / Swayam [www. edx.com](http://www.edx.com), [www.coursera.com](http://www.coursera.com)

### Elective Group II – Information Security

| Programme :BCA CBCS– Revised Syllabus w.e.f.-Year2022 –2023                                                                                                                                                                                                                                                                                                                                                                                                         |                              |                               |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Course Code                  | Course Title                  |       |
| V                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Information Security 504-2-A | Information Security Concepts |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Prepared by                  | Mr. Dhankumar Wadar           |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Credits                      | Evaluation                    | Marks |
| DSE                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                            | 60(UE)+40(IA)                 | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                           |                              |                               |       |
| To make students to: <ul style="list-style-type: none"> <li>Introduce the learner to concepts involved in Information Security domain</li> <li>Theoretical understanding of Information Security Concepts</li> </ul>                                                                                                                                                                                                                                                |                              |                               |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                             |                              |                               |       |
| After completing the course the students shall be able to<br><b>CO1:</b> To understand the basic concepts of information security.<br><b>CO2:</b> To understand the application of Physical Security.<br><b>CO3:</b> To understand the basic concepts of network security.<br><b>CO4:</b> To understand the application of operating system security and database security.<br><b>CO5:</b> To understand the concept of web application, standards and cyber space. |                              |                               |       |

| Unit                                 | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Sessions (in Hrs) | COs Number | Teaching Methodology | Cognition Level | Evaluation Tools |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------|----------------------|-----------------|------------------|
| <b>Information Security Concepts</b> | Information Security, Need for Information Security<br>Cyber Security and Information Security<br><b>CIA</b> -Confidentiality, Integrity and Availability of Information,<br>Information classification<br>Risk, Threats, Vulnerabilities<br>Cyber Crimes<br>Data Security Identification, Authentication and Authorization,<br>Digital Signature, Cryptography, substitution and transposition ciphers, block cipher, stream cipher, Security Principles and Models | 9                 | CO1        | Lecture with Ppt     | Understand      | Short Answers    |

|                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |     |                    |            |               |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|--------------------|------------|---------------|
| <b>Physical Security</b>                                   | Physical Security and Facility Requirement, Perimeter Security, Fire Protection, Fire Suppression, Power Protection, General Environmental Protection, Equipment Failure Protection<br>Environmental Security(Critical Infrastructure Security)<br>Data Backup, Business Continuity and Disaster Recovery                                                                                                                                                                        | 8  | CO2 | Lecture with Ppt   | Apply      | Case Study    |
| <b>Network Security</b>                                    | <b>Network Security:</b><br>Secure Network design, Firewalls-Design and Types of Firewalls, Personal Firewalls,, IDS, email security, WLAN Security, VPNs, Types and Sources of Network Threats                                                                                                                                                                                                                                                                                  | 8  | CO3 | Lecture with Ppt   | Understand | Short answers |
| <b>Operating System Security</b>                           | Operating System Security and Application Security<br>Windows, Linux/UNIX, file permissions in UNIX<br><b>Database Security:</b> MS SQL                                                                                                                                                                                                                                                                                                                                          | 8  | CO4 | Lecture with ppt   | Understand | Short answers |
| <b>Web Application Security &amp; Compliance Standards</b> | <b>Web Application Security, Cloud Security</b><br>Web Application Vulnerabilities, Secure Coding Techniques, Continuous Security Testing and Assessments<br>Cloud Computing, Benefits, Security challenges<br><b>Compliance Standards :</b><br>IT Act, ISO 27001, ITIL Framework<br><b>Other Standards/Best practices – NIST CSF, SOC 2,</b><br>What's new in the Cyber World<br>Cyber Threats, Types<br>Security Operations Center<br>Cyber Forensics<br>AI and Cyber Security | 12 | CO5 | Lectures with PPTs | Understand | Short Answer  |

**Reference Books:**

| Sr.No. | Name of the Author                    | Title of the Book                                                                                                                                       | Year | Publisher Company |
|--------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------|
| 1      | Linda Volonino &<br>Reynaldo Anzaldua | Computer Forensics:<br>Principles and Practices                                                                                                         | 2006 | Pearson           |
| 2      | Jason Andress                         | The Basics of<br>Information<br>Security:<br>Understanding the<br>Fundamentals of<br>InfoSec in Theory<br>and Practice 1st<br>Edition Kindle<br>Edition | 2011 | Syngress          |

**Online Resources:**

| Online Resources No. | Website address                                                                                                                                       |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.javatpoint.com/principle-of-information-system-security">https://www.javatpoint.com/principle-of-information-system-security</a> |
| 2                    | <a href="https://www.javatpoint.com/cyber-security-and-information-security">https://www.javatpoint.com/cyber-security-and-information-security</a>   |
| 3                    | <a href="https://www.geeksforgeeks.org/what-is-information-security/">https://www.geeksforgeeks.org/what-is-information-security/</a>                 |

**MOOCs:**

| Resources No. | Website address                                        |
|---------------|--------------------------------------------------------|
| 1             | NPTEL / Swayam                                         |
| 2             | <a href="http://www.coursera.com">www.coursera.com</a> |

| Programme :BCA CBCS– Revised Syllabus w.e.f.-Year2022 –2023                                                                                                                                                                                                                                                                                                                                                                                                             |                              |                                     |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Course Code                  | Course Title                        |       |
| VI                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Information Security 604-2-B | Information Security Administration |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Prepared by                  | Mr. Dhankumar Wadar                 |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Credits                      | Evaluation                          | Marks |
| DSE                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3                            | 60(UE)+40(IA)                       | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                               |                              |                                     |       |
| <ul style="list-style-type: none"> <li>Introduce the learner to concepts involving security administration</li> <li>Introduce the learner about setup of LAN, connection and its setup.</li> </ul>                                                                                                                                                                                                                                                                      |                              |                                     |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                              |                                     |       |
| <p>After completing the course the students shall be able to</p> <p><b>CO1:</b> To understand the setup, manage and security of a client.</p> <p><b>CO2:</b> To understand the setup, manage and security of LAN.</p> <p><b>CO3:</b> To understand the connection of a LAN to the Internet.</p> <p><b>CO4:</b> To understand sharing an Internet connection and resources over a LAN.</p> <p><b>CO5:</b> To understand Setup support servers and Hosting a Website.</p> |                              |                                     |       |

| Unit                                 | Content                                                                                                                                                                          | Sessions (in Hrs) | COs Number | Teaching Methodology | Cognition Level | Evaluation Tools      |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------|----------------------|-----------------|-----------------------|
| <b>Setup a Client</b>                | Introduction to client-side devices, Setup, Manage and Secure a Desktop PC<br>Setup, Manage and Secure a Mobile Device<br>Monitoring and managing the Client OS and Applications | 9                 | CO1        | Lecture with Ppt     | Understand      | Quiz<br>Short Answers |
| <b>Setup a LAN</b>                   | Introduction to LAN devices, Simulate a LAN, Setup, Manage and Secure a Local Area Network<br>Firewalls, Zero Trust, Segmentation                                                | 9                 | CO2        | Lecture with Ppt     | Understand      | Case Study            |
| <b>Connect a LAN to the Internet</b> | Introduction to WAN devices, Setup, Manage and Secure a Connection to the Internet                                                                                               | 6                 | CO3        | Lecture with Ppt     | Understand      | Quiz<br>Short answers |

|                                                                |                                                                                                                                                                                                                                                                                             |    |     |                    |            |                   |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|--------------------|------------|-------------------|
| <b>Share an Internet Connection &amp; resources over a LAN</b> | Introduction to Internet Connection sharing, Introduction to NAT and PAT Setup, Manage and Secure a Proxy Server, Implementing Security policies, Login Security<br><b>Share resources over a LAN:</b> Setup, Manage and Secure a Print Server, Setup, Manage and Secure a File server      | 10 | CO4 | Lecture with ppt   | Understand | Quiz              |
| <b>Setup support servers &amp; Hosting a Website</b>           | <b>Setup support servers:</b> Setup, Manage and Secure a Mail Server, Setup, Manage and Secure a FTP Server, Setup, Manage and Secure a Boot Server, Setup, Manage and Secure a DNS Server<br><b>Host a Website:</b> Introduction to website hosting, Setup, Manage and Secure a Web Server | 11 | CO5 | Lectures with PPTs | Understand | Quiz Short Answer |

#### Reference Books:

| Sr.No. | Name of the Author | Title of the Book                                  | Year | Publisher Company |
|--------|--------------------|----------------------------------------------------|------|-------------------|
| 1      | Christopher Negus  | Red Hat Linux Bible: Fedora and Enterprise Edition | 2003 |                   |
| 2      | Mark Stamp         | Information Security, Principles and Practice      | 2011 | Wiley India       |

#### Online Resources:

| Online Resources No. | Website address                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 1                    | <a href="https://www.tutorialspoint.com/communication_technologies">https://www.tutorialspoint.com/communication_technologies</a> |
| 2                    | <a href="https://www.pcweenie.com/book/export/html/23">https://www.pcweenie.com/book/export/html/23</a>                           |

#### MOOCs:

| Resources No. | Website address                                                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1             | <a href="https://www.mooc-list.com/tags/security-management">https://www.mooc-list.com/tags/security-management</a>                                                       |
| 2             | <a href="https://www.futurelearn.com/courses/digital-security-policy-and-management-sc">https://www.futurelearn.com/courses/digital-security-policy-and-management-sc</a> |
| 3             | <a href="http://www.coursera.com">www.coursera.com</a>                                                                                                                    |

### Elective Group III – Data Science

| Programme: BCA CBCS –Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                         |                                 |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Course Code             | Course Title                    |       |
| V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Data Science<br>504-3-A | Statistical Programming using R |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Prepared by             | Dr.M.K.Patil                    |       |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Credits                 | Evaluation                      | Marks |
| DSE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                       | 60(UE)+40(IA)                   | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                         |                                 |       |
| <ul style="list-style-type: none"> <li>To teach the Beginners of R Programming of the a master level.</li> <li>A variety of topics will be covered that are important for Data science to prepare the students for real life prediction of data engineering.</li> <li>To impart knowledge of the concepts related to Probability and Application on data sets.</li> <li>It also gives the idea how data is managed in various environments with emphasis on Predictions measures as implemented in data sets.</li> </ul> |                         |                                 |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                         |                                 |       |
| <b>CO1:</b> Remember the definitions of concepts and their Implementation in R.<br><b>CO2:</b> Understand the concept of data and statistical techniques for its Implementation.<br><b>CO3:</b> Design different data behaviors and their Predictions.<br><b>CO4:</b> Analyzing Data set & Studying Historical Data.<br><b>CO5:</b> Convert the historical Data into Prediction Model using R                                                                                                                            |                         |                                 |       |

| Unit No. | Content                                                                                                                                                                               | Session (Hrs.) | COs Number   | Teaching Methodology                                         | Cognition Level | Evaluation Tools          |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|--------------------------------------------------------------|-----------------|---------------------------|
| 1        | <b>Introduction of Probability</b><br>Concept, Types of Probability, Permutation and Combination concept, Addition and Multiplication Theorem, Condition Probability, Bayes's Theorem | 8              | CO 1<br>CO 2 | Lecture with PPTs                                            | Understand      | Problems and its Solution |
| 2        | <b>Random Variable</b><br>Concept, Discrete and Continuous Random Variable, Probability density function, Mathematical Expectation and their Theorem                                  | 5              | CO 1<br>CO 2 | Problem Illustration                                         | Apply (Analyze) | Problems and its Solution |
| 3        | <b>Data Distribution</b><br>Distribution, Types of Data distribution, Exponential distribution, Binomial distribution, Normal                                                         | 7              | CO 3         | Concept Explanation, Mathematical Problems, and its Solution | Analyze         | Problems and its Solution |

|   |                                                                                                                                                                                                                                                                                                                                                                            |    |      |                                                              |          |                           |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------|--------------------------------------------------------------|----------|---------------------------|
|   | distribution, Poisson distribution, Random number generation, Monte Carlo Simulation.                                                                                                                                                                                                                                                                                      |    |      |                                                              |          |                           |
| 4 | <b>Testing of Hypothesis</b><br>Procedure of Testing Hypothesis, Standard Error and Sampling distribution, Estimation, Student's t-distribution, Chi-Square test and goodness of fit, F-test and analysis of variance. Factor analysis.                                                                                                                                    | 5  | CO4  | Concept Explanation, Mathematical Problems, and its Solution | Evaluate | Problems and its Solution |
| 5 | <b>Introduction to R programming language</b><br>Getting R, Managing R, Arithmetic and Matrix Operations, Introduction to Functions, Control Structures. Working with Objects and Data: Introduction to Objects, Manipulating Objects, Constructing Data Objects, types of Data items, Structure of Data items, Reading and Getting Data, Manipulating Data, Storing Data. | 5  | CO 5 | Concept Explanation, Mathematical Problems, and its Solution | Create   | Problems and its Solution |
| 6 | <b>Graphical Analysis using R</b><br>Basic Plotting, Manipulating the plotting window, Box Whisker Plots, Scatter Plots, Pair Plots, Pie Charts, Bar Charts.                                                                                                                                                                                                               | 5  | CO 5 | Software Demonstration and use of R Language                 | Evaluate | Problems and its Solution |
| 7 | <b>Advanced R</b><br>Statistical models in R, Correlation and regression analysis, Analysis of Variance (ANOVA), creating data for complex analysis, Summarizing data, and case studies.                                                                                                                                                                                   | 10 | CO 5 | Software Demonstration and use of R Language                 | Evaluate | Problems and its Solution |

|                        |                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Text Books</b>      | Fundamentals of Statistics" Seven Edition By S.C.Gupta                                                                                                                                                                                                                                                                                                                                   |
| <b>Reference Books</b> | 1."Fundamentals of Statistics" Seven Edition By S.C.Gupta<br>2."R Programming Fundamentals by KaelenMedeiras<br>3." Reinforcement Learning e-book.<br>4. Learning R Programming Guide on line<br><b>Suggested MOOC :</b> Please refer these websites for MOOCS:<br>NPTEL / Swayam <a href="http://www.edx.com">www. edx.com</a> , <a href="http://www.coursera.com">www.coursera.com</a> |



| <b>Programme: BCA CBCS –Revised Syllabus w.e.f. - Year 2022 – 2023</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                 |                                     |              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------------|--------------|
| <b>Semester</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Course Code</b>              | <b>Course Title</b>                 |              |
| <b>VI</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Data Science<br/>604-3-B</b> | <b>Introduction to Data Science</b> |              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Prepared by</b>              | <b>Dr.M.K.Patil</b>                 |              |
| <b>Type</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Credits</b>                  | <b>Evaluation</b>                   | <b>Marks</b> |
| DSE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3                               | 60(UE)+40(IA)                       | 100          |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                 |                                     |              |
| <ul style="list-style-type: none"> <li>To teach the Beginners of Data analysis through R /Python Programming of the a master level.</li> <li>A variety of topics will be covered that are important for Data science in order to prepare the students for real live Project Analysis</li> <li>To impart knowledge of the concepts related to Machine Learning and implement and variety Application on data sets.</li> <li>It also gives the idea how data is managed in various environments with emphasis on Analysis measures as implemented.</li> </ul> |                                 |                                     |              |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                 |                                     |              |
| <b>CO1:</b> Remember the definitions of concepts and their Programming skills.<br><b>CO2:</b> Understand the fundamentals of Data Science, methods, techniques, and its implementation<br><b>CO3:</b> Design different Model, test for its validity, and apply to different domain area.<br><b>CO4:</b> Design different Model, test for its validity, and apply to different domain area.<br><b>CO5</b> Analysing Data set and Comparing different Model.Convert the analysis in Modern approaches.<br><b>CO6:</b> Write R/Python coding for Analysis      |                                 |                                     |              |

| <b>Unit No.</b> | <b>Content</b>                                                                                                                                                                                             | <b>Session (Hrs .)</b> | <b>COs Number</b> | <b>Teaching Methodology</b> | <b>Cognition Level</b> | <b>Evaluation Tools</b>   |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------|-----------------------------|------------------------|---------------------------|
| 1               | <b>Association Rule</b><br>Mining Frequent Patterns, Associations, and Correlations: Basic Concepts and a Road Map, Association Rules, the Apriori Algorithm<br>Classification and Prediction              | 5                      | CO 1<br>CO 2      | Lecture with PPTs           | Understand             | Problems and its Solution |
| 2               | <b>Classification</b><br>Classification, Issues Regarding Classification, Classification by Decision Tree Induction, Bayesian Classification, Rule-Based Classification, Metrics for Evaluating Classifier | 5                      | CO 2<br>CO 3      | Problem Illustration        | Apply (Analyze)        | Problems and its Solution |

|   |                                                                                                                                                                                                                                                                                                                                           |    |              |                                                              |          |                           |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------------|--------------------------------------------------------------|----------|---------------------------|
|   | Performance, Holdout Method and Random Sub sampling                                                                                                                                                                                                                                                                                       |    |              |                                                              |          |                           |
| 3 | <b>Prediction</b><br>Prediction, Issues Regarding Prediction, Accuracy and Error Measures, Evaluating the Accuracy of a Classifier or Predictor. Clustering : Cluster Analysis, Agglomerative versus Divisive Hierarchical Clustering, Distance Measures in Algorithmic, Evaluation of Clustering                                         | 5  | CO 3<br>CO4  | Concept Explanation, Mathematical Problems, and its Solution | Analyze  | Problems and its Solution |
| 4 | <b>Linear Regression</b><br>Prediction using Linear Regression, Gradient Descent, Linear Regression with one variable, Linear Regression with multiple variables, Polynomial Regression, Feature Scaling/Selection                                                                                                                        | 5  | CO 3<br>CO 4 | Concept Explanation, Mathematical Problems, and its Solution | Evaluate | Problems and its Solution |
| 5 | <b>Logistic Regression</b><br>Classification using Logistic Regression, Logistic Regression vs. Linear Regression, Logistic Regression with one variable and with multiple variables                                                                                                                                                      | 5  | CO 3<br>CO 4 | Concept Explanation, Mathematical Problems, and its Solution | Create   | Problems and its Solution |
| 6 | <b>Deep Learning</b><br>History, Scope and specification, why deep learning now, building block of neural network, neural networks, Deep learning hardware. Backward and forward neural networks, XOR model, cost function estimation (maximum likelihood), units, activation functions, layers, , normalization, hyper-parameter tuning, | 10 | CO 5<br>CO 6 | Software Demonstration and use of R Language                 | Evaluate | Problems and its Solution |

|   |                                                                                                                              |    |              |                                              |          |                           |
|---|------------------------------------------------------------------------------------------------------------------------------|----|--------------|----------------------------------------------|----------|---------------------------|
|   | Convolution neural networks, architecture                                                                                    |    |              |                                              |          |                           |
| 7 | <b>Case study</b><br>Iris Data set ,Loan Data set, Titanic survival Data set ,Share Market Data set, Covide -19 Data set etc | 10 | CO 5<br>CO 6 | Software Demonstration and use of R Language | Evaluate | Problems and its Solution |

|                        |                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Text Books</b>      | <b>An Introduction to Machine Learning Springer by GopinathRebala</b>                                                                                                                                                                                                                                                                                                                      |
| <b>Reference Books</b> | 1. Fundamentals of Statistics" Seventh Edition By S.C.Gupta<br>2. <b>An Introduction to Machine Learning Springer byGopinathRebala</b><br>3. <b>Deep Learning MIT Press by John D.Kelleher.</b><br><b>Suggested MOOC :</b> Please refer these websites for MOOCS:<br>NPTEL / Swayam <a href="http://www.edx.com">www. edx.com</a> , <a href="http://www.coursera.com">www.coursera.com</a> |

### Elective Group III – Information System

| Programme: BCA CBCS –Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                                                                                          |                               |                         |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                 | Course Code                   | Course Title            |       |
| V                                                                                                                                                                                                                                                                                                                        | Information System<br>504-4-A | E-Commerce              |       |
|                                                                                                                                                                                                                                                                                                                          | Prepared by                   | Dr.Devendra Puntambekar |       |
| Type                                                                                                                                                                                                                                                                                                                     | Credits                       | Evaluation              | Marks |
| DSE                                                                                                                                                                                                                                                                                                                      | 3                             | 60(UE)+40(IA)           | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                |                               |                         |       |
| <ul style="list-style-type: none"> <li>To thoroughly understand the information technology for supporting E-commerce;</li> <li>To understand the necessary infrastructure and functional components to develop Ecommerce systems;</li> <li>To understand the design and application of E-commerce systems.</li> </ul>    |                               |                         |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                  |                               |                         |       |
| <b>CO1:</b> Recognize the impact of Information and Communication technologies, especially of the Internet in business operations<br><b>CO2:</b> Recognize the fundamental principles of e-Business and e-Commerce<br><b>CO3:</b> Use tools and services of the internet in the development of a virtual e-commerce site |                               |                         |       |

| Unit | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Sessions (Hrs) | COs Number | Teaching Methodology            | Cognition Level | Evaluation Tools                                   |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|---------------------------------|-----------------|----------------------------------------------------|
| 1    | <b>Introduction to E-Commerce:</b><br>Definition, E-commerce fundamentals, different types of E-commerce<br>E-Commerce Infrastructure - The Internet and World Wide Web, Web system, Internet basics, Characteristics of Internet, Components of Internet – Uniform Resource Locators, Internet Protocol, Hypertext Transfer Protocol (HTTP), Internet Service Provider (ISP), Types of ISP, domain name, domain name types<br>E-commerce vs Traditional Commerce, Networking Categories, Mobile Commerce | 8              | CO 1       | Lecture with Ppts<br>Quiz       | Understand      | Quiz<br>End Term<br>Internals:<br>Short<br>Answers |
| 2    | <b>Business Models for e-commerce:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 8              | CO 2       | Lecture with Ppts<br>Case Study |                 | Case Study ,<br>Newspaper<br>Article               |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |          |                                                         |                 |                                                                                         |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------|---------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------------|
|   | Business-to-Consumer (B2C), Consumer-to-Consumer (C2C), Business-to-Business (B2B)<br><b>Electronic Data Interchange</b><br>Requirement of EDI, types of EDI, Advantages and Disadvantages of EDI                                                                                                                                                                                                                                                                                                                    |   |          | Psychometric Tools                                      | Apply (Analyse) | End Term: Applied Questions                                                             |
| 3 | <b>E-commerce Payment System:</b><br>Limitations of traditional payment system, requirement of e-payment system, Internet payment systems - Credit card payment (e.g., SET protocol), E-cash, E-check, smart card, Electronic Funds Transfer, Digital Token Based E-Payment Systems, Modern Payment Systems, Steps for Electronic Payment, Payment Security, Net Banking                                                                                                                                             | 8 | CO 2     | Lecture with PPTs<br>Case Study                         | Understand      | Case Study with Presentations<br>End Term Exams: Case based Questions/Applied Questions |
| 4 | <b>Applications of E-Commerce:</b><br>E-commerce in banking, retailing, online publishing, online marketing, e-advertising, e-branding.                                                                                                                                                                                                                                                                                                                                                                              | 6 | CO2      | Lectures with PPTs<br><br>Group Activity<br>Video Cases | Evaluate        | Group Activity<br><br>End Term Exam: Short case and situation based questions           |
| 5 | <b>E-commerce Security:</b><br>Security issues, Privacy issues, Computer Security, security threats, security tools, Denial-of-Service attacks, Viruses, Unauthorized access to a computer network, Vulnerability of Internet Sites requirements, malicious code, intruders, attacking methods, Cryptography- encryption and decryption, public key encryption, private key cryptography, message digest, digital signature, digital certificate, firewalls, SSL. Firewall – Packet filtering, Application gateways. | 9 | CO2      | Lecture<br>Case Activity                                | Create          | Case Presentation Activity<br>End Term: Theory Applied                                  |
| 6 | <b>Implementation of E-Commerce:</b><br>WWW.EBAY.COM - B2C Website – Registration, Growth of eBay, PayPal – New Trend in Making                                                                                                                                                                                                                                                                                                                                                                                      | 6 | CO2, CO3 | Lectures with PPTs<br>Flip Classroom                    | Create          | Activity<br>End Term: Theory Applied                                                    |

|  |                                                         |  |  |  |  |  |
|--|---------------------------------------------------------|--|--|--|--|--|
|  | Payments Online, National<br>Electronic Funds Transfer. |  |  |  |  |  |
|--|---------------------------------------------------------|--|--|--|--|--|

**References :**

- E-commerce - C.S.V. Murthy, Himalaya Publishing House
- E-commerce A Managerial Perspective - P.T. Joseph, Prentice Hall Of India
- Frontiers of Electronics Commerce - Kalakota and Whinston, Pearson Education

| Programme: BCA CBCS –Revised Syllabus w.e.f. - Year 2022 – 2023                                                                                                                                                                                                                                                                                  |                               |                         |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------|-------|
| Semester                                                                                                                                                                                                                                                                                                                                         | Course Code                   | Course Title            |       |
| VI                                                                                                                                                                                                                                                                                                                                               | Information System<br>604-4-B | Knowledge Management    |       |
|                                                                                                                                                                                                                                                                                                                                                  | Prepared by                   | Dr.Devendra Puntambekar |       |
| Type                                                                                                                                                                                                                                                                                                                                             | Credits                       | Evaluation              | Marks |
| DSE                                                                                                                                                                                                                                                                                                                                              | 3                             | 60(UE)+40(IA)           | 100   |
| <b>Course Objectives:</b>                                                                                                                                                                                                                                                                                                                        |                               |                         |       |
| <ul style="list-style-type: none"> <li>The objective of the course is to provide the basic skills of managing knowledge in organizations. Knowledge is an asset for retaining the competitive advantage of the organization. This course develops the capabilities of towards managing students to manage knowledge in organizations.</li> </ul> |                               |                         |       |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                          |                               |                         |       |
| <b>CO1:</b> Will be able to understand the concepts of Knowledge and knowledge management .<br><b>CO2:</b> Can be able to design and develop Knowledge management systems for Business applications .                                                                                                                                            |                               |                         |       |

| Unit | Content                                                                                                                                                                                                                                                  | Sessions (Hrs) | COs Number | Teaching Methodology                                  | Cognition Level | Evaluation Tools                                                         |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|-------------------------------------------------------|-----------------|--------------------------------------------------------------------------|
| 1    | <b>Introduction:</b><br>Definition, Scope and Significance of Knowledge Management , Difficulties of Knowledge Management, Techniques of KM – Implementation of KM, Organizational knowledge, Characteristics and Components of Organizational Knowledge | 9              | CO 1       | Lecture with Ppts<br>Quiz                             | Understand      | Quiz<br>End Term<br>Internals:Short Answers                              |
| 2    | <b>Drivers of knowledge Management:</b><br>Pillars of knowledge Management, KM framework , Supply Chain of KM , Formulation of KM strategy.                                                                                                              | 8              | CO 2       | Lecture with Ppts<br>Case Study<br>Psychometric Tools | Apply (Analyse) | Case Study , Newspaper Article<br>End Term: Applied Questions            |
| 3    | <b>Technology and KM:</b><br>Technology components of KM – IT & KM , Ecommerce and KM                                                                                                                                                                    | 7              | CO 2       | Lecture with PPTs<br>Case Study                       | Understand      | Case Study with Presentations<br>End Term Exams: Case based Questions/Ap |

|   |                                                                                                                                                                                                                                                                                                                                            |   |     |                                                         |          |                                                                               |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|---------------------------------------------------------|----------|-------------------------------------------------------------------------------|
|   |                                                                                                                                                                                                                                                                                                                                            |   |     |                                                         |          | plied Questions                                                               |
| 4 | <b>Total Quality Management and KM:</b><br>TQM and KM , Bench marking and KM.                                                                                                                                                                                                                                                              | 6 | CO2 | Lectures with PPTs<br><br>Group Activity<br>Video Cases | Evaluate | Group Activity<br><br>End Term Exam: Short case and situation based questions |
| 5 | <b>Implementation of KM:</b><br>Discussion on Roadblocks to success, Implementing a KM programme , Critical Success Factors in KM , Implementation of KM                                                                                                                                                                                   | 7 | CO2 | Lecture<br>Case Activity                                | Create   | Case Presentation Activity<br>End Term: Theory Applied                        |
| 6 | <b>KM and Organizational Restructuring:</b><br>The Mystique of Learning, Organization:- Outcomes of learning, Learning and Change – Innovation, continuous Improvements, Corporate Transformation.<br><b>Case studies in Knowledge Management :</b> Knowledge management in Health Care, Knowledge Management in Human Resource Management | 8 | CO2 | Lectures with PPTs<br>Flip Classroom                    | Create   | Activity<br>End Term: Theory Applied                                          |

#### References (Books, Websites etc.):

1. MadhukarShukla:Competing Through Knowledge-Building a learning Organisation(Response Books, New Delhi.
2. Tiwana, The Knowledge Management Toolkit: Practical Techniques for building a Knowledge Management Systmes, 2/e, Pearson Edu.
3. Honey Cutt : “Knowledge Management Strategies”, PHI, New Delhi.
4. A wad, KM, Pearson Edn, 2007.
5. Barnes, Knowledge Management Systems, 1/e, Thomson 2006.
6. IkudiroNonka&Hiroataka Takeuchi, “ The Knowledge – Creating Company”, Oxford University Press, London.